

Heterogeneous Attention to Inflation and Monetary Policy

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Abstract

We study how heterogeneous attention to inflation affects monetary policy transmission. Using household-level surveys, we find that highly educated households pay more attention to inflation. A Heterogeneous Agent New Keynesian model in which education generates differences in permanent income and endogenous attention to inflation shows that relative to rational expectations, monetary tightening causes a milder recession when expectations are stable. Less-educated households misperceive real labor income and increase labor supply, thereby incurring larger welfare losses, a channel we confirm in the data. A policy experiment shows that weaker central bank responses to inflation increase attention and worsen the inflation-output trade-off.

Keywords: Inattention, HANK, Monetary Policy, Inflation Expectations

JEL Codes: D84, D91, E21, E71, E52

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1. Introduction

Inflation expectations matter for the macro economy but the way they are generally introduced into state-of-the-art macroeconomic models is not in line with micro-level evidence. Micro-level data shows that inflation expectations systematically differ between household groups, indicating an important dimension of household heterogeneity in understanding the macroeconomic impact of monetary policy (see [Weber et al. \(2022\)](#)). In this paper, guided by the empirical estimates of the inflation expectations process from household-level data, we incorporate an endogenous inflation expectation formation process, driven by the level of attention households pay towards inflation, into a Heterogeneous Agent New Keynesian model and find that the resulting distributional consequences of monetary policy are larger than under a rational expectations framework.

Firstly, we show empirically that households do not have full information rational expectations (FIRE) about inflation but are rather inattentive to inflation. The level of inattention thereby differs significantly between household groups and leads to heterogeneous inflation expectations. These inflation expectations then transmit to heterogeneous households' labor supply decisions, which ultimately drive consumption-saving decisions. Based on [Pfäuti \(2024\)](#) we estimate the level of attention to inflation, which is measured by the Kalman gain placed on forecast errors, across different household groups. We employ data from the Survey of Consumer Expectations from the New York Fed (SCE) for the US and the Consumer Attitudes, Sentiments and Expectations in Australia Survey (CASIe) for Australia and find robust cross-country evidence that highly educated households, i.e. those who attended colleges, pay significantly more attention to inflation than less educated households, i.e. those whose education is below the college-level. Other socio-demographic characteristics like age, home-ownership or gender either drive less heterogeneity in attention levels or provide less robust results¹.

Secondly, to quantify the aggregate implications of our empirical findings, we develop a Heterogeneous Agent New Keynesian (HANK) model with heterogeneity in income, wealth, and attention to inflation and where households face occasionally binding borrowing constraints. Educational attainment generates permanent differences in labor productivity and, therefore permanent income across households, providing a direct link between the empirical evidence and the model. More educated households are endowed with higher permanent productivity, while less educated households have lower permanent productivity. Idiosyncratic Bewley-type productivity shocks generate additional transitory income fluctuations and drive ex-post income heterogeneity.

To account for the empirically found heterogeneous attention to inflation, we draw on [Gabaix \(2014\)](#) and assume that households are inattentive to parts of the economy and face a constraint on how much information they can process. Specifically, households have a “sparse” representation of the world in the sense that they only observe variables which are of first order importance to their decision-making. We assume that they are inattentive to inflation and hence the price level, while being fully attentive to the rest of the economy². Households then have

1 Our findings of heterogeneous attention levels between educational levels is also in line with the evidence in [D'Acunto et al. \(2023\)](#) who find heterogeneity in inflation forecasts across different IQ levels

2 This framework can of course be extended by assuming inattention with respect to other variables. Since we specifically want to study the implications of non-rational expectations about prices, we assume inattention only

to decide once and for all how much attention they want to allocate towards inflation, which ultimately affects their subjective inflation expectations. This misperception of inflation due to only observing it partially, leads to suboptimal consumption-saving and labor supply choices and a loss in utility which in equilibrium is equal to the costs of paying attention. The model is calibrated to match our empirical evidence of education-specific attention levels. We close the model with monopolistically competitive firms and a Taylor-type monetary policy rule.

Through the lens of this model, we find that in response to a contractionary monetary policy shock, less educated, i.e., low-income households who are at or close to their borrowing constraint, increase their hours worked in response to a monetary tightening by more than highly educated, and thus high-income households. Observing nominal wages and due to the misperception of inflation rates as a result of low attention, low-income households perceive a larger fall in their real labor income following a contractionary monetary policy shock than actually materializes. Amid a lack of savings buffers, an increase in labor supply allows these low-income households to smooth their consumption along the business cycle. High-income households experience similar dynamics, however, due to larger savings buffers and as a result of consumption-smoothing and higher inflation attention, and thus lower misperception of real income, their labor supply response is more muted. This channel is absent in representative agent models since a representative household is sufficiently wealthy to behave in line with their Euler equation and for whom the substitution effect dominates the income effect of labor supply. The dominant income effect is key for the monetary policy trade-off in our model. Due to the increased labor supply of low-income and inattentive households, the recession induced by a contractionary monetary policy is milder than in a standard FIRE economy, and the policy maker faces a more preferable inflation-output trade-off, where the latter is measured as the ratio of standard deviations of inflation to output. However, given the household's disutility from working, low-income households encounter large welfare losses. Thus, the distributional consequences of monetary policy are larger in an economy with inattentive agents than in an economy under FIRE³. To strengthen the argument, we vary the level of idiosyncratic risk, which presents the lack of safety nets in the economy, which increases the share of borrowing-constrained households. The higher the risk, the stronger the labor supply response, and as a result, the differences in welfare costs between households are amplified even more.

Armed with an understanding of the mechanism through which heterogeneous inflation expectations affect the transmission of monetary policy, we study policy design. Lowering the Taylor rule coefficient in the central bank's monetary policy rule leads to an increase in attention levels and a less preferable inflation-output trade-off for the central bank once the change in attention level is accounted for. So, if a central bank responds systematically less strongly to the deviations of inflation from target, households increase their levels of attention to inflation due to the increase in the volatility of inflation. This in turn makes the increase in labor supply after the monetary policy tightening weaker, and the output falls by more in response to the tightening worsening the inflation-output trade-off. To put the strength of the channel in

towards inflation and the aggregate price level. In additional simulations of the quantitative model we find that assuming inattention to interest rates, in addition to inflation and price levels, leads to the same conclusions for the comparison of inattention to inflation and rational inflation expectations.

3 Distributional effects under FIRE are assessed in, for example, [Auclert \(2019\)](#) or [Gornemann et al. \(2021\)](#); for an empirical analysis see [Coibion et al. \(2017\)](#).

perspective, we compare calibrations of the model for the current period with the 1970-1980s, which featured a relatively lower coefficient in front of inflation in the Taylor rule and a lower supply of highly educated households. While a lower inflation coefficient in the Taylor rule leads to an increase in attention and thus a weaker labor supply response, a larger share of less educated households implies a stronger labor supply response due to inattention. We find that the increase in attention in 1970-1980s in comparison to the current period due to a less reactive to inflation central bank implied a worse inflation-output trade-off, while higher share of less educated households with lower levels of attention led to an improvement in the sacrifice ratio. On balance, we find that the latter channel played a larger role with the sacrifice ratio being smaller by 20% and therefore the trade-off being better in the 1970-1980s period in comparison to the current period.

To verify our inattention to labor supply channel in the data, we explore the positive relationship between labor supply across the educational distribution and the stability of inflation expectations in response to monetary policy shocks. Using the local projection framework of [Jordà \(2005\)](#) and the monetary policy shocks of [Bauer and Swanson \(2023\)](#), we find that after a contractionary monetary policy shock, the probability of being employed increases for less educated households more after an increase in inflation expectations compared to highly educated households.

The model is solved using the sequence-space Jacobian method by [Auclert et al. \(2021\)](#). To simulate our model with non-rational inflation expectations, we solve for the fixed point between different attention levels and volatilities of inflation to conduct policy counterfactuals: more precisely, we vary the Taylor rule coefficient in front of the inflation. The fixed point arises as a result of an inverse relationship between attention levels and inflation volatility following our analytical results employing [Gabaix \(2014\)](#). With counterfactually fixed levels of attention from the baseline specification, we find that a decrease in the Taylor rule coefficient from its baseline value improves the central bank's inflation-output trade-off. Once the fixed point between attention levels and inflation volatility is solved, an increase in attention levels eliminates the benefits of a less stronger response to inflation deviations from its target and implies a worse trade-off.

Related Literature. We add to four strands of literature. First, we provide new empirical evidence of household-level inattention to inflation for both US and Australia. Early contributions that study cross-sectional variations in inflation expectations include [Malmendier and Nagel \(2016\)](#), [Madeira and Zafar \(2015\)](#) and [Ehrmann and Tzamourani \(2012\)](#) who show that cohorts build their inflation expectations depending on their lifetime inflation experiences or whose expectations are shaped by their demographic characteristics introducing a lot of heterogeneity across households, a fact complemented by recent contributions from [Coibion et al. \(2020\)](#), [Weber et al. \(2023\)](#) and [Pfäuti \(2024\)](#) who observe that households' inattention varies with economic conditions. [Link et al. \(2023\)](#) find that attention to macroeconomic variables is strongly persistent at the individual level, specifically, more attentive households are more likely to adjust inflation expectations during a shock to inflation as the cost of acquiring new information is lower for an attentive household than for an inattentive household which translates to an adjustment of expectations - a finding consistent with theories of inattention ([Sims, 2003](#); [Gabaix, 2014](#)). In a recent paper [D'Acunto et al. \(2025\)](#), whose findings complement [Das et al. \(2020\)](#) and

D’Acunto et al. (2023) in studying inflation expectations based on socio-economic status, find that inflation expectations are lower among highly educated and high-income households. In addition, the authors postulate that differences in inflation expectations between educational groups are likely driven by their attention to inflation. We confirm this hypothesis by providing direct empirical evidence for education-specific attention. More broadly, we add to this literature by firstly, estimating heterogeneous attention levels following the framework of Pfäuti (2024) and secondly, by studying the relation between inflation expectations and household labor supply as a new transmission channel. Our results of a positive labor supply channel in response to high inflation expectations complement the work by Coibion et al. (2023) and Jiang et al. (2024) who find that individuals form pessimistic views about real income and total spending if they expect future inflation to increase. Although the findings in our paper are also complementary to the observed inflation induced income-specific labor supply response by households in Stantcheva (2024), our paper emphasizes the effects on the labor market from inflation expectation rather than inflation itself.

Second, our paper adds to the growing literature that incorporates information frictions in macroeconomic models. Lately, these frictions have also been incorporated into models with heterogeneous agents and incomplete markets to not only match macroeconomic moments but also reconcile them with microeconomic evidence on marginal propensities to consume. One application of non-FIRE in heterogeneous economies is Angeletos and Huo (2021) who analyze the consequences of noisy information for agents with different wealth levels. They show that “the habit-like sluggishness generated by informational frictions is amplified when the agents with the highest marginal propensity to consume are also the ones with the most cyclical income”. In a closely related paper to ours, Broer et al. (2021a) study heterogeneous expectations in a model with information frictions and find that the accuracy of households’ economic forecasts is mostly positively related to wealth and employment status. If households differ across these dimensions, they also acquire different information sets about the economy which ultimately results in heterogeneous forecasts. Similarly to our paper, they find that the choice to be better informed increases with households’ wealth levels. In a methodologically related paper to ours, Guerreiro (2022) builds a HANK model with households who have endogenous but heterogeneous beliefs about their cyclical income. He shows that if households are heterogeneously exposed to business cycle shocks which affect their beliefs about future income, then business cycle fluctuations are amplified. Compared to Pfäuti and Seyrich (2022), whose model assumes cognitive discounting on the household side in the sense of Gabaix (2020), we implement a cognitive discounting parameter that is endogenously chosen by the household and is group-specific. We add to this literature by applying the sparsity-based model by Gabaix (2014) or rather the dynamic programming extension in Gabaix (2016) to a HANK model, allowing agents with heterogeneous permanent income levels to endogenize their attention choice towards inflation, ultimately resulting in heterogeneous inflation expectations. We thereby partially speak to Moll (2025) who emphasizes the necessity of introducing non-rational expectations about prices into heterogeneous macroeconomic models.

Third, we contribute to the heterogeneous agent literature studying monetary policy transmission (see also Kaplan et al., 2018; Auclert, 2019; Acharya et al., 2023; Bayer et al., 2024). The closest paper to ours on this front is Auclert et al. (2020). In their paper, the authors study

monetary policy transmission in a HANK model in which households have sticky expectations. Contrary to them, our model features heterogeneous inflation expectations, whereas in their framework, all households update their expectations and beliefs about their value of illiquid assets and the state of the economy infrequently, but with the same probability. Furthermore, the information choice in our framework is endogenized and source-dependent while theirs is not. In a partial equilibrium heterogeneous agent model, [Laibson et al. \(2024\)](#) show that present bias increases households' marginal propensity to consume and also amplifies the effect of monetary policy, leading to an increase in households' consumption, which however also decelerates monetary transmission. Additionally, [Angeletos and La'O \(2020\)](#) and [Eusepi and Preston \(2018\)](#) study optimal monetary policy and inflation expectations when agents are imperfectly informed. [Pfäuti and Seyrich \(2022\)](#) show that cognitive discounting generates amplification of monetary policy through indirect effects which has strong implications for the business cycle due to unequal exposure of households to shocks. We add to this literature by quantifying the effects of heterogeneity and information frictions for inflation expectations in a sticky-price model and documenting the transmission channel of monetary policy through labor supply.

Lastly, our results on the labor supply channel relate our paper to studies of labor supply in heterogeneous agent environments and in response to monetary policy shocks, see [Faia and Shabalina \(2025\)](#), [Eeckhout and Sepahsalari \(2020\)](#), [Gornemann et al. \(2021\)](#), [Graves et al. \(2023\)](#), [Cantore et al. \(2026\)](#), [Berger et al. \(2025\)](#) and [Broer et al. \(2020\)](#). More precisely, we study the interaction between inflation expectations (that react to monetary policy) and labor supply, uncovering a new transmission channel of monetary policy under heterogeneity through a supply-driven labor market. Our work is complementary to [Pilossoph and Ryngaert \(2024\)](#) who show that employed workers with high inflation expectations are more likely to search for and transfer to a new job compared to households with low inflation expectations. The households' job search thereby transmits inflation expectations to nominal wage demands. We add to this by additionally uncovering effects at the extensive margin and by studying the implications for monetary policy in a HANK environment.

The paper is structured as follows. Section 2 presents the empirical estimation strategy and results of the estimated attention levels to inflation. Section 3 introduces the attention choice problem in an analytical heterogeneous agent model without transitory income risk. Then, Section 4 presents our quantitative HANK model which features heterogeneity in attention, ex-ante and ex-post income levels. We use the model to quantify the structural implications of heterogeneous and endogenous inattention to inflation on the aggregate economy. Section 5 shows model results and discusses the implications for the design of monetary policy. Section 6 confirms our theoretical results with our empirical evidence of monetary policy driven responses of labor supply using survey data. Section 7 concludes.

2. Empirical Analysis

This section presents our empirical strategy to estimate heterogeneous attention to inflation from micro-level data. We use two sources of data, the NY Fed SCE for the USA and CASiE for Australia. We use quarterly data for June 2013 to December 2023 for the USA and January 1995 to December 2023 for Australia. Highly educated households hold a college degree, while less

educated households have a lower than college degree⁴. To estimate attention levels from the data, we follow Pfäuti (2024) and extend the method by incorporating household heterogeneity.

2.1. Cross-Sectional Variation

To better understand the heterogeneity in the data, we first analyze the cross-sectional variation in households' inflation expectations for both countries. We run the following regression

$$FE_{j,t} = \beta_0 + \sum_j \beta_j D_j + \epsilon_j \quad (1)$$

where $FE_{j,t} = \pi_t - E_{j,t-4}\pi_t$ is the absolute or squared forecast error inflation of an individual household j measured as the deviation of last year's inflation expectations $E_{j,t-4}\pi_t$ to current inflation π_t . D_j are dummy variables that capture the following household characteristics: gender (female, male), income level (low, mid, high), occupation level (full-time, part-time, unemployed for the USA and occupational categories for Australia), home ownership (rent, own), age level (below 40, 40-60, above 60), education level (high-school, some college, college for the USA and high-school, advanced diploma, trade certificate or certificate III/IV, non trade certificate/University or higher for Australia), and dummy for if the household works full-time and if it is self-employed. Additionally, for Australia, we include state, regional, and change jobs dummies. Errors are clustered at the year-quarterly level. Also, we bin households based on their self-employment status, sex, income level, occupational level, educational level, homeownership, age level, and regions they live in⁵. Using the survey data data, we construct quarter-on-quarter CPI Inflation as $\pi_t = 100 \times \left(\frac{P_t - P_{t-1}}{P_{t-1}} \right)$, where P_t is the quarterly consumer price index in t , and quarterly inflation expectations $E_t\pi_{t+1} = \frac{E_t\pi_{t+4}}{4}$ where $E_t\pi_{t+4}$ are one-year-ahead inflation expectations governed by the surveys.

The results of estimating (1) are displayed in Table 1. We regress the absolute (columns 2 and 4) or squared (columns 3 and 5) inflation forecast errors on the household characteristics specified in column 1. Columns 2 and 3 show the results for the USA, while columns 4 and 5 show the results using Australian data. In general, each household makes an average forecast error, independent of its specific group, denoted by the constant. If a household belongs to either the income, educational, sex, or occupational groups, the error is changed by the magnitude of the coefficient shown in the respective columns. To fix ideas, let us focus on column 2: on average a household in the USA makes a significant absolute inflation forecast error of 3.42% per quarter. If the household, however, has a college degree, this error significantly decreases by 1.14%, such that the error decreases to 2.28%. If the household is self-employed, the unconditional error diminishes by the smaller magnitude of 0.25%, such that the forecast error remains higher than that of a college-educated household. Similar reasoning is applicable to the other categories. Those characteristics in Table 1 however are not mutually exclusive. For robustness, we also check across other demographics (home ownership (own or rent), the age level (below 40, 40-60, above 60), region (Mid, West, South, North), and dummy variables for working full-time), however, they produce less significant results across our samples. In

⁴ See Appendix A.1 and A.2 for descriptive statistics of the data.

⁵ Self-employment dummy is omitted when bins are constructed using Australian survey as it becomes omitted. Dropping one of the other dummies and including self-employment dummy instead does not change the results.

Table 1: Cross-Sectional Distribution of Inflation Expectations

	USA		Australia	
	Abs. Errors	Squared Errors	Abs. Errors	Squared Errors
Income				
Mid-Income	-0.46*** (0.04)	-3.72*** (0.46)	-0.25* (0.08)	-1.47* (0.61)
High-Income	-0.67*** (0.04)	-4.95*** (0.48)	-0.44*** (0.09)	-2.48*** (0.64)
Education				
Some College	-0.59*** (0.06)	-4.95*** (0.64)	0.00 (0.05)	0.02 (0.46)
College	-1.14*** (0.06)	-8.59*** (0.58)	-0.13** (0.04)	-0.65* (0.28)
Sex				
Male	-0.76*** (0.03)	-5.35*** (0.34)	-0.21*** (0.04)	-1.02** (0.30)
Occupation				
Self-Employed	-0.25*** (0.05)	-2.00*** (0.45)	-0.23** (0.07)	-1.65*** (0.43)
Demographics	X	X	X	X
Constant	3.42*** (0.10)	21.56*** (1.03)	1.83*** (0.17)	7.38*** (1.38)
N	18,893	18,893	7,810	7,810
R-squared	0.11	0.07	0.06	0.03

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The table shows the results for the regression of inflation expectation forecast errors on different demographics. Errors are clustered at the year-quarter level. Results are shown for both USA and Australia (AUS) using quarterly data. Demographics includes the following indicator variables as regressors: home ownership (own or rent), the age level (below 40, 40-60, above 60), region (Mid, West, South, North or states for Australia) and a dummy variable for working full-time. Additionally, for Australia the demographics include occupations, dummy if one changed jobs and regional dummy.

general, we see that education drives most of the heterogeneity in inflation expectations between households: being a highly educated household significantly reduces the magnitude of the average individual forecast error robustly across our samples. The results complement the empirical findings in several empirical studies (Das et al., 2020; D’Acunतो et al., 2025; Weber et al., 2022) and are robust to stricter winsorizing, the inclusion of time fixed effects and macro controls. Next, we show how education affects the attention to inflation, being an important driver of inflation expectations and forecast errors. We thereby group households into less educated (if their education is below college level) and highly educated (if they have a college degree) groups.

2.2. Attention to Inflation

Assume that households perceive the following law of motion for inflation

$$\pi_t = \rho\pi_{t-1} + \zeta_t \quad (2)$$

where π_t is the inflation rate in t , $\rho \in (-1, 1]$ is the autocorrelation coefficient, and $\xi_t \sim i.i.d.N(0, \sigma_\xi^2)$ is the inflation innovation in t . Each household j , based on their educational level, belongs to a group g . In every period, each household receives a noisy signal of inflation which according to them is generated as

$$s_{j,t} = \pi_t + \varepsilon_{j,t} \quad (3)$$

where $\varepsilon_{j,t} \sim i.i.d.N(0, \sigma_{\varepsilon_g}^2)$ is the noise in the signal. The noise term is perceived to have different variances σ_g^2 across household groups due to limited attention⁶: different types of households pay different amount of attention to inflation and the higher their attention, the lower the variation in the noise, $\varepsilon_{j,t}$. Applying the standard Kalman filter allows us to generate a conditional forecast of future inflation as

$$\begin{aligned} E[\pi_{t+1}|\mathcal{I}_{j,t}] &= \rho E[\pi_t|\mathcal{I}_{j,t}] = \rho E[\pi_t|\mathcal{I}_{j,t-1}] + \rho\gamma^g (s_{j,t} - E[\pi_t|\mathcal{I}_{j,t-1}]) \\ &= \rho E[\pi_t|\mathcal{I}_{j,t-1}] + \rho\gamma^g (\pi_t - E[\pi_t|\mathcal{I}_{j,t-1}]) + v_{j,t} \end{aligned} \quad (4)$$

where $E[\pi_t|\mathcal{I}_{j,t}]$ is the nowcast for inflation once the household receives new information, $E[\pi_t|\mathcal{I}_{j,t-1}]$ is the household's prior mean, $v_{j,t}$ is the noise. The inflation forecast is a linear combination of the household's prior mean, the product of the Kalman gain, and the difference between the realized inflation and the previous period inflation forecast. When the household forecasts inflation it corrects its forecast by the error multiplied by the amount of attention it pays towards inflation: the lower the level of attention, the less strong the update and the more anchored the prior beliefs. Averaging across j for each group and introducing different means across the groups allows us to estimate group-specific attention from the data via the following specification

$$E_{g,t}\pi_{t+1} = \beta_g + \beta_1 E_{g,t-1}\pi_t + \beta_2^g I_g (\pi_t - E_{g,t-1}\pi_t) + v_{g,t} \quad (5)$$

where β_g are the potentially different mean expectations of the households, I_g are type- g dummies for less and highly educational groups, $\beta_1 = \rho$, and $\gamma^g = \frac{\beta_2^g I_g}{\beta_1}$ is the group-specific level of attention. The mean inflation expectation of households in group g is a linear combination of its own lag and the latest observed forecasting error after the agent observed its signal about inflation.

The estimated attention levels using specification (5) for the two educational groups of both countries are shown in Table 2. γ^1 shows the attention to inflation of less educated households and γ^2 shows the attention for highly educated households. We find that there is a significant difference across attention levels between the household groups. In particular, highly educated households pay more attention to inflation than less educated households. These results are robust for both estimators, the Blundell-Bond⁷ and the pooled OLS estimator. Table 2 confirms our hypothesis that highly educated households are more attentive to inflation than less educated households. Further, the results are robust to stricter winsorizing and using an AR(2)-process as a perceived inflation process as well as to the inclusion of macro controls. We have also estimated the same attention specification for other demographic groups based on

6 This is different to Pfäuti (2024) who doesn't distinguish between household characteristics and the observed volatility of the shock is the same across households. For the analysis of individual or group-specific forecasts see also Vellekoop and Wiederholt (2019) or Broer and Kohlhas (2024).

7 The Blundell-Bond estimator is applied since the dependent variable in specification (5) includes a lag on the right-hand side. See also Pfäuti (2024).

occupation, self-employment status, sex, income level, home-ownership status, age, different states within the country, urban/rural and job-switcher dummies. None of these characteristics produce robust results with similar or larger heterogeneity in attention levels than heterogeneity across educational groups. When estimating attention jointly across a variety of characteristics, education is the characteristic that stands out and produces significant and robust results⁸.

Table 2: Attention to Inflation of Less and Highly Educated Households

	USA		AUS		
	Blundell Bond	Pooled OLS	Blundell Bond	Pooled OLS	Annual Blundell Bond
Less educated γ^1	0.08***	0.06**	0.05**	0.05	0.14**
Highly educated γ^2	0.20**	0.16**	0.08***	0.05	0.22***
Observations	78	78	218	218	218
Robust to AR(2) specification	Yes	Yes	Yes	Yes	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows levels of attention across household education groups by estimating $E_{g,t}\pi_{t+1} = \beta_g + \beta_1 E_{g,t-1}\pi_t + \beta_2^g I_g (\pi_t - E_{g,t-1}\pi_t) + v_{g,t}$ on quarterly data. Attention of less educated households is given by γ^1 , and attention of highly educated households is given by γ^2 .

3. The Household's Attention Choice Problem

To quantify the results from the empirical analysis we propose a Heterogeneous Agent New Keynesian (HANK) model with inattentive households and both permanent and transitory income heterogeneity. The permanent income heterogeneity is thereby driven by the educational levels of the households and is exogenously determined⁹, while the transitory income heterogeneity will stem from idiosyncratic risk, as standard in the HANK literature. To provide intuition, we first introduce a reduced form of the full HANK model generalized in Section 4 by abstracting from idiosyncratic risk¹⁰ and focusing on permanent income and attention heterogeneity. We present the general attention choice problem the household faces, following the framework by Gabaix (2014) or more specifically, Gabaix (2016) for dynamic programming problems.

Consider an economy populated by a continuum of households $j \in [0, 1]$. Each household belongs to an educational group $g \in \{1, 2\}$ and gets group-specific labor income $Y_{g,t} \equiv s_j W_t n_t$, where s_j is the educational level, W_t is the nominal wage, n_t are fixed labor hours the household has to work and which are equal for both household groups. If the household is in the low educational group, it belongs to $g = 1$ and gets income Y_1 , otherwise the household belongs to the group of the highly educated households $g = 2$, and receives income Y_2 , where $Y_2 > Y_1$

⁸ In Appendix B.1 we also analyze if there is heterogeneity between the reactions of households to supply and demand shocks. We do not find any significant results in group-specific reactions to demand or supply reactions and therefore continue with the analysis using the aggregate inflation and aggregate consumption basket.

⁹ Similar to Guvenen et al. (2014) wage inequality arises from differences in the household's ability to acquire a certain education respectively skill as well as from idiosyncratic shocks. Contrary to Guvenen et al. (2014) we don't endogenize the educational respectively skill accumulation. Doing so would lead to the same results we have in the present paper.

¹⁰ The role of which is assessed in Section 5.2.2

since $s_2 > s_1$. Hence, less educated households are ex-ante low-income households and highly educated households are ex-ante high-income households. Households also hold a risk-free nominal asset a_t^n with nominal interest rate i_t^a between periods $t - 1$ and t . Borrowing constraints prevent these households from taking negative bond positions. The household gets utility from consumption c_t and has a discount factor β .

At every point in time, the household forms beliefs about the behavior of future variables relevant to its decision problem, in particular, the aggregate price index, nominal labor income and the nominal interest rate. To isolate the effects driven by non-rational expectations about inflation, we assume that households have rational expectations about their future nominal income $E_t(\{Y_{g,t+h}\}_{h=1}^\infty)$, they fully observe the correct nominal interest rate and therefore have rational beliefs $E_t(\{i_{t+h}^a\}_{h=1}^\infty)$. We also make the simplifying assumption that agents know the current aggregate price index P_t . Thus, the only variables to which the household may be inattentive to and forms subjective beliefs about is future inflation π_{t+h} , $h = 1, 2, \dots$ and consequently the future aggregate price index. In other words, the household takes the aggregate sequences as given, but the sequence of perceived inflation depends on their (endogenous) attention. Fully rational and inattentive agents both operate within the same objective environment, their perceptions of it differ, however. Formally, a household in this economy solves the following perceived maximization problem

$$\begin{aligned} V_j(m_t) &= \max_{c_t, a_t^n} \{u(c_t) + \beta \tilde{E}[V_j(m_{t+1})]\} \\ \text{s.t. } P_t c_t + a_t^n &= Y_{g,t} + (1 + i_t^a) a_{t-1}^n, \quad a_t^n \geq 0 \end{aligned} \quad (6)$$

where $\tilde{E} \equiv \gamma^\delta E_t$ is the subjective expectation operator defined in [Definition 1](#), E_t is the traditional rational expectation operator and m_t describes the macroeconomic state¹¹ as

$$m_t = (a_{t-1}^n, \{\pi_{t+j}\}_{j=0}^\infty, \{i_{t+j}^a\}_{j=0}^\infty, \{Y_{g,t+j}\}_{j=0}^\infty) \quad (7)$$

with $\gamma^\delta \in [0, 1]$ being the household's degree of attention. If a household has FIRE, the attention parameter is $\gamma^\delta = 1$, while $\gamma^\delta = 0$ indicates that the household is fully inattentive. Thus, if $\gamma^\delta \in [0, 1)$ the household has some level of inattention and $\{\gamma^\delta \pi_{t+j}\}_{j=0}^\infty$ denotes the inflation path perceived under inattention. If $\gamma^\delta = 1$ the perceived inflation path is the same as the actual inflation path $\{\pi_{t+j}\}_{j=0}^\infty$. Consumption taken under the subjective model is given by c_t^s . We define the subjective expectations of a household as follows

Definition 1. Subjective Beliefs. Suppose the agent observes variable z_t and forms expectations about $\{z_{t+h}\}_{h=1}^\infty$. Then, the expectations about the future variable are:

$$\tilde{E}(z_{t+h}) = \gamma^\delta E_t(z_{t+h}) \text{ if } \gamma^\delta \in [0, 1) \quad (8)$$

$$\tilde{E}(z_{t+h}) = E_t(z_{t+h}), \text{ if } \gamma^\delta = 1 \quad (9)$$

where E_t is the rational expectation operator, γ^δ is the attention to variable z_t and we define $\tilde{E} \equiv \gamma^\delta E_t(z_{t+h})$ as the subjective beliefs operator.

¹¹ Given that the household fully perceives labor income and the nominal interest, we could also just take the state variables to be $m_t = (a_{t-1}^n, \{\pi_{t+j}\}_{j=0}^\infty)$

Subjective expectations $\tilde{E}$ are the expectations taken with respect to the household's subjective model (with partial inattention to macro variables) under the agent's discounting of future variables of γ^g . Thus, if the household is partially inattentive to variable z_{t+h} , they form expectations according to their subjective expectations $\tilde{E}(z_{t+h}) = \gamma^g E_t(z_{t+h})$. With increasing attention, the quality of subjective beliefs about inflation increases, possibly aligning with fully rational expectations. Notice that the agent is rational about its non-rationality, such that the policy of the behavioral agent is the policy of the rational agent under its inattention, c_t^s . The policy of a fully rational agent is c_t^r . Combining the budget constraint and the utility function of the household in (6) allows us to analyze the implications of attention to inflation as follows.

The *rational* household who observes the variables fully rational and pays full attention to the entire state of the economy has realized utility (i.e., utility evaluated at the objectively true environment with rational expectations)

$$v_{j,t}(c_t^r, m_t) \equiv u(c_t^r) + \beta E_t[V_j^r(Y_{g,t} + (1 + i_t^a)a_{t-1}^n - P_t c_t^r, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)] \quad (10)$$

The *inattentive* household is inattentive to inflation and has perceived utility (i.e., utility evaluated at the subjective environment with subjective expectations)¹²

$$\tilde{v}_{j,t}^s(c_t^s, m_t) \equiv u(c_t^s) + \beta \tilde{E}[V_j^s(Y_{g,t} + (1 + i_t^a)a_{t-1}^n - P_t c_t^s, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)] \quad (11)$$

However, evaluating the policy choice under inattention at the objectively true environment (i.e., with rational expectations), gives the inattentive household's realized utility

$$v_{j,t}(c_t^s, m_t) \equiv u(c_t^s) + \beta E_t[V_j(Y_{g,t} + (1 + i_t^a)a_{t-1}^n - P_t c_t^s, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)] \quad (12)$$

Being inattentive leads to an expected loss in realized utility for the inattentive household

$$\mathcal{L} = \mathbb{E}(v_{j,t}(c_t^r, m_t) - v_{j,t}(c_t^s, m_t)) \quad (13)$$

measured as the difference between realized utilities under the optimal policy c_t^r with rational expectations and the optimal policy c_t^s with inattention. $\mathbb{E}$ is the unconditional objective expectations operator. Hence, the household wants to minimize its expected loss in utility before any shocks are realized. Further, the agent can only process a limited amount of information and, therefore, faces a cognitive constraint, $\chi\gamma^g$, where $\chi > 0$ measures the amount of sparsity. If $\chi = 0$, the agent is the traditional rational agent and can acquire new information at 0 cost. Minimizing the expected loss in utility subject to the cognitive constraint¹³

$$\min_{\gamma^g} -\frac{1}{2} \frac{\partial^2 v}{\partial c^2} \sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \left(\frac{\partial c}{\partial \pi_h} \frac{\partial c}{\partial \pi_{h'}} \sigma_{\pi_h \pi_{h'}} \right) (1 - \gamma^g)^2 + \chi \gamma^g \quad (14)$$

where $\frac{\partial c}{\partial \pi_h}$ denotes the household's change in consumption due to a change in inflation at

12 More precisely, utility is given by $v_{j,t}(c_t^s, m_t) = v_{j,t}(c_t, \gamma^g m_1, \gamma^g m_2, \gamma^g m_3, \dots)$. If we, for example, assume $m_t = \pi_t$, then utility is $v_{j,t}(c_t, \gamma^g \pi_1, \gamma^g \pi_2, \gamma^g \pi_3, \dots)$. In Section 4 this will be extended by a default inflation level to which households anchor their expectations: $v_{j,t}(c_t, \gamma^g \pi_1 + (1 - \gamma^g)\pi_1^d, \gamma^g \pi_2 + (1 - \gamma^g)\pi_2^d, \gamma^g \pi_3 + (1 - \gamma^g)\pi_3^d, \dots)$

13 Following Gabaix (2014) we approximate the household's utility function with a second order Taylor approximation; see Appendix C.1 for a detailed derivation.

horizon h evaluated at the zero-inflation steady state, $\sigma_{\pi_h \pi_{h'}} = \mathbb{E}(\pi_h \pi_{h'})$ the auto-covariance of the aggregate inflation level at horizon h and h' , and $\frac{\partial^2 v}{\partial c^2}$ is the disutility from misoptimized consumption evaluated at the steady state with $v \equiv v(c, m)$ such that $\frac{\partial c}{\partial \pi_h} (1 - \gamma^g) \pi_{t+h}$ denotes the error due to inattention, gives optimal attention of the household to inflation as follows

Proposition 1. Optimal Attention. *The optimal level of attention $\gamma^g \in [0, 1]$ of a household towards inflation is given by*

$$\gamma^g = \max \left\{ 0, 1 - \frac{\chi}{\Lambda} \right\} \quad (15)$$

where the cost-of-inattention factor is $\Lambda := -\frac{\partial^2 v}{\partial c^2} \sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \left(\frac{\partial c}{\partial \pi_h} \frac{\partial c}{\partial \pi_{h'}} \sigma_{\pi_h \pi_{h'}} \right)$.

Proof. See Appendix C.1 □

Lemma 1. Group-level Attention. *Since the partial derivatives of the steady-state utility and the policy functions in (15) are identical for all agents sharing the same educational level $s_j \in \{s_1, s_2\}$ which selects households into their income group $g \in \{1, 2\}$, all households within group g solve the same objective and select the same optimal attention parameter γ^g .*

The optimal level of attention (15) depends on the state of the economy and the levels of income and wealth of the household. Depending on their income level, agents either value or do not value additional information about inflation. If a household is low-income (due to being less educated) and is close to the borrowing constraint, i.e., is hand to mouth (HtM), it has a high marginal propensity to consume (MPC) and is not able to smooth consumption and therefore does not acquire more information. However, if the household is high-income (due to being highly educated), it benefits from acquiring more information since the disutility from misoptimized consumption has direct effects on their savings, and large forecasting mistakes are costly for their wealth and income accumulation, making information acquisition more valuable for them. In addition, attention decreases with increasing wealth as a result of the decrease in the marginal utility of consumption.

When the cost of cognition χ increases, it becomes too costly for households to gather new information about the state of the economy. As a result, their expectations stay close or at their default value of inflation (usually the last observed inflation or perceived expectation), receiving no update.

Additionally, attention increases with perceived inflation volatility. Highly varying inflation creates a bigger taste for attention as it leads to high variation in levels of real income, making correct inflation forecasts more valuable and increasing attention levels. This is independent of households' income level.

4. Quantitative Model

In this section we generalize the model from Section 3 to a canonical one-asset HANK model with incomplete markets and heterogeneous beliefs. In particular, we allow households to differ in three dimensions of heterogeneity. First, each household is characterized by the idiosyncratic, transitory income (this introduces ex-post heterogeneity in income). Second, we assume that

households additionally differ in their education levels, which drive their permanent income levels (this characterizes ex-ante heterogeneity in income). The latter allows us to match the education-specific attention we found in the data. Third, we assume that households are inattentive to the aggregate price and inflation level and that households endogenously decide how much attention they want to allocate towards these objects. Based on that, they optimize their consumption-saving and labor decisions. The firm optimizes production and labor unions optimize welfare of their members as in [Ferriere and Navarro \(2025\)](#) or [Auclert et al. \(2021\)](#), while the central bank sets the nominal interest rate according to a Taylor-type rule.

4.1. Household Problem

In the following, we present the household problem of our model. The timing is as follows. First, the household makes the attention choice by evaluating its loss in utility from misoptimized consumption due to being inattentive. Second, the agent simultaneously decides on labor supply and consumption-saving. The labor supply margin is an important insurance margin that is especially used by low-income households and is crucial to understand for the monetary policy transmission and business cycle dynamics, as shown in [Cantore et al. \(2026\)](#), [Stantcheva \(2024\)](#) or [Faia and Shabalina \(2025\)](#).

4.1.1. Baseline Problem

The economy is populated by a continuum of households $j \in [0, 1]$. Each household has preferences over consumption c_t , hours worked n_t , gets nominal income $s_j e_t W_t n_t$ and faces a nominal budget as well as a borrowing constraints for nominal assets. Specifically, each household solves the following Bellman equation

$$\begin{aligned} V_j(S_t) &= \max_{c_t, n_t, a_t^n} \{u(c_t, n_t) + \beta \tilde{E}[V_j(S_{t+1})]\} \\ \text{s.t. } P_t c_t + a_t^n &= s_j e_t W_t n_t + (1 + i_t^a) a_{t-1}^n \\ a_t^n &\geq 0, \quad u(c_t, n_t) = \frac{c_t^{1-\sigma} - 1}{1-\sigma} - \varphi \frac{n_t^{1+\nu}}{1+\nu} \end{aligned} \tag{16}$$

where W_t is the nominal wage, P_t is the aggregate price index, e_t are Bewley-type idiosyncratic income shocks creating transitory income inequality in the economy, s_j are educational levels, and $S_t = (a_{t-1}^n, s_j, e_t, \{\pi_{t+j}\}_{j=0}^\infty, \{i_{t+j}^a\}_{j=0}^\infty, \{W_{t+j}\}_{j=0}^\infty)$ describes the macroeconomic state. σ, ν, β_j are the elasticity of intertemporal substitution, the inverse Frisch elasticity, and discount factors respectively, where latter evolve stochastically following a Markov chain $\pi_{\beta, \beta'}$ between states $[\beta, \beta - \Delta_\beta]$ (see [Krusell and Smith \(1998\)](#)). Households hold only one risk-free asset which are the nominal savings a_t^n with nominal interest rate i_t^a between periods $t-1$ and t . We assume that they have rational expectations with respect to nominal wages and interest rate, while they pay limited attention to inflation and thus the aggregate price index. In additional simulations of the model, we extend this assumption by allowing households to also have limited attention to the nominal interest rate (see Appendix E.5) and show that the results are robust. $\tilde{E}$ denotes the subjective expectations with respect to the aggregate and indiosyncratic states of the household and is described below.

4.1.2. Beliefs

At every point in time, the household forms beliefs about the future behavior of variables relevant to its decision-making. In the baseline model, we maintain the assumption that households have rational expectations about their nominal income and that they also fully observe the nominal interest rate and have FIRE about it. Thus, the only variables relevant to their decision-making to which the agent is inattentive to are the future price and inflation levels. We assume that households observe all current and past prices and only form subjective beliefs about future inflation rates and consequently the price index¹⁴. In the model with borrowing constraints, this assumption also guarantees, that the constraints are not violated due to misperception of current prices. We follow the same definition for the path of expectations for inflation that we use in our empirical estimation, see equation (5). We specify subjective beliefs in the following.

Definition 2. Subjective Beliefs, HANK. Suppose the agent observes variable z_t and forms expectations about $\{z_s\}_{s>t}$. Then, the expectations about the future variable are:

$$\tilde{E}(z_s) = \gamma^s E_t z_s + (1 - \gamma^s) \tilde{E}_{t-1} z_s \quad \forall s > t, \text{ if } \gamma^s \in [0, 1) \quad (17)$$

$$\tilde{E}(z_s) = E_t(z_s), \text{ if } \gamma^s = 1 \quad (18)$$

where E_t is the traditional rational expectation operator, γ^s is the attention to variable z_t and we define $\tilde{E} \equiv \gamma^s E_t z_s + (1 - \gamma^s) \tilde{E}_{t-1} z_s$ as the subjective beliefs operator.

Definition 2 formalizes a similar idea as in the Kalman gain, namely that households learn about the state of the economy if they receive a new signal about inflation updating their beliefs about future inflation. If agents decide to pay attention to inflation, they update their old inflation expectations and learn from new information. With increasing attention, the quality of subjective beliefs about inflation increases, eventually minimizing the importance of the previous inflation expectations, which is totally mitigated under $\gamma^s = 1$. If agents decide not to pay attention, their beliefs about future inflation are anchored at their old beliefs. For the aggregate price index, the expectational process follows the definition in [Definition 2](#) and is derived using subjective beliefs about inflation and the relationship between inflation and the price level, see [Appendix D.3](#) for more details.

Numerical Solution. We solve the model using the sequence-space Jacobian method by [Auclert et al. \(2020\)](#) which allows us to introduce behavioral frictions into the model. For that, we define the expectation matrix as

$$E = \begin{pmatrix} 1 & \gamma^s & & \gamma^s & & \gamma^s & & \dots \\ 1 & 1 & \gamma^s + (1 - \gamma^s)\gamma^s & & \gamma^s + (1 - \gamma^s)\gamma^s & & \dots \\ 1 & 1 & 1 & \gamma^s + (1 - \gamma^s)(\gamma^s + (1 - \gamma^s)\gamma^s) & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

where the columns represent the time period of the variable to which the household pays

¹⁴ In [Appendix D.4](#) and [E.8](#) we provide intuition about the implications of households misperceiving the current (and not only future) prices. Our results would most likely be amplified.

attention to and the rows represent the periods in which the expectation is formed. The lower triangular matrix of ones captures the fact that the household has full information about the current and past value of the input variable. γ^g is the level of attention of a household in educational, respectively, income group g towards the input variable (in our case we only focus on π_t and P_t) which affects the consumption-savings and labor supply choices. If the household holds FIRE, $E(r, l) = 1 \forall r, l$, i.e. has rational expectations E_t and pays full attention to the economy with $\gamma^g = 1$. Under inattention, the household has beliefs deviating from FIRE and $\gamma^g \in [0, 1)$ given by Definition 2. Using the fact that a change in expectations at time τ for the household is equivalent to a news shock

$$J_{t,s}^{o,i} = \sum_{\tau=0}^{\min\{s,t\}} (E_{\tau,s} - E_{\tau-1,s}) J_{t-\tau,s-\tau}^{FI,o,i} \quad (19)$$

where o is the output variable, i is the input variable, the relationship between behavioral Jacobians and FIRE Jacobians can be obtained in the following.

Proposition 2. Behavioral Jacobian. *In the outlined setting, behavioral Jacobians with respect to input i are related to FIRE Jacobians in the following form*

$$J_{t,s}^{o,i} = \begin{cases} \gamma^g J_{t,s}^{FI,o,i}, & t = 0, s > 0 \\ J_{t,s}^{FI,o,i}, & s = 0 \\ \gamma^g J_{t,s}^{FI,o,i} + (1 - \gamma^g) J_{t-1,s-1}^{o,i}, & t > 0, s > 0 \end{cases} \quad (20)$$

where J^{FI} are FIRE Jacobians.

Proof. See Appendix D.1 □

The Jacobian $J_{t,s}^{o,i}$ describes the marginal change in period t of the output variable o to a period- s -change in the input variable i . For consumption, the Jacobian with respect to a change in inflation would be given by $J_{t,s}^{C,\pi} = \delta C_t / \delta \pi_s$ which is the partial response in consumption in t to a change in inflation in s . Inattention thereby enters the household's beliefs via the expectation matrix, affecting the household's Jacobian as seen in equation (19). Note that we assume $\gamma^g \neq 1$ only for π_s and $P_s \forall s > 0, t > 0$, but keep the assumption of FIRE for every other input variable. In line with the literature, we assume that the aggregate price in period t is fully observed. Next, we endogenize γ^g and assume that households solve an attention allocation problem.

4.1.3. The Attention Choice

We assume that households choose their attention levels optimally, facing a cognitive capacity constraint. In the present HANK model, households do not only choose consumption (as in the small model in Section 3) but they also contemporaneously optimize their labor supply. As in Gabaix (2014) we call these variables the “action” variables. In the following, we assume that households are inattentive to inflation, and thus the aggregate price index, when they choose consumption and labor. This allows us to isolate the loss in utility from consumption and labor directly. Being inattentive leads to a suboptimal consumption and labor choice resulting in a

loss in utility, evaluated under the objectively true model, compared to the choice under FIRE. The household wants to minimize the loss by choosing its optimal level of attention γ^g .

The *rational* household who observes the variables fully rational and pays full attention to the entire state of the economy has realized utility

$$v_{j,t}(c_t^r, n_t^r, S_t) \equiv u(c_t^r, n_t^r) + \beta E_t[V_j^r(s_j e_t W_t n_t^r + (1 + i_t^a) a_{t-1}^n - P_t c_t^r, e_{t+1}, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{W_{t+j+1}\}_{j=0}^\infty)] \quad (21)$$

The *inattentive* household is inattentive to inflation and has realized utility

$$v_{j,t}(c_t^s, n_t^s, S_t) \equiv u(c_t^s, n_t^s) + \beta E_t[V_j(s_j e_t W_t n_t^s + (1 + i_t^a) a_{t-1}^n - P_t c_t^s, e_{t+1}, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{W_{t+j+1}\}_{j=0}^\infty)] \quad (22)$$

Being inattentive leads to policy choices, c_t^s and n_t^s , and an expected loss in realized utility

$$\mathcal{L} = \mathbb{E}(v_{j,t}(c_t^r, n_t^r, S_t) - v_{j,t}(c_t^s, n_t^s, S_t)) \quad (23)$$

where c_t^r and n_t^r is the optimal policies under full information and c_t^s and n_t^s are the policy functions under inattention when households form expectations according to their subjective expectations $\tilde{E}$. The household wants to minimize its expected loss in utility and faces a cognitive constraint on how much information to process, $\chi^g \gamma^g$ where $\chi^g > 0$. Replacing the household's utility function by a linear quadratic approximation allows us to state the household's attention optimization problem

$$\min_{\gamma^g} -\frac{1}{2} \sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \left(\left(\frac{\partial^2 v}{\partial c^2} \frac{\partial c}{\partial \pi_h} \frac{\partial c}{\partial \pi_{h'}} + \frac{\partial^2 v}{\partial n^2} \frac{\partial n}{\partial \pi_h} \frac{\partial n}{\partial \pi_{h'}} \right) \sigma_{\pi_h \pi_{h'}} \right) (1 - \gamma^g)^2 + \chi^g \gamma^g \quad (24)$$

where $\sigma_{\pi_h \pi_{h'}}$ is the standard deviation of inflation, $\frac{\partial^2 v}{\partial c^2}$ the disutility from misoptimized consumption with $v \equiv v(c, n, S)$ and $\frac{\partial^2 v}{\partial n^2}$ is the disutility from misoptimized labor supply. The first term in equation (24) is the leading term of the second order Taylor approximation of the expected utility loss of a household in equation (23). The second term, $\chi^g \gamma^g$, is the linear psychological cost attention creates, whereby under $\chi^g = 0$ the household is the rational agent. Note, that we allow for varying costs of attention among educational groups as those are necessary to match our empirical evidence of lower attention among less educated households, see Section 5.3.1. Solving equation (24) gives the household's optimal level of attention to inflation

Proposition 3. Optimal Attention, HANK. *The optimal level of attention to inflation of a household is given by*

$$\gamma^g = \max \left\{ 0, 1 - \frac{\chi^g}{\sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \left(\left(\frac{\partial^2 v}{\partial c^2} \frac{\partial c}{\partial \pi_h} \frac{\partial c}{\partial \pi_{h'}} + \frac{\partial^2 v}{\partial n^2} \frac{\partial n}{\partial \pi_h} \frac{\partial n}{\partial \pi_{h'}} \right) \sigma_{\pi_h \pi_{h'}} \right)} \right\} \quad (25)$$

Proof. See Appendix D.2 □

Lemma 2. Group-level Attention, HANK. *Let each household belong to an income group $g \in \{1, 2\}$ defined by their educational level s_j which can be partitioned into two disjoint state-space regions, s_1 and s_2 . Since the partial derivatives of the steady-state utility and the policy functions in (25) are identical for*

all agents sharing the same s_j , all households within group s_j solve the same objective and select the same optimal attention parameter γ^s .

If $\gamma^s = 0$, the household “does not think about π ” and sets a default value for inflation and doesn’t update its expectations, and when $\gamma^s = 1$ the household pays full attention, perceives the true value of inflation and prices and updates its inflation expectations if necessary. Given our definition of expectations, attention accumulates over time and includes new information about the state of inflation and the aggregate price index in period s . It thereby depends positively on the volatility of inflation, $\sigma_{\pi_h \pi_{h'}}^2$, the change of group-specific consumption given a change in inflation, $\frac{\partial c}{\partial \pi_h}$, and the disutility from misoptimized consumption, $\frac{\partial^2 v}{\partial c^2}$. Again, a similar argument as in the small model can be applied here: attention, and therefore also the correct expectations of inflation, increases with inflation fluctuations in the economy, decreases with an increase in information acquisition costs and increases with the disutility from misoptimized consumption-saving decisions¹⁵.

4.2. The Rest of the Model

4.2.1. Production

Monopolistically competitive firms produce output by combining total labor input and capital using a Cobb–Douglas production function $y_t = z_t k_{t-1}^\nu L_t^{1-\nu}$, where y_t is the variety produced, z_t is the total factor productivity, ν is the capital share, k_t is capital and L_t is labor input. Competitive final good producers aggregate varieties using the CES aggregator, such that optimal demand for each variety is given by $p_t = \left(\frac{Y_t}{y_t}\right)^{\frac{1}{\eta}} P_t$, where P_t is the aggregate price index and is normalized to 1 in the steady state, η is the elasticity of substitution across varieties. Monopolistically competitive firms choose prices, p_t , labor demand, capital demand, k_t , and investment, I_t , to maximize the sum of future discounted real profits, which recursively reads as follows

$$J_t(k_{t-1}) = \max_{p_t, k_t, I_t, L_t} \left\{ \frac{p_t}{P_t} y_t - W_t L_t - I_t - \frac{\zeta}{2} \left(\frac{I_t}{k_{t-1}} - \delta \right)^2 k_{t-1} - \frac{\eta}{2\kappa} (\ln(1 + \pi_t))^2 Y_t + \frac{E_t J_{t+1}(k_t)}{1 + r_{t+1}} \right\}$$

$$\text{s.t. } k_t = (1 - \delta)k_{t-1} + I_t; \quad (26)$$

$$p_t = \left(\frac{Y_t}{y_t}\right)^{\frac{1}{\eta}} P_t; \quad y_t = z_t k_{t-1}^\nu L_t^{1-\nu}$$

where the first constraint is the capital accumulation equation, δ is the depreciation rate of capital, ζ is the investment adjustment cost parameter, $\frac{\eta}{2\kappa} (\ln(1 + \pi_t))^2 Y_t$ is the quadratic price adjustment cost necessary to study monetary policy and $w_t = \frac{W_t}{P_t}$. Investment adjustment costs are not necessary for the main mechanism, but they dampen investment volatility that is large otherwise, and as shown in Auclert et al. (2020), investment dynamics are an important driver

¹⁵ The aggregate loss function is derived by weighing the expected loss (24) with the ergodic distribution across households, shown in Appendix D.2 in (D.27).

of monetary policy transmission in HANK with deviations from FIRE. The first order condition with respect to prices gives the price Phillips curve

$$\ln(1 + \pi_t) = \kappa \left(mc_t - \frac{1}{\mu_p} \right) + E_t \frac{Y_{t+1}}{Y_t} \ln(1 + \pi_{t+1}) \Psi_{t,t+1} \quad (27)$$

where $\mu_p = \frac{\eta}{\eta-1}$ and $\Psi_{t,t+1}$ is the stochastic discount factor and equal to $\frac{1}{1+r_{t+1}}$ and $\pi_t \equiv P_t/P_{t-1} - 1$. κ is the coefficient for the slope of the Phillips curve.

4.2.2. Labor Unions

There is a continuum of monopolistically competitive labor unions that hire labor from households, create union-specific labor varieties and sell them to competitive labor packers. Labor unions face wage adjustment costs à la [Rotemberg \(1982\)](#) and maximize welfare of its members. Competitive labor packers aggregate labor varieties k using CES aggregator such that each union k choosing wage, $w_{k,t}$ and hours worked $n_{k,t}$ faces a demand for its varieties: $l_{k,t} = \left(\frac{w_{k,t}}{w_t} \right)^{-\epsilon_w} L_t$.

The problem of each union

$$\max_{w_{k,t}, n_{k,t}} \sum_{s=0}^{\infty} E_t \tilde{\beta}^s \left\{ \int \left\{ \frac{c_{i,t+s}^{1-\sigma} - 1}{1-\sigma} - \varphi \frac{n_{i,k,t+s}^{1+\nu}}{1+\nu} \right\} dD(e_t, a_{t-1}^n) - \frac{\epsilon_w}{2\kappa_w} \log \left(\frac{w_{k,t}}{w_{k,t-1}} \right)^2 \right\} \quad (28)$$

$$\text{s.t. } l_{k,t} = \int e_t n_{k,t} dD_t = \left(\frac{w_{k,t}}{w_t} \right)^{-\epsilon_w} L_t, \quad \int_k n_{k,t} dk = n_t \quad (29)$$

leads to a wage Phillips curve

$$\log(1 + \pi_{w,t}) = \kappa_w \left(\int \left(\varphi n_t^\nu - \frac{\epsilon_w - n_t/L_t}{\epsilon_w} w_t e_{i,t} c_{i,t}^{-\sigma} \right) dD(e_t, a_{t-1}^n) \right) L_t + \tilde{\beta} E_t [\log(1 + \pi_{w,t+1})] \quad (30)$$

where $\tilde{\beta}$ is the average discount factor across households: $\tilde{\beta} = \beta - \frac{\Delta\beta}{2}$, ϵ_w is the elasticity of substitution across varieties, κ_w is the slope of the wage Phillips curve and $\pi_{w,t} = (1 + \pi_t) \frac{w_t}{w_{t-1}} - 1$. The labor supply equation of each household in turn incorporates the Lagrange multiplier $\varphi n_t^\nu = w_t e_t c_t^{-\sigma} - \lambda_t e_t$, where the Lagrange multiplier is pinned down from the labor market clearing condition $\sum_j m_j \int s_j e_t n_{k,t} dD_t = L_t$. The wage rigidity is not essential for the transmission in the model, but implies a more muted wage response in line with the empirical estimates in [Christiano et al. \(2005\)](#), and, as pointed out in [Auclert et al. \(2023\)](#), is an important ingredient in heterogeneous agent models. For the responses without wage rigidity, see Appendix E.4.

4.2.3. Asset Market and Equilibrium Conditions

Let v_t denote the price of equity and d_{t+1} the firm dividend. The real return on equity is $\frac{d_{t+1} + v_{t+1}}{v_t}$. The no-arbitrage condition is $v_t = \frac{E_t(d_{t+1} + v_{t+1})}{1+r_{t+1}}$. The asset that households use for their savings is the equity index. Thus, the return on households' assets is $(1 + i_{t-1}^a) = \frac{d_t + v_t}{v_{t-1}} (1 + \pi_t)$ ¹⁶.

The aggregate supply of goods is equal to the aggregate demand of goods, hence, $Y_t =$

¹⁶ This includes the return on capital and profits due to the monopolistic power of the firms.

$C_t + I_t + \psi_t^p$, where consumption is aggregated across households as $C_t = \sum_j m_j \int c_t D(e_t, a_{t-1}^n)$, where m_j are the shares of households with different educational levels, and ψ_t^p are the price adjustment costs. Finally, asset market clearing implies $\frac{A_t}{P_t} = v_t$, where again the aggregation is obtained through $A_t = \sum_j m_j \int a_t^n D(e_t, a_{t-1}^n)$.

4.2.4. Policy

Monetary policy follows a Taylor-type rule, which endogenously responds to macroeconomic conditions as $i_t^a = \rho_r i_{t-1}^a + (1 - \rho_r)(r_t^* + \phi_\pi \pi_t + \phi_y(Y_t - Y_{ss})) + \varepsilon_t^r$, where i_t^a is the monetary policy interest rate, ε_t^r is the monetary policy shock, ρ_r is the smoothing parameter, ϕ_π is the weight on inflation π_t , ϕ_y is the weight on output gap, $(Y_t - Y_{ss})$, with Y_{ss} as the steady state value of output Y_t , r_t is the real interest rate, r_t^* is the natural interest rate, which is equal to the real interest rate in the steady state. The Fisher equation is given by $1 + r_t = \frac{1+i_{t-1}^a}{1+\pi_t}$.

Definition 3. Competitive Equilibrium. *A competitive equilibrium of the economy satisfies the following definition: the sequence $[c_t, a_t^n, n_t]_{t=0}^\infty$ solves households' consumption-saving and labor supply decisions in equation (16), given the distribution of idiosyncratic shocks, $P(e_{t+1}|e_t)$ and discount factor shocks $P(\beta_{t+1}|\beta_t)$ and the sequence of prices i_t^a, W_t, π_t, P_t and the attention choices. The policy functions resulting from the consumption-saving and attention problem solve a fixed point equilibrium. Aggregate asset holdings and consumption of the households are equal to the product of the individual optimal functions and the distribution of households across educational levels, idiosyncratic shocks and assets. Firms choose prices, labor demand, and capital inputs to solve discounted profit optimization, given in equation (26). Labor unions choose wages and labor hours according to equations (28) and (29). Labor and asset market clearing conditions and the aggregate resource constraints are satisfied. Monetary policy determines the short term interest rate according to the Taylor rule.*

4.2.5. Calibration

For the calibration of the model, except for the attention parameters whose calibration procedure we specify below, we follow the literature and assume that the steady state is common knowledge. This means that the steady state is solved under FIRE. The parameters for the calibration are shown in Table 3. To calibrate the model to the US economy we follow Auclert et al. (2021) for most of the parameters¹⁷. The investment adjustment cost parameter is chosen to achieve an investment standard deviation of 3.5 relative to the standard deviation of output under the shock calibration shown in Table 4. The real interest rate is set to 1% annually.

The shares of households who are highly educated and less educated, 0.56 and 0.44, respectively, are taken from our survey data that we use to estimate our attention parameters. Similarly, the attention parameters for those educational groups, 0.20 and 0.08, are taken from our own estimates in Table 2. Education specific productivities are chosen to match the skill premium of 70% estimated in Bengali et al. (2025) and such that $s_1 m_1 + s_2 m_2 = 1$ (normalization). Δ_β is chosen to match the average marginal propensity to consume of 0.23 from Boehm et al. (2025). Additional parameters of the Markov transition matrix ω and q are chosen as in Auclert et al.

¹⁷ $\delta, K/Y, \kappa, \kappa_w, \mu_w, \sigma, v, \rho_z, \sigma_z, A/Y$ are set as in Auclert et al. (2021).

Table 3: Calibration

Parameter	Description	Value
<i>Production Function</i>		
δ	Capital depreciation	0.02
K/Y	Capital to output ratio	10.0
κ	Slope of the price Phillips curve	0.1
ξ	Investment adj. cost parameter	10.0
r	Real interest rate	1% annually
<i>Labor Unions</i>		
κ_w	Slope of the wage Phillips curve	0.1
μ_w	Wage markup	1.1
<i>Households</i>		
σ	Elasticity of intertemporal substitution	0.5
ν	Inverse Frisch elasticity	1
ρ_z	Autocorrelation of earnings	0.966
σ_z	Conditional cross-sectional std of log earnings	0.92 (unconditional std 0.238)
A/Y	Total wealth to GDP	14.0
s_1, s_2	Education-specific productivities	1.22, 0.72
m_1, m_2	Shares of households at each education level	0.56, 0.44
γ_1, γ_2	Attention parameters	0.22, 0.11
Δ_β, ω, q	Discount factor shock parameters	0.1, 0.49, 0.01
<i>Monetary and Fiscal Policy</i>		
ϕ_π	Coefficient on inflation in Taylor rule	1.87
ϕ_y	Coefficient on output gap in Taylor rule	1.04
ρ_r	Smoothing parameter in Taylor rule	0.92

Table 4: Shock Process Parameters

Shocks	Monetary Policy	Markup	TFP
ρ	0.15	0.89	0.95
σ	0.24	0.14	0.46

Notes: The table shows the persistence and volatility of shocks used for the specification of the shock processes. The values are taken from [Smets and Wouters \(2007\)](#).

(2025) and imply that a new draw of β occurs every 25 years, which in the spirit of [Krusell and Smith \(1998\)](#) reflects generational turnover.¹⁸

Taylor rule parameters are set as in [Aastveit et al. \(2024\)](#) as they estimate a regime-switching Taylor rule, whose estimates for the second regime are used in the counterfactual in Section 5.3.2 to reflect the Taylor rule parameters relevant to the 1970-1980s.

Attention. For our counterfactual analysis of optimal attention as a function of policy changes,

¹⁸ $\Pi = \begin{pmatrix} (1-q)\omega & \omega q \\ q(1-\omega) & (1-q) + q\omega \end{pmatrix}$.

we do the following. We derive optimal attention analytically and find that it is an inverse function of the volatility of inflation, see equation (25). We simulate the model under three shocks: monetary policy, markup and technology shocks. Table 4 shows the values for the shock process parameters we use in the calibration. The values are taken from Smets and Wouters (2007). We find attention for different values of ϕ_π by solving a fixed point between the attention levels of two types of households (highly educated and less educated) and the volatility of inflation. So, for the baseline value of the parameter, $\phi_\pi = 1.87$, attention is calibrated to match the empirical results shown in Table 2 (parameters γ^1 and γ^2) through the calibration of the cost of attention parameter χ^g . For the other values of the ϕ_π , we guess the volatility for inflation σ_π , solve for attention levels given the volatility of inflation and the calibrated value of costs of attention from the baseline, then we simulate the model with the found levels of attention under the above specified calibration of shocks, verify that the guessed volatility of inflation is close to the equilibrium volatility of inflation given ϕ_π and if not, we update the guess of inflation volatility and repeat the process until convergence.

5. Model Results

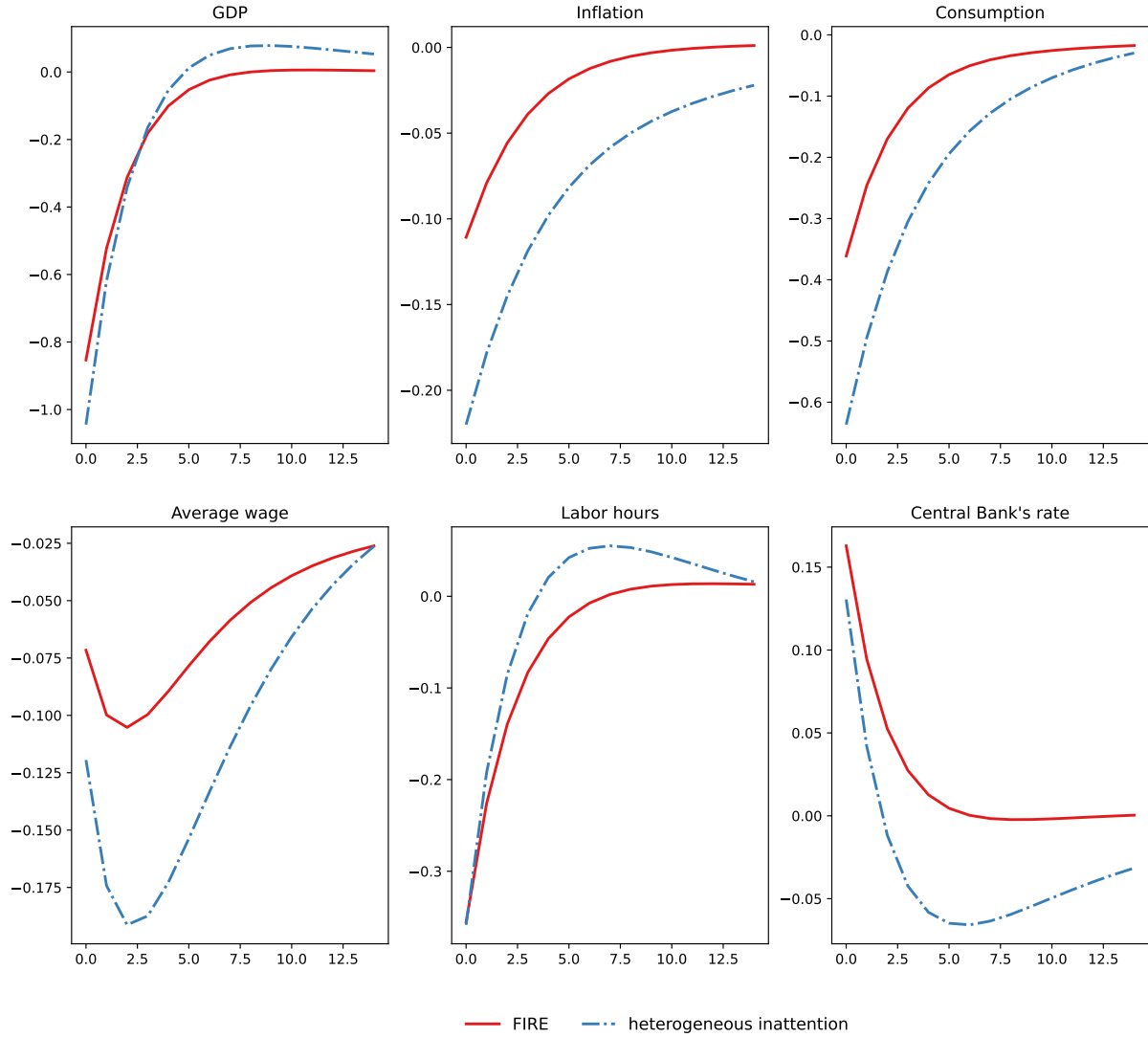
The following section presents model results for our quantitative HANK model with inattention to inflation. We first present impulse responses to a contractionary monetary policy shock¹⁹. We compare results to HANK and RANK economies, both under FIRE and as well as heterogeneous inattention, and vary the variance of idiosyncratic risk to highlight the insurance mechanism through labor supply. Then we conduct two counterfactual exercises: first, we vary the Taylor rule inflation coefficient to study the inflation-output trade-off under different policy rules in a model with households who are heterogeneous in their attention, income, and wealth levels, and second, we study the change in the monetary policy trade-off since the 1970-1980s period that was driven by the change in the skill premium and supply of college-educated workers, providing more evidence for the permanent income channel we assume in the model.

5.1. Heterogeneous Inattention and Monetary Policy

To study the effects of heterogeneous inattention across income levels, we allow households to differ in their education, respectively, permanent income levels. We compare impulse responses in our model with heterogeneous inattention ($\gamma^1 \neq \gamma^2 \in [0, 1)$) to the traditional FIRE ($\gamma^g = 1 \forall g = [1, 2]$) economy. Figure 1 shows the impulse response functions to a 25 basis points contractionary monetary policy shock. Across both economies, the shock decreases inflation and induces a recession on the aggregate level: output, consumption, the average wage, and labor hours decrease on impact. The average wage decreases due to the decrease in labor demand, which accompanies the decrease in consumption and production. As a result of a decrease in wages, labor supply and consumption also fall. Due to wage rigidity, the wage level adapts more slowly. However, in an economy with heterogeneous inattention, households misperceive the fall in inflation to be smaller than it actually is, such that the perceived real wage decreases by more and the perceived real interest rate increases by less than they actually do.

¹⁹ Impulse responses to TFP and markup shocks can be found in Appendix E.2.

Figure 1: Impulse Response Functions to a Monetary Policy Shock



Notes: The figure shows in percentage deviations from the steady state the impulse responses of output, consumption, average wage, and labor hours to a contractionary 25 bps monetary policy shock with 0.15 persistence. Inflation and Central Bank's interest rate are shown in deviations from the steady state. The red line shows the impulse response function under full information rational expectations, the dashed blue line shows the results under heterogeneous inattention.

The reason being that due to inattention to prices, the household does not realize that a decrease in inflation or the price level actually implies a less severe decrease in their real wage as they instead fully observe their nominal wage but only partially capture their real income. Therefore, households perceive themselves to be poorer than they actually are, leading them to work more to compensate for their perceived loss in real income, thus supplying more labor. The larger perceived drop in labor income also leads to a larger decrease in consumption, when compared to the FIRE economy, which dominates the more muted intertemporal substitution effect that stimulates consumption. This is another manifestation of larger indirect effects compared to a direct effect in models with heterogeneous households which we further decompose into its

components in Appendix E.1. The effect is amplified by two forces: persistence due to wage stickiness and persistence due to inattention²⁰.

In the HANK model with inattention, the increase in the household's marginal propensity to consume additionally incentivizes the agents to increase their labor supply, and due to labor capital complementarity, it also incentivizes firms to increase investments, stimulating the overall economy, making the recession less deep in the medium run. Due to a higher labor supply, real wages decrease by more in the model with inattention and the larger decrease in consumption under inattention leads to a more pronounced fall in inflation following the contractionary monetary policy shock. Over time, households update their expectations following Definition 2 and observe their actual real income more correctly, adjust their labor supply and the economy recovers.

Welfare Analysis Figure 2 extends this analysis and decomposes the aggregate effect into its education-specific group components. The figure shows the responses of consumption, labor hours and the welfare for highly educated households in the top panel and the less educated households in the bottom panel. Welfare is evaluated by calculating the implied consumption equivalent variation as the change in the permanent consumption of a household needed to make them just as well off as if the shock has not happened.²¹

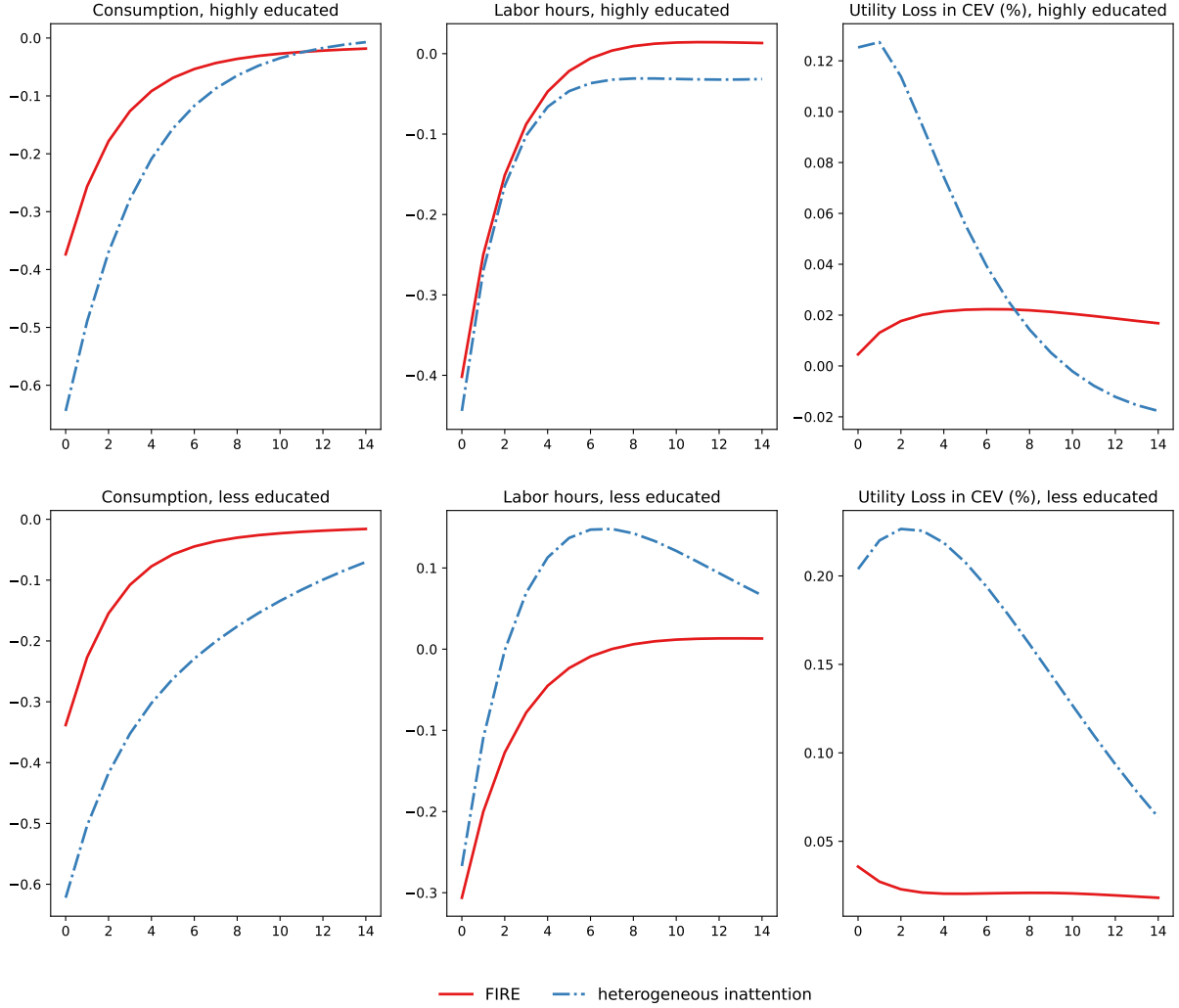
Figure 2 shows that there are significant differences in education-specific responses. In particular, we uncover a novel channel in models with inattention. Generally, if real wages decrease, households face income and substitution effects of labor supply. As we discussed previously, the income effect, realized by an increase in labor supply to compensate for the perceived loss in income, dominates in models with inattentive households, leading to an increased response of labor hours. As less educated, and thus low-income, households are less attentive to inflation and learn slower about inflation (see Definition 2), they perceive the fall in real wages to be larger than under FIRE. As these households lack the smoothing mechanism that high-income, respectively highly educated, households have, they smooth consumption along the business cycle by varying labor supply. The increase in labor supply, however, leads to a substantial welfare loss among low earners due to a high disutility of working, in addition to the transitory income risk and borrowing constraints they face. The distributional consequences of monetary policy are thus underestimated under rational inflation expectations. Larger welfare losses of low-income households in response to contractionary monetary policy shocks are well documented, see for example, Coibion et al. (2017), and see for instance Auclert (2019) and Gornemann et al. (2021) for a theoretical analysis. Our results support these findings and extend them to include inattention showing that it amplifies the distributional effects.

Further, in the economy with heterogeneous inattention, the consumption drop is mostly concentrated among the more constrained households. Their lower attention, larger perceived drop in wages and lack in consumption smoothing mechanism leads to a more persistent decrease in consumption.

20 Under flexible wages, as shown in Appendix E.4, the nominal wage level adjusts more quickly to new labor market equilibrium. However the labor supply channel is still prominent, however more muted.

21 We plot λ_t that solves: $\frac{((1+\lambda_t)C)^{1-\sigma}}{1-\sigma} / \frac{(C)^{1-\sigma}}{1-\sigma} - 1 = \frac{\Delta W_t}{W_{ss}}$, where ΔW_t is the change in welfare that also accounts for the changes in the hours worked: $\Delta W_t = \frac{C_t^{1-\sigma}}{1-\sigma} - \phi \frac{n_t^{1+\nu}}{1+\nu} - \left(\frac{C^{1-\sigma}}{1-\sigma} - \phi \frac{n^{1+\nu}}{1+\nu} \right) = \frac{C_t^{1-\sigma}}{1-\sigma} - \phi \frac{n_t^{1+\nu}}{1+\nu} - W_{ss}$.

Figure 2: Heterogeneity across Educational Levels



Notes: The figure shows in percentage deviations from the steady state the impulse responses of consumption, labor hours and the consumption equivalent welfare variation to a 25 bps monetary policy shock with 0.15 persistence. The red line shows the impulse response function under full information rational expectations, the dashed blue line shows the results under heterogeneous inattention. The top panel shows the results for highly educated households, the lower panel shows the results for less educated households.

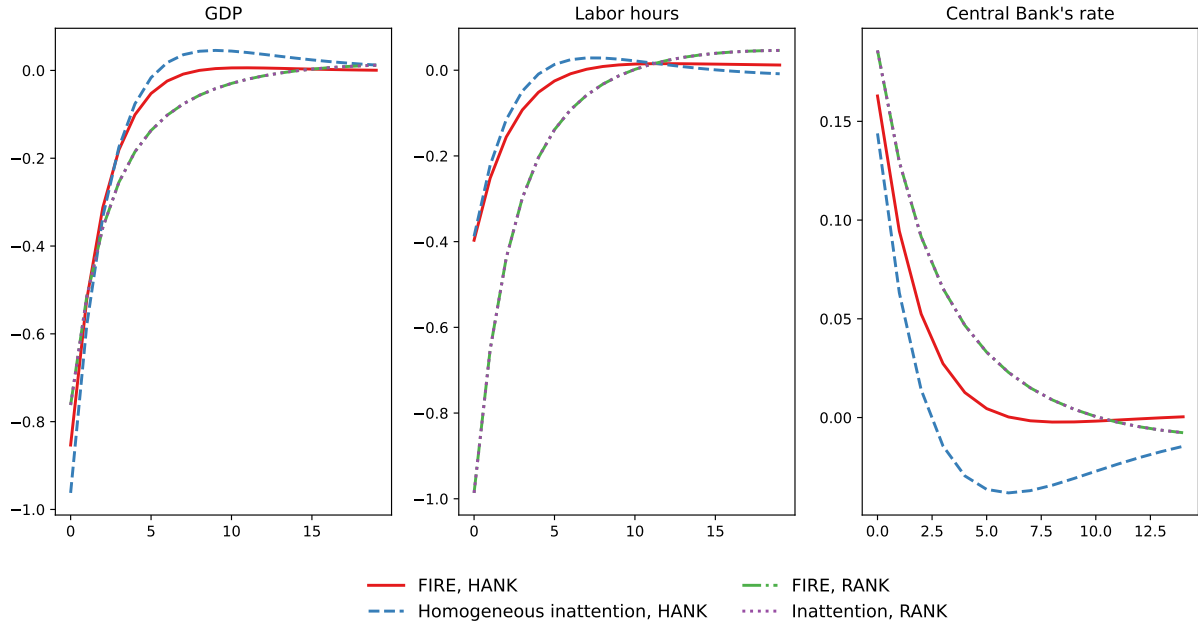
5.2. Inspecting the Mechanism

5.2.1. Comparison to RANK

To analyze the labor supply due to inattention channel, we compare our HANK model to a standard RANK model in Figure 3. We compare impulse responses from the HANK model with homogeneous inattention ($\gamma^1 = \gamma^2 \neq 1$) and FIRE ($\gamma^1 = \gamma^2 = 1$) to those of the RANK model²².

²² Full response functions for the model under homogeneous attention are shown in E.7

Figure 3: Comparison to RANK



Notes: The figure shows in percentage deviations from the steady state the impulse responses of GDP, and labor hours to a 25 bps monetary policy shock with 0.15 persistence. Central Bank's rate is shown in deviations from the steady state. The red line shows the impulse response function under full information rational expectations and in HANK, the dashed blue line shows the results under homogeneous inattention in HANK, the dash-dotted green line shows the results under full information rational expectations in RANK, and the dotted purple line shows the results with inattention in RANK.

The role of inattention differs across models. Inattention has a negligible impact in RANK models relative to its FIRE benchmark. The reason is that RANK models systematically underestimate the indirect effects. In these models, a single representative agent optimizes against aggregate consumption and labor supply and does not face borrowing constraints. Thus, monetary tightening that raises the real interest rates, allows the agent to increase saving and shift consumption along the business cycle. As aggregate consumption drops, labor demand falls sharply. Without liquidity constraints, the representative agent substantially reduces labor hours as real wages adjust, such that the difference between responses under FIRE and the model with inattention is negligible. The initial decline in labor hours is much larger and the recovery is substantially slower implying that the model implies a much more persistent contraction in labor input. Consequently GDP recovers more gradually and remains negative for longer.

In contrast, HANK economies incorporate uninsurable income risk and HtM households. These households lack wealth buffers and spend out of the current income rather than adjusting based on real interest rates. To smooth their consumption and self-insure against (perceived) falling income, they increase their labor supply relative to unconstrained agents. This precautionary labor response mitigates the overall fall in total labor hours in the HANK economies and speeds up the aggregate GDP recovery. Inattention, even homogeneous, tends to amplify the dynamics. A HtM household with high MPC that is inattentive to the aggregate inflation, drives their decision far more by current liquid cash flows than by forward looking forecasts. Hence, they supply more labor and increase output, compared to the HANK FIRE economy. The central

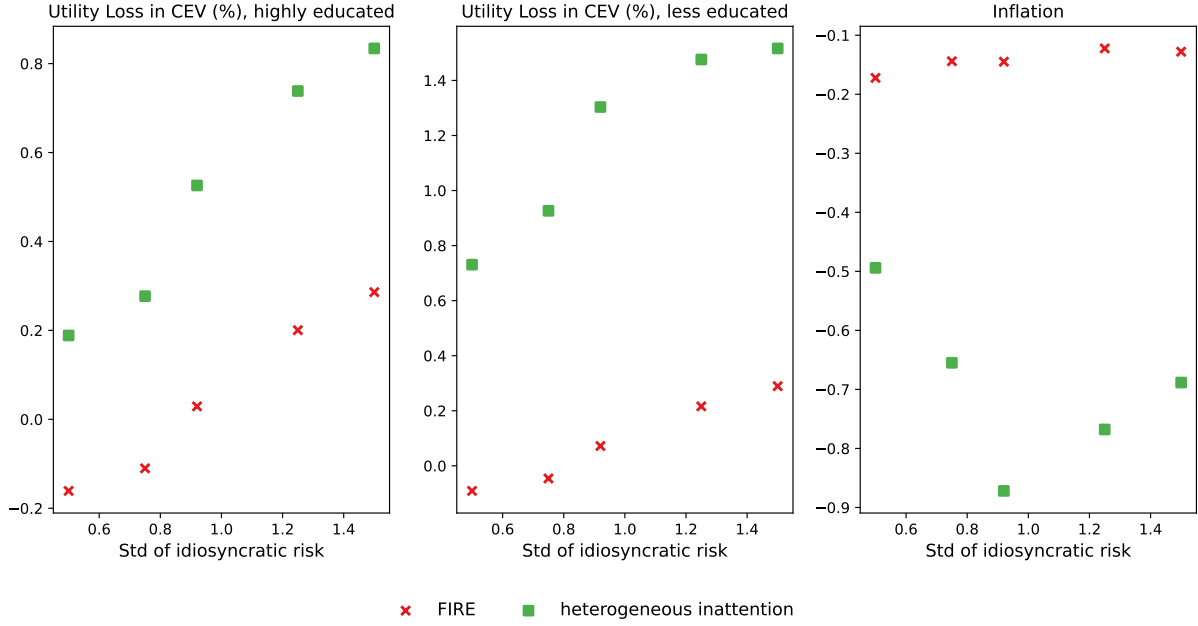
bank rate reflects these different real economic dynamics. If output and hours recover faster or overshoot in HANK, the policy rule has less reason to keep the initial policy stance in place, so the rate reverses sooner. In the RANK economy the real contraction is more persistent, such that the policy response is correspondingly more persistent.

5.2.2. The Role of Transitory Income Inequality

As our model features both transitory and permanent income heterogeneity, we now analyze the effect from transitory income across households and inflation. This analysis strengthens our understanding of the labor supply due to inattention channel, since changes in risk shift the share of constraint households that lack liquid buffers to smooth their consumption. Idiosyncratic risk thereby reflects the safety nets present in the economy which have a direct effect on household's transitory income. Figure 4 shows the cumulative welfare losses in consumption equivalent terms for both educational groups to a 25 basis point contractionary monetary policy shock as a function of the variance of idiosyncratic risk in the model under FIRE and under heterogeneous inattention. The baseline value of risk, as shown in Table 3, is 0.92. The first and second graphs show how idiosyncratic risk amplifies the difference in welfare costs between inattentive less and highly educated households. The household's utility loss increases with the volatility of the idiosyncratic income risk, independent of the expectational framework. The loss thereby is two to three times higher for less educated households. This result arises because the number of households at the borrowing constraint or close to it increases with the level of idiosyncratic risk and these households are also those who supply labor precautionary to smooth their consumption and therefore endure large welfare losses due to labor disutility. Inattention amplifies these dynamics. Compared to the welfare under FIRE, households have bigger utility losses if they pay less attention to inflation, which again reinforces their increased need for labor supply as discussed above. Since the less educated household is less attentive than the highly educated, the utility losses are accordingly bigger.

The role of idiosyncratic risk to inflation differs from these dynamics. As we see in Figure 4, the deflation following a monetary contraction is significantly stronger under heterogeneous inattention than under FIRE. Under FIRE higher idiosyncratic risk dampens the deflationary impact since higher precautionary buffers decrease consumption and therefore aggregate demand falls. Under heterogeneous inattention, the relationship between risk and inflation is non-monotonic and systematically lower because information frictions systematically distort aggregate price setting dynamics. More precisely, under FIRE price setters and households immediately absorb macro shocks, adjust their expectations fully, and re-optimize demand. Hence, they forecast future deflationary pressure more precisely and price it in instantly. An increase in idiosyncratic income risk additionally creates a stronger precautionary savings motive across the agents (by either hoarding assets or supplying labor), dampening the aggregate demand decrease and pushing inflation upward. If on the other side households are inattentive, information updating is costly and households underreact to policy shocks in real time. This leads, together with nominal stickiness, to sluggish wage and price setting. As a result, the fall in inflation is more persistent resulting in a larger cumulative decrease in inflation when idiosyncratic risk rises from the low to middle values. After that the former channel through better insured households

Figure 4: The Role of Idiosyncratic Risk



Notes: The figure shows the change of consumption equivalent welfare variation as a function of the change of the standard deviation of idiosyncratic risk for highly educated (left panel) and less educated (middle panel) households under full information rational expectations (in red), and heterogeneous inattention (in green). Each dot shows a cumulative loss in consumption equivalent terms. Right panel shows cumulative responses of inflation as a function of idiosyncratic risk under full information rational expectations (in red) and under inattention (in green).

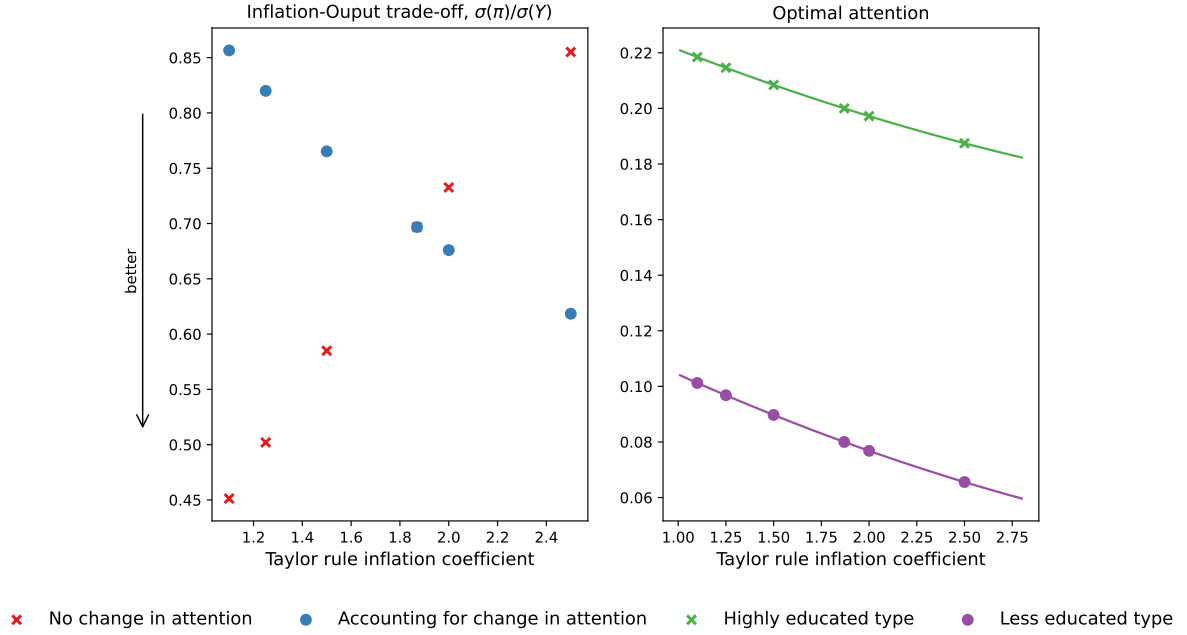
holding larger liquid buffers dominates. The simulations of the economy under high and low transitory income risk in Appendix E.6 underline these results.

5.3. Policy Counterfactuals

5.3.1. The Role of Monetary Policy

To analyze how attention to inflation affects the design of monetary policy we simulate the model for different Taylor-rule coefficients, ϕ_π . We use our baseline calibration of the model and change the Taylor rule coefficient in it. We then reassess volatility of inflation σ_ϕ under the shock processes specification reported in Table 4. In line with the expression for optimal attention (25), we recompute attention levels of both types of households in the economy given the new volatility of inflation. Given the new attention levels we recompute the inflation volatility and repeat the procedure until convergence. The results are shown in Figure 5. The figure shows both the inflation-output trade-off in terms of the ratio between inflation volatility and volatility of output on the left and the optimal attention level of the households for each value of the Taylor rule coefficient on the right. We assess the trade-off when first, households have a constant level of attention, when the central bank changes its reaction to inflation (i.e., without the fixed point), and second, when the households change their attention level in response to the central bank's policy (i.e., with the fixed point). The optimal attention levels consistent with the ϕ_π values are evaluated for less educated and highly educated households shown in the right panel.

Figure 5: The Role of Monetary Policy



Notes: The figure shows the ratio between inflation volatility and output volatility σ_y/σ_π for different values of ϕ_π (panel on the left side) with the optimal level of attention that is consistent with the volatility of inflation (blue dots) and with fixed as in the baseline attention (red crosses). Right panel shows optimal attention for highly educated (green crosses) and less educated (purple dots) households.

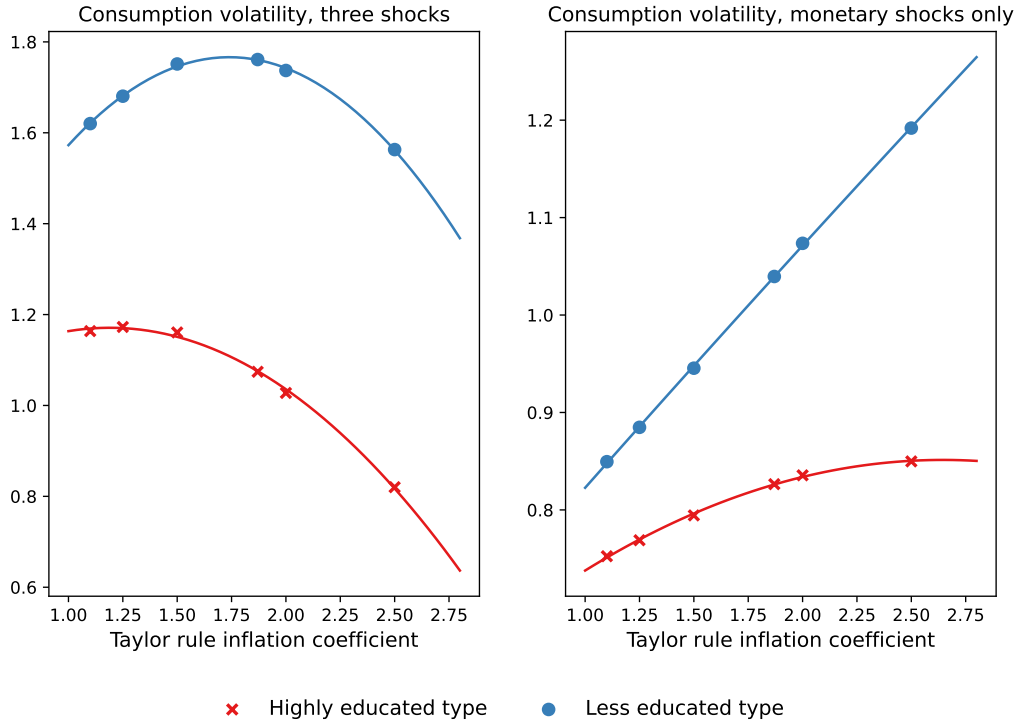
If attention to inflation is kept counterfactually fixed, i.e. there is no change in attention, then the central bank faces a better inflation-output trade-off with smaller inflation Taylor rule coefficient such that inflation remains volatile. This is because the central bank does not induce a deep recession each time inflation deviates from its target. The inflation-output trade-off worsens under a constant level of attention, the more strongly the central bank reacts to inflation. However, once the change in household's attention is accounted for, the opposite is true and the inflation-output trade-off becomes better the more aggressive the central bank reacts to inflation. A larger inflation volatility leads to higher levels of attention among households. Both types of households increase their attention rapidly with a decrease in ϕ_π which is in line with the theory of endogenous attention and equation (25). Agents adapt their attention to changes in the economic environment, and the more volatile inflation becomes, the more attentive they become. The richer type households remain more attentive than the poorer type, irrespective of the Taylor rule coefficient. As less educated households have larger costs of paying attention²³, their attention levels drop faster compared to highly educated households in response to a higher ϕ_π .

Figure 6 shows the effect of an active monetary policy on consumption volatility for highly and less educated households when, first, simulating the model with three shocks (TFP, markup and monetary policy shock) and second, with only a monetary policy shock²⁴. Across both panels,

23 As they lose from inattention more in terms of welfare, the model rationalizes their lower levels of attention found in the data through larger costs of paying attention.

24 The aggregate impulse responses to the TFP and markup shocks can be found in Appendix E.2.

Figure 6: The Role of Monetary Policy



Notes: The figure shows consumption volatility for different values of ϕ_π with the optimal level of attention that is consistent with the volatility of inflation. Values stand for the standard deviation of consumption relative to output. Consumption volatility of highly educated is shown in red and for less educated is shown in blue. Left panel shows consumption volatility when simulating three shocks (TFP, markup and monetary) and the right panel shows the results only for monetary shocks.

less educated, and thus low-income, households experience significantly higher consumption volatility than highly educated households, who are also high-income. This reflects lower liquid wealth, higher MPCs, and lower attention to inflation among less educated households. As the central bank becomes more aggressive towards inflation, consumption volatility increases monotonically for both groups if the economy is driven by the monetary policy shock only. The reason is that aggressive monetary policy requires stronger interest rate changes in response to monetary disturbances, which amplifies consumption volatility. Less educated households cannot insure against strong monetary reactions due to their limited wealth buffers, and therefore their consumption varies a lot. Highly educated households are less affected by aggressive reactions to inflation as they have sufficient wealth buffers. If the economy is hit by all three shocks simultaneously, consumption volatility becomes hump-shaped. Strong inflationary responses by the central bank stabilize the households' consumption across both educational groups. This helps stabilize inflation expectations and real economic demand against productivity and markup disturbances, ultimately lowering overall consumption risk for households. Notice that the hump is much more pronounced for less educated households since they prefer output stabilization more, whereas highly educated households prefer a larger stabilization of inflation. These results are in line with [Del Negro et al. \(2025\)](#), who highlight the benefits of stronger inflation stabilization for the rich.

5.3.2. Inflation-Output Trade-off in the 1970-1980s vs. the Current Times

As technology progressed quickly since the 1970-1980s in the US, the relative demand for highly educated workers increased rapidly. However, the supply of these households did not keep up with the demand, and therefore producers were willing to pay higher wages to these types of households, compared to less educated workers, i.e. the skill premium increased strongly. At the same time, monetary policy became more aggressive towards inflation. To put the findings in Section 5.3.1 in a historical perspective, we study how monetary policy trade-offs changed in response to structural changes in the economy experienced in this period of time. Table 5 compares the key parameters of the 1970-1980s with the current period where we recalibrate the skill premium and relative supply of highly educated households and the Taylor rule. The upper part of Table 5 shows the results with the lower skill premium and a less reactive to inflation central bank, the middle part of the table shows the results with the lower skill premium and the Taylor rule of the current period, and the lower part of the table shows the results using the baseline calibration from Tables 3. The parameters for the skill premium and relative supply of highly educated households is taken from Bengali et al. (2025), differential Taylor rules are taken from Aastveit et al. (2024), who estimate regime-switching Taylor rules, with one regime being more prevalent in the current period and the second being regime dominating in the 1970-1980s.

As discussed in Section 5.3.1, when the central bank reacts less strongly to inflation, the attention of households increases, and this leads to a smaller response of their labor supply to a monetary tightening and a worse inflation-output trade-off. At the same time, the relatively larger share of less educated households in the 1970-1980s implies that the response of labor supply could be stronger. Comparing the middle and lower parts in Table 5 shows that the relative supply and the differential skill premium in the 1970-1980s improved the monetary policy trade-off. Instead, the relatively lower reaction to inflation of the central bank led to an increase in attention levels, and through the weakening of the labor supply response driven by higher attention, this led to a worse inflation-output trade-off (see the comparison of the top and middle parts of Table 5). On balance, our exercise implies a better trade-off in the 1970-1980s, highlighting the role of a change in attention through the change in educational attainment levels in the economy. Note, that we assumed that the cost of acquiring information remained constant throughout the exercise as we lack evidence on the change in these costs in the data.

The lower reaction of the central bank to inflation leads to a larger volatility of hours worked, output and consumption for both highly educated and less educated households (see the last column in Table 5, increase from 0.71 to 1.01 for highly educated and from 1.21 to 1.44 for less educated households). The lower skill premium and relative supply of highly educated households (i.e. lower inequality) brought the volatilities of consumption of highly educated and less educated households closer to each other through the decrease in the volatility of consumption of less educated households.

This counterfactual exercise, together with the results in Section 5.3.1, highlights the sensitivity of the monetary policy trade-off to changes in households' attention to inflation, which we find to be substantial.

Table 5: Trade-Offs in the 1980s vs. current period

	Rel. Supply	Skill premium	ρ	ϕ_π	ϕ_y	Attention	Trade-off	Cons. std MP shocks	Cons. std 3 shocks
1970-1980s									
Highly Educated	0.33	1.4	0.72	1.94	1.44	0.12	0.57	0.54	1.01
Less Educated	0.66	1.0	0.72	1.94	1.44	0.03	0.57	0.61	1.44
1970-1980s with the current Taylor rule									
Highly Educated	0.33	1.4	0.92	1.87	1.04	0.11	0.40	0.64	0.71
Less Educated	0.66	1.0	0.92	1.87	1.04	0.02	0.40	0.87	1.21
Current period									
Highly Educated	0.56	1.7	0.92	1.87	1.04	0.20	0.70	0.83	1.07
Less Educated	0.44	1.0	0.92	1.87	1.04	0.08	0.70	1.04	1.76

Notes: The coefficient for the change in relative supply and the skill premium of highly educated (i.e. college educated) households is taken from [Bengali et al. \(2025\)](#), the coefficient for the change in the Taylor rule parameters is taken from [Aastveit et al. \(2024\)](#). The relative supply of highly educated households is reported in the second column, skill premium in the third, columns 4-6 show Taylor rule coefficients, optimal attention parameters are reported in the 7th column. Trade-off shows the ratio $\sigma(y)/\sigma(\pi)$, where a lower ratio means a better trade-off. Standard deviations of consumption relative to GDP for monetary policy shocks only and for three shocks (TFP, markup, monetary) are shown in columns 9-10.

6. Empirical Evidence of Labor Supply and Inflation Expectations

We provide empirical supportive evidence for the found transmission channel from our quantitative model using survey data. Specifically, we uncover the channel of education-specific inflation expectations and labor supply by studying the extensive margin of labor supply across educational groups²⁵. To increase the number of observations we run the analysis on the household level using quarterly data from the NY Fed SCE for the period of June 2013 to December 2023 we describe in Appendix A.

Specifically, we analyze whether the probability of having a specific employment status changes for households after they expect an increase in future inflation and if this effect differs by their educational level. Formally, we estimate

$$\begin{aligned}
Employmentstatus_{j,t} = & \beta_j + \beta_1 \cdot LowEdu_j + \beta_2 E_{j,t} \pi_{t+1} + \beta_3 (LowEdu_j \times E_{j,t} \pi_{t+1}) + \beta_4 \cdot BS_t \\
& + \beta_5 (LowEdu_j \times BS_t) + \beta_6 (E_{j,t} \pi_{t+1} \times BS_t) + \beta_7 (LowEdu_j \times E_{j,t} \pi_{t+1} \times BS_t) \\
& + \sum_j \Gamma_{j,t} + \varepsilon_{j,t}
\end{aligned} \tag{31}$$

where $Employmentstatus_{j,t}$ denotes the employment status of household j in t where a household can be unemployed or employed where latter measures it the household works full-time or part-time, $LowEdu_j$ is a dummy for being less educated or highly educated, $E_{j,t} \pi_{t+1}$ are quarterly one-year-ahead inflation expectations, BS_t is the standardized exogenous monetary policy shock by [Bauer and Swanson \(2023\)](#) and $\Gamma_{j,t}$ are additional control variables which include year-quarter fixed effects, the household's quarterly one-year-ahead income expectations and a dummy for being male. The introduction of an exogenous shock allows us to claim causality. In general, the expression (31) tells us how likely the household will

²⁵ Due to data limitations in the NY Fed SCE we are unable to study the intensive margin of labor supply as actual hours worked are not being reported.

be employed or unemployed if it expects an increase in inflation due to a change in monetary policy and if there is a difference between educational groups.

To examine both short- and medium-run employment dynamics, we additionally study whether a household changes their employment status from one quarter to another after an increase in inflation expectations. We estimate

$$\begin{aligned}
Transition_{j,t+h} &= \beta_j + \beta_1 \cdot LowEdu_j + \beta_2 E_{j,t} \pi_{t+1} + \beta_3 (LowEdu_j \times E_{j,t} \pi_{t+1}) + \beta_4 \cdot BS_t \\
&+ \beta_5 (LowEdu_j \times BS_t) + \beta_6 (E_{j,t} \pi_{t+1} \times BS_t) + \beta_7 (LowEdu_j \times E_{j,t} \pi_{t+1} \times BS_t) \\
&+ \sum_j \Gamma_{j,t} + v_{j,t}
\end{aligned} \tag{32}$$

where $Transition_{j,t+h} = Employment_{j,t+h+1} - Employment_{j,t+h}$ is the quarter-on-quarter change in the status of the employment of the household at horizon $h \in \{0, 1\}$ such that $Transition_{j,t}$ measures a contemporaneous change and $Transition_{j,t+1}$ the one-period-ahead change. Estimating equation (32) allows us to analyze a dynamic layer of labor supply and to measure if a shock induced increase in inflation expectations causes a household to leave their unemployment and move into full- or part-time employment over the next quarter and vice versa. Table 6 shows the estimation results of regression (31) in columns (2)–(3) and the estimates for the regression of regression (32) in (4)–(15) where the effects for the contemporaneous transition, $Transition_{j,t}$, are shown in (4)–(9) and for the one-period-ahead transition, $Transition_{j,t+1}$, are shown in (10)–(15). Errors are clustered at the household level. From columns (2)–(3) in the regression table, we can infer that if the household is less educated, they are 13.5 percentage points less likely to be employed and 1.1 percentage points more likely to be unemployed than highly educated households. Inflation expectations significantly affect the employment status of households, as an increase in expected inflation is associated with a 0.7 percentage point reduction in the probability of employment and a 0.2 percentage point increase in the probability of unemployment. The difference between educational groups thereby is insignificant. Further, a one standard deviation contractionary monetary policy shock, evaluated at zero inflation expectations, leads to a heterogeneous response between households: the likelihood of less educated households to be unemployed is 0.2 percentage points higher than that of the highly educated households. Both households are more likely to be unemployed after a contractionary monetary policy, although the effect is insignificant for less educated agents. Most importantly, monetary policy shocks have an additional heterogeneous effect on households via inflation expectations: an increase in inflation expectations after a contractionary monetary policy shock decreases the probability of a highly educated household to be employed relative to a less educated household where the latter is 0.2 percentage point more likely to be employed after an increase in inflation expectations, however, the decreased likelihood of being unemployed among the less educated is significant. In general, we observe that higher inflation expectations have a negative connotation for households, as they associate them with a higher probability of unemployment. This relationship is amplified by changes in inflation expectations due to monetary policy shocks that cause heterogeneous effects in labor supply decisions between educational groups. The full effects of a transmission of monetary policy to employment are mostly experienced after 2 quarters, as shown in columns (4)–(15). Columns (4)–(8) in Table 6 underline the previous dynamics, in particular the contemporaneous effect of the contractionary monetary policy shown by the estimates of β_7 in equation (32). Less educated unemployed households are 2.2 percentage points more likely to work full-time after an increase in inflation expectations while they are 1.3 percentage points less likely to work part-time, which is the opposite effect of highly educated households. A similar dynamic can be observed for those who work part-time and switch to full-time employment: highly educated households are 2.1 percentage points less likely to change employment whereby less educated households are 1.9 percentage points more likely to do so. For the other groups, the effects are muted and insignificant. Although there is a significant likelihood for less educated households to become part-time employed after being full-time employed,

Table 6: Employment Status and Labor Market Transitions

	Employment Status			Contemporaneous Transitions					One-Period-Ahead Transitions						
	Employed (2)	Unemployed (3)		U → F (4)	U → P (5)	F → P (6)	F → U (7)	P → F (8)	P → U (9)	U → F (10)	U → P (11)	F → P (12)	F → U (13)	P → F (14)	P → U (15)
Less Educated	-0.135*** (0.007)	0.011*** (0.002)		-0.013 (0.033)	-0.009 (0.037)	-0.022 (0.022)	-0.059* (0.034)	-0.031 (0.020)	0.003 (0.039)	-0.009 (0.042)	0.026 (0.044)	-0.006 (0.026)	-0.051 (0.044)	-0.019 (0.025)	-0.014 (0.050)
$E_t\pi_{t+1}$	-0.007*** (0.002)	0.002*** (0.001)		0.007 (0.011)	0.014 (0.010)	-0.001 (0.007)	-0.001 (0.008)	-0.005 (0.007)	-0.010 (0.008)	0.013 (0.011)	0.008 (0.011)	0.010 (0.008)	-0.004 (0.012)	-0.003 (0.007)	-0.000 (0.008)
Less Educated $\times E_t\pi_{t+1}$	0.000 (0.002)	0.001 (0.001)		-0.001 (0.013)	-0.004 (0.012)	0.004 (0.007)	-0.005 (0.011)	0.007 (0.008)	0.015 (0.012)	-0.007 (0.013)	0.005 (0.013)	-0.007 (0.010)	-0.008 (0.013)	0.002 (0.009)	0.006 (0.011)
BS	-0.059*** (0.015)	0.011* (0.006)		0.016 (0.144)	-0.172 (0.125)	-0.133 (0.092)	0.209 (0.151)	-0.155* (0.091)	0.072 (0.181)	-0.080 (0.111)	0.129 (0.108)	0.031 (0.075)	-0.073 (0.066)	0.039 (0.088)	0.130 (0.129)
Less Educated $\times$ BS	0.000 (0.004)	0.002 (0.001)		-0.008 (0.034)	0.079 (0.056)	-0.015 (0.016)	-0.007 (0.028)	0.006 (0.021)	-0.075** (0.031)	0.039 (0.052)	-0.049 (0.033)	-0.021 (0.021)	0.076** (0.030)	-0.010 (0.019)	0.091** (0.038)
$E_t\pi_{t+1} \times$ BS	-0.002 (0.001)	0.001 (0.001)		-0.024* (0.012)	0.008 (0.015)	-0.008** (0.003)	-0.006 (0.006)	-0.021*** (0.008)	0.001 (0.005)	-0.003 (0.010)	-0.007 (0.008)	-0.001 (0.006)	0.008 (0.012)	-0.007** (0.003)	0.016 (0.011)
Less Educated $\times E_t\pi_{t+1} \times$ BS	0.002 (0.002)	-0.001* (0.001)		0.022 (0.014)	-0.013 (0.017)	0.009** (0.004)	0.004 (0.009)	0.019** (0.010)	0.007 (0.012)	0.002 (0.014)	0.011 (0.011)	0.011 (0.008)	-0.025** (0.012)	0.009** (0.005)	-0.038** (0.017)
Constant	0.743*** (0.014)	0.039*** (0.006)		0.498*** (0.133)	0.609*** (0.120)	0.597*** (0.089)	0.385*** (0.145)	0.754*** (0.074)	0.580*** (0.171)	0.670*** (0.111)	0.346*** (0.092)	0.339*** (0.089)	0.780*** (0.059)	0.516*** (0.092)	0.437*** (0.129)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Quarter FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	71,597	71,597	430	386	908	408	1,090	294	282	263	574	272	679	190	

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

U = unemployed; F = full-time; P = part-time.

Notes: Control variables include the quarter-on-quarter one-year-ahead income expectations and a dummy for male.

this change becomes insignificant in the next period, as shown in column (11). Overall, changes in inflation expectations affect labor supply, but the impact is delayed. In columns (13)–(15) we observe that less educated households are less likely to be unemployed by 2.5 or 3.8 percentage points and are 0.9 percentage points more likely to become full time employed relative to highly educated households. This transition from one labor market status to another is merely driven by the exogenous shift in inflation expectations.

Our result on the extensive margin of labor supply pushes our main transmission channel of labor supply: households who expect inflation to increase in the future change their labor supply. Estimates of β_7 show that a contractionary monetary policy on average decreases the unemployment probability but increases the labor supply among less educated households, which is consistent with previous literature on income heterogeneity (which we assume here to be driven by educational levels), see, for example, [Broer et al. \(2021b\)](#). The results also agree well with the theory put forward by [Blundell et al. \(2016\)](#), [Cantore et al. \(2026\)](#) or [Graves et al. \(2023\)](#) that labor supply works as a mechanism to smooth consumption. This is especially true for less educated and thus low-income households who cannot self-insure against income risk or only hold a limited amount of liquid assets and therefore have a larger incentive to increase their labor supply if they expect inflation to be higher in the future. Additionally, given that low-income households are less attentive to inflation, they associate higher inflation with a decrease in their real income, which they would like to self-insure against, as put forward in our model but also in [Jiang et al. \(2024\)](#), [Coibion et al. \(2023\)](#) and [Stantcheva \(2024\)](#). This channel is usually missed in standard RANK models in which the substitution effect dominates the income effect of labor supply.

7. Conclusion

This paper studies how attention varies across the household income distribution and its implications for monetary policy transmission. We show empirical cross-country evidence for varying attention levels between demographic groups, specifically educational levels. We find that less educated households pay significantly less attention to inflation than highly educated households. To quantify the implications of the empirical results for macro dynamics, we introduce an endogenous attention choice into a one-asset HANK model and allow households to be inattentive to inflation. Households thereby differ in how much attention they pay to inflation based on their educational levels, which ultimately characterizes their permanent income levels. We calibrate our model to match our empirical evidence of inattention that we find in the data. Counterfactual exercises show that compared to fully rational expectations, monetary policy has a better inflation-output trade-off with stable expectations. However, the better trade-off is achieved through a larger decrease in welfare among less educated, and thus low-income households, exacerbating the distributional consequences of monetary policy. A more muted response to inflation by the central bank worsens the trade-off due to a rapid rise in attention levels, leading to a rapid fall in labor supply.

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A. Data

Here we describe the data used in Section 2.

A.1. USA Descriptive Statistics

For the USA we use quarterly data for the period June 2013 to December 2023 from the Survey of Consumer Expectations (SCE) provided by the New York Fed and the St. Louis Fed FRED data to construct quarterly CPI Inflation. We drop the values above 50% and below -50% for one-year-ahead inflation expectations and above 100% and below -100% for income expectations. For shocks, we consider the monetary policy shock series by [Bauer and Swanson \(2023\)](#) (which were recently updated and extended to the period of December 2023 and can be found [here](#)) and oil supply news shocks by [Känzig \(2021\)](#) which we standardize around the mean. Additionally, we construct a set of dummies for cross-sectional characteristics we need for our analysis.

Table 7 shows the summary statistics of the data used. Income is categorized into three groups: low-income households are those households that had a pre-tax income in the last 12 months of less than \$40,000, mid-income households hold \$40,000 – \$99,999 and high-income households have a total pre-tax income level of $\geq$ \$100,000. CPI Inflation is the annualized, quarterly CPI inflation rate constructed using the US consumer price index from the St. Louis Fed FRED database (CPIAUCSL). For inflation expectations, we use responses to the following question from the SCE: “What do you expect the rate of inflation/deflation to be over the next 12 months?” Respondents are asked to assign some probabilities. Across the quarterly data, there are 41,693 full-time, 10,813 part-time and 3,450 unemployed households.

Table 7: Summary Statistics, USA

Variable	Median	25%	75%	1%	99%	N
Inflation Expectations (one-year-ahead)	3.61	2.00	7.05	-20.01	41.57	71,596
CPI Inflation (qoq)	0.62	0.36	0.84	-0.98	2.35	71,596
Bauer & Swanson shocks	0.02	-0.02	0.05	-0.41	0.14	71,596
Oil supply news shocks	0.06	-0.69	0.56	-2.85	1.92	68,723
Male dummy	Yes	No	Yes	No	Yes	71,596
Income level	\$40-99k	< \$40k	$\geq$ \$100k	< \$40k	$\geq$ \$100k	70,918
Self-employed	No	No	No	No	Yes	71,596
Education	College	Some College	College	High School	College	71,379
Home-owner	Yes	No	Yes	No	Yes	71,596
Age	40-60	< 40	> 60	< 40	> 60	71,596
Full-time worker (dummy)	Yes	No	Yes	No	Yes	71,596
Part-time worker (dummy)	No	No	No	No	Yes	71,596
Unemployed worker (dummy)	Yes	No	Yes	No	Yes	71,596

For the attention estimation in Section 2.2 and the model simulation in Section 5 we group educational levels into two groups where less educated households have a lower than college degree and highly educated households have a college degree.

A.2. Australia Descriptive Statistics

Table 8 shows the summary statistics for data in the Consumer Attitudes, Sentiments and Expectations in Australia Survey (CASIe). We drop the values above 50% and below -50% for one-year-ahead inflation expectations. We use quarterly data for the period from 1995 to 2023. Additionally, we construct a similar set of dummies for cross-sectional characteristics as we do for the US data. To quantify heterogeneous inflation expectations from the data we use responses to the following question: “By what percentage do you think prices will have gone up by this time next year?” Income is again categorized into three categories: low-income households are those households that had a pre-tax income in the last 12 months of less than \$40,000, mid-income households have an annual income of \$40,000 – \$89,999 and high-income households have a total pre-tax income level of $\geq$ \$90,000.

Table 8: Summary Statistics, AUS

Variable	Median	25%	75%	1%	99%
Inflation expectations	5.0	2.0	8.0	-10.0	30.0
CPI inflation	2.5	1.7	3.1	-0.3	7.3
Male	Yes	No	Yes	No	Yes
Income level	\$40-89k	$\leq$ \$40k	$\geq$ \$90k	$\leq$ \$40k	$\geq$ \$90k
Self-employed	No	No	No	No	Yes
Education	school or below	school or below	above school	school or below	above school
Home-owner	Yes	Yes	Yes	No	Yes
Age	$\geq$ 45	34-44	$\geq$ 45	18-33	$\geq$ 45
Full-time workers	Yes	No	Yes	No	Yes

B. Additional Results to Section 2

B.1. Shock-specific Inflation Expectations

In addition to analyzing the effects of the aggregate price index, we test if increases in the prices of different goods or consumption baskets affect households in different groups heterogeneously by analyzing the effects of different types of shocks. To do so, we further derive the inflation forecast errors as functions of structural shocks by starting from equation (4) and rewriting it as

$$E_{j,t}\pi_{t+1} = \gamma^g E_t \pi_{t+1} + (1 - \gamma^g) E_{j,t-1} \pi_{t+1} \quad (\text{B.1})$$

Next, rewriting inflation as a function of structural shocks (in case of linearity)

$$\pi_{t+1} = \sum_{s=-\infty}^{\infty} J_{t+1-s} u_{t+1-s} \quad (\text{B.2})$$

where u_t is a vector of structural shocks that are uncorrelated across time and with each other, and J_t represents impulse response functions (IRFs) of inflation to them, gives the forecast error

$$\pi_{t+1} - E_{j,t}\pi_{t+1} = \sum_{s=-\infty}^{\infty} J_{t+1-s} u_{t+1-s} - \gamma^g E_t \pi_{t+1} - (1 - \gamma^g) E_{j,t-1} \pi_{t+1}$$

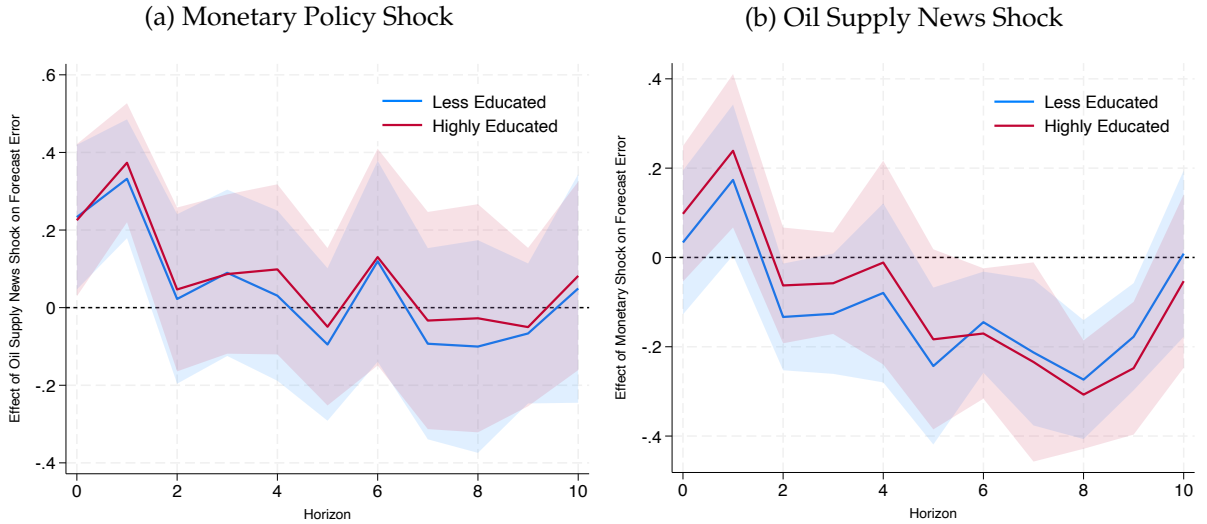
$$= \sum_{s=-\infty}^0 J_{t+1-j} u_{t+1-j} + \sum_{s=1}^{\infty} J_{t+1-s} u_{t+1-s} \left[1 - \gamma^g \sum_{k=0}^{s-1} (1 - \gamma^g)^k \rho^k \right] \quad (\text{B.3})$$

The first term denotes unpredictable at time t future shocks that affect inflation (i.e. shocks that occur in periods $t + 1$ or later) and the second term shows how forecast errors depend on past shocks - for example the coefficient in front of the shock in period t , u_t , is simply $J_t(1 - \gamma^g)$. Shock-specific attention can then be estimated by regressing the forecast error on past shocks

$$e_{j,t+1} = \beta_j + \beta_1^g I_g^m u_t^m + v_{j,t} \quad (\text{B.4})$$

where $e_{j,t+1} = \pi_{t+1} - E_{j,t}\pi_{t+1}$ is the forecast error of inflation, m stands for the shock of interest (for example, monetary policy shock) and $\gamma^g = 1 - \frac{\hat{\beta}_1^g I_g^m}{J_t^m}$ is the group- g specific attention level. J_t^m can be estimated or taken from other studies. Note, that this equation allows us to estimate attention using different shock series²⁶. Figure 7 shows the responses of the forecast errors over 10 quarters to standardized [Bauer and Swanson \(2023\)](#) monetary policy and [Känzig \(2021\)](#) oil supply news shocks. The blue line shows the response of less educated households, and the red line shows the response of highly educated households. Inferring from the figure we don't find any heterogeneity in the household's reactions to and of these shocks. Both educational groups do not react statistically different from each other to changes in either of the shocks. Hence, we can infer that there is no heterogeneous effect on inflation expectations due to households' heterogeneous consumption goods baskets, which allows us to look at aggregate consumption baskets and price levels as we do in the main analysis of the paper.

Figure 7: Responses of Inflation Forecast Errors to Shocks



Notes: The figure shows in percentage points the impulse responses of inflation forecast errors to standardized [Bauer and Swanson \(2023\)](#) monetary policy shocks in (a) and standardized [Känzig \(2021\)](#) oil supply news shocks in (b). Responses of highly educated households are shown in red, responses of less educated households are shown in blue. The 95% confidence intervals are shown as shaded areas.

²⁶ The regression can either be estimated with each shock individually or with all shocks together. These shocks are structural shocks with the assumption that they are uncorrelated with each other and the regression results will be equivalent.

C. Appendix to Section 3

In the following we apply the sparse max framework by [Gabaix \(2014\)](#), specifically the dynamic programming extension in [Gabaix \(2016\)](#), to our small model. We extend the state space representation in the original work using the sequence space notation.

C.1. Endogenous Attention

Under FIRE, households optimize policy decisions by being fully attentive to all variables that are relevant to their decision making. Conversely, if the household is inattentive to parts of the economy, they adapt a “sparse” representation of the world and anchor expectations to default baselines, which deviate systematically from their fully rational counterparts based on prior attention allocations. The agent endogenously determines their optimal consumption path by balancing the benefit of updating information against the cognitive cost of adjusting attention away from these defaults, given their subjective model. Consequently, while fully rational and inattentive agents operate within an identical objective environment, their representations of it differ fundamentally. *Note: Note: In the following we mostly suppress state arguments for brevity.*

The *rational* household pays full attention to the entire state of the economy and chooses consumption c_t^r by solving the objectively true dynamic programming problem

$$\begin{aligned} V_j^r(m_t) &= \max_{c_t, a_t^n} \{u(c_t) + \beta E_t[V_j^r(m_{t+1})]\} \\ \text{s.t. } P_t c_t + a_t^n &= Y_{g,t} + (1 + i_t^a) a_{t-1}^n \end{aligned} \quad (\text{C.1})$$

where the state is given by $m_t = (a_{t-1}^n, \{\pi_{t+j}\}_{j=0}^\infty, \{i_{t+j}^a\}_{j=0}^\infty, \{Y_{g,t+j}\}_{j=0}^\infty)$ ²⁷ and $V_j^r(\cdot)$ is the value function evaluating the rational policy and represents the maximum utility achievable under the true objective reality. Once the rational agent takes their consumption decision c_t^r (and hence also their savings decision a_t^n), their realized utility evolves according to

$$\begin{aligned} V_j^r(m_t) &= u(c_t^r) + \beta E_t[V_j^r(m_{t+1}^r)] \\ \text{s.t. } P_t c_t^r + (a_t^n)^r &= Y_{g,t} + (1 + i_t^a) a_{t-1}^n \end{aligned} \quad (\text{C.2})$$

where $m_{t+1}^r = ((a_t^n)^r, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)$ since assets accumulate through the choice of consumption in $t + 1$. The realized utility of the *rational* household can be rewritten as

$$v_{j,t}(c_t^r, m_t) \equiv u(c_t^r) + \beta E_t[V_j^r(Y_{g,t} + (1 + i_t^a) a_{t-1}^n - P_t c_t^r, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)] \quad (\text{C.3})$$

The *inattentive* agent, on the other hand, solves a subjective dynamic programming problem at t subject to their budget constraint, using their subjective expectations

$$\begin{aligned} V_j^s(m_t) &= \max_{c_t, a_t^n} \{u(c_t) + \beta \tilde{E}[V_j^s(m_{t+1})]\} \\ \text{s.t. } P_t c_t + a_t^n &= Y_{g,t} + (1 + i_t^a) a_{t-1}^n \end{aligned} \quad (\text{C.4})$$

where $\tilde{E}$ is the subjective expectation operator we define in [Definition 1](#) with the attention parameter $\gamma^s \in [0, 1)$. The perceived utility of the *inattentive* household can be written as

$$\tilde{v}_{j,t}^s(c_t^s, m_t) \equiv u(c_t^s) + \beta \tilde{E}[V_j^s(Y_{g,t} + (1 + i_t^a) a_{t-1}^n - P_t c_t^s, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)]$$

²⁷ Note that the choice of the state is only for notational completeness. Since the household is only inattentive to inflation, the action variable and the utility loss function will only be distorted by the inattention to inflation, and thus the state could have also been denoted as $m_t = (a_{t-1}^n, \{\pi_{t+j}\}_{j=0}^\infty)$.

respectively, under [Definition 1](#),

$$\tilde{v}_{j,t}^s(c_t^s, m_t) \equiv u(c_t^s) + \beta E_t[V_j^s(Y_{g,t} + (1 + i_t^a)a_{t-1}^n - P_t c_t^s, \{\gamma^s \pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)] \quad (C.5)$$

Once the inattentive agent takes their consumption decision c_t^s (with savings a_t^s), their actual realized utility evolves according to true macroeconomic state and the objective expectations operator E_t

$$\begin{aligned} V_j(m_t) &= u(c_t^s) + \beta E_t[V_j(m_{t+1}^s)] \\ \text{s.t. } P_t c_t^s + (a_t^n)^s &= Y_{g,t} + (1 + i_t^a)a_{t-1}^n \end{aligned} \quad (C.6)$$

where $m_{t+1}^s = ((a_t^n)^s, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)$ with assets $(a_t^n)^s$ determined by consumption choice and $V_j(\cdot)$ denotes the true value function representing the maximum utility achievable under the true objective reality when choosing the inattentive policy. Thus, the realized utility for the *inattentive* household is

$$v_{j,t}(c_t^s, m_t) \equiv u(c_t^s) + \beta E_t[V_j(Y_{g,t} + (1 + i_t^a)a_{t-1}^n - P_t c_t^s, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)] \quad (C.7)$$

Notice that the household's savings are a direct result of consumption. The inattentive household who chooses c_t^s ultimately also chooses $(a_t^n)^s$ and the rational household who chooses c_t^r accumulates $(a_t^n)^r$.

Taking policy c_t^s instead of c_t^r leads to a loss in the ex-ante realized utility of the household. Following [Gabaix \(2014\)](#), the utility loss $\mathcal{L}$ is defined as the expected difference between the utility under the rational policy c_t^r and the utility under the inattentive policy c_t^s , both evaluated at the true economic macroeconomic state

$$\mathcal{L} = \mathbb{E}(v_{j,t}(c_t^r, m_t) - v_{j,t}(c_t^s, m_t)) \quad (C.8)$$

where $\mathbb{E}$ is the unconditional expected objective operator, such that the household evaluates the expected loss over the probability distribution of future macroeconomic states before shocks are realized. To quantify (C.8), we follow [Gabaix \(2014\)](#) and replace $v_{j,t}(\cdot)$ by a second order Taylor approximation around the default state which we normalize to be the deterministic steady state. Resorting to the recursive nature of the optimization problem allows us to rewrite the realized utility of the rational household (C.3) as

$$\begin{aligned} v_{j,t}(c_t^r, m_t) &\approx v_j + v_c \hat{c}_t^r + v_{a^n} \hat{a}_{t-1}^n + E_t \sum_{h=1}^\infty v_Y \hat{Y}_{g,t+h} + E_t \sum_{h=1}^\infty v_\pi \hat{\pi}_{t+h} + E_t \sum_{h=1}^\infty v_{i^a} \hat{i}_{t+h}^a \\ &+ \frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^r)^2 + \frac{\partial^2 v}{\partial c \partial a^n} \hat{c}_t^r \hat{a}_{t-1}^n + E_t \sum_{h=1}^\infty \frac{\partial^2 v}{\partial c \partial Y_g} \hat{c}_t^r \hat{Y}_{g,t+h} + E_t \sum_{h=1}^\infty \frac{\partial^2 v}{\partial c \partial \pi_h} \hat{c}_t^r \hat{\pi}_{t+h} \\ &+ E_t \sum_{h=1}^\infty \frac{\partial^2 v}{\partial c \partial i^a} \hat{c}_t^r \hat{i}_{t+h}^a + t.i.o.c. \end{aligned} \quad (C.9)$$

where v_j is the utility in the steady state. We use the notation $\hat{x}_t = x_t - x$ as the deviation of a variable x_t from its steady state value x . Following [Gabaix \(2014\)](#), we evaluate derivatives at the zero-inflation steady state $\pi = 0$, i.e., we define for $c_t = (c_t^r, c_t^s)$ first order derivatives by $v_x \equiv \frac{\partial v_{j,t}(c_t, m_t)}{\partial x_t} \Big|_{c_t=c, m_t=m}$, second order derivatives $\frac{\partial^2 v}{\partial x^2} \equiv \frac{\partial^2 v_{j,t}(c_t, m_t)}{\partial x_t^2} \Big|_{c_t=c, m_t=m}$, cross-partials by $\frac{\partial^2 v}{\partial x \partial y} \equiv \frac{\partial^2 v_{j,t}(c_t, m_t)}{\partial x_t \partial y_t} \Big|_{c_t=c, m_t=m}$ and $v_j \equiv v_{j,t}(c_t, m_t) \Big|_{c_t=c, m_t=m}$ is the utility function evaluated at the steady state. *t.i.o.c.* stands for "terms independent of c ". Notice that steady state consumption c is the same for both inattentive and rational household ($c = c^r = c^s$). Since consumption is of first order importance for the maximization of the

household and we assume a zero-inflation steady state, the approximated utility function of the *rational* household, where $\frac{\partial c}{\partial x} \equiv \frac{\partial c_t}{\partial x_t} |_{c_t=c, m_t=m}$, is given by

$$\begin{aligned} v_{j,t}(c_t^r, m_t) \approx & v_j + v_c \hat{c}_t^r + \frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^r)^2 + \frac{\partial^2 v}{\partial c^2} \hat{c}_t^r E_t \left(\frac{\partial c}{\partial a^n} \hat{a}_{t-1}^n + \sum_{h=1}^{\infty} \frac{\partial c}{\partial Y_g} \hat{Y}_{g,t+h} \right. \\ & \left. + \sum_{h=1}^{\infty} \frac{\partial c}{\partial \pi_h} \pi_{t+h} + \sum_{h=1}^{\infty} \frac{\partial c}{\partial i^a} \hat{i}_{t+h}^a \right) + t.i.o.c. \end{aligned} \quad (C.10)$$

The approximated utility function of the *inattentive* household (C.7) is given similarly by

$$\begin{aligned} v_{j,t}(c_t^s, m_t) \approx & v_j + v_c \hat{c}_t^s + \frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^s)^2 + \frac{\partial^2 v}{\partial c^2} \hat{c}_t^s E_t \left(\frac{\partial c}{\partial a^n} \hat{a}_{t-1}^n + \sum_{h=1}^{\infty} \frac{\partial c}{\partial Y_g} \hat{Y}_{g,t+h} \right. \\ & \left. + \sum_{h=1}^{\infty} \frac{\partial c}{\partial \pi_h} \pi_{t+h} + \sum_{h=1}^{\infty} \frac{\partial c}{\partial i^a} \hat{i}_{t+h}^a \right) + t.i.o.c. \end{aligned} \quad (C.11)$$

Notice that c is the optimal choice when we maximize steady state utility, $\max_c v_j$, such that $v_c = 0$. Following Lemma 2 in Gabaix (2014) the expected loss function is then given by

$$\begin{aligned} \mathcal{L} &= \mathbb{E}(v_{j,t}(c_t^r, m_t) - v_{j,t}(c_t^s, m_t)) \\ &= \mathbb{E} \left(\frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^r)^2 + \frac{\partial^2 v}{\partial c^2} \hat{c}_t^r E_t \left(\frac{\partial c}{\partial a^n} \hat{a}_{t-1}^n + \sum_{h=1}^{\infty} \frac{\partial c}{\partial Y_g} \hat{Y}_{g,t+h} + \sum_{h=1}^{\infty} \frac{\partial c}{\partial \pi_h} \pi_{t+h} + \sum_{h=1}^{\infty} \frac{\partial c}{\partial i^a} \hat{i}_{t+h}^a \right) + t.i.o.c. \right. \\ &\quad \left. - \frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^s)^2 - \frac{\partial^2 v}{\partial c^2} \hat{c}_t^s E_t \left(\frac{\partial c}{\partial a^n} \hat{a}_{t-1}^n + \sum_{h=1}^{\infty} \frac{\partial c}{\partial Y_g} \hat{Y}_{g,t+h} + \sum_{h=1}^{\infty} \frac{\partial c}{\partial \pi_h} \pi_{t+h} + \sum_{h=1}^{\infty} \frac{\partial c}{\partial i^a} \hat{i}_{t+h}^a \right) - t.i.o.c. \right) \\ &= \mathbb{E} \left(\frac{1}{2} \frac{\partial^2 v}{\partial c^2} ((\hat{c}_t^r)^2 - (\hat{c}_t^s)^2) + (\hat{c}_t^r - \hat{c}_t^s) \frac{\partial^2 v}{\partial c^2} \sum_{h=1}^{\infty} \frac{\partial c}{\partial \pi_h} E_t \pi_{t+h} \right) \end{aligned} \quad (C.12)$$

where we use in the second to fourth line the fact that both types of households face the exact same true macroeconomic environment and the same past assets such that all pure macro terms, asset terms, and their cross-interactions cancel out identically. Hence, the loss simplifies purely to terms involving consumption differences and the cross-partials tied to consumption.

To resolve the cross-partial term in (C.12), we use the fact that $\hat{c}_t^r$ is the optimal policy under the true state, which is derived from the first-order condition $v_c(c_t^r, m_t) |_{c_t=c, m_t=m} = 0$ from maximizing the rational agent's problem. Expanding this first-order condition (to get a linear condition) and evaluating around the steady state gives

$$v_c + \frac{\partial^2 v}{\partial c^2} \hat{c}_t^r + \frac{\partial^2 v}{\partial c^2} E_t \left(\sum_{h=1}^{\infty} \frac{\partial c}{\partial \pi_h} \pi_{t+h} \right) \stackrel{!}{=} 0 \quad (C.13)$$

such that

$$\frac{\partial^2 v}{\partial c^2} E_t \left(\sum_{h=1}^{\infty} \frac{\partial c}{\partial \pi_h} \pi_{t+h} \right) = - \frac{\partial^2 v}{\partial c^2} \hat{c}_t^r \quad (C.14)$$

Substituting condition (C.14) into the loss function (C.12) and using the fact that $\hat{c}_t^s = c_t^s - c$ and $\hat{c}_t^r = c_t^r - c$ gives

$$\mathcal{L} = \mathbb{E} \left(- \frac{1}{2} \frac{\partial^2 v}{\partial c^2} (c_t^s - c_t^r)^2 \right) \quad (C.15)$$

Since policies c_t^s and c_t^r are unknown functions of the state variable we use the Implicit Function Theorem and linearize the policy functions around the steady state. The rational agent finds its policy c_t^r from its

first-order condition by maximizing its utility under rational expectations (C.3), $v_c(c_t^r, m_t) = 0$. Expanding the rational policy function around the steady state gives

$$c_t^r \approx c + E_t \sum_{h=1}^{\infty} \frac{\partial c}{\partial \pi_h} \pi_{t+h} + \text{other} \quad (\text{C.16})$$

The inattentive agent finds its policy c_t^s from its first-order condition by maximizing its perceived utility under subjective expectations (C.5), $\tilde{v}_c(c_t^s, m_t) = 0$. Expanding the inattentive policy function around the steady state yields

$$c_t^s \approx c + \gamma^g E_t \sum_{h=1}^{\infty} \frac{\partial c}{\partial \pi_h} \pi_{t+h} + \text{other} \quad (\text{C.17})$$

such that the gap between the policy choices can be rewritten as

$$c_t^s - c_t^r = -(1 - \gamma^g) \sum_{h=1}^{\infty} E_t \left(\frac{\partial c}{\partial \pi_h} \right) \pi_{t+h} \quad (\text{C.18})$$

where the "other" terms disappear as they are equivalent for both the rational and the inattentive agent. Substituting (C.18) into (C.15) yields the expected loss function

$$\mathcal{L} = -\frac{1}{2} \frac{\partial^2 v}{\partial c^2} \sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \left(\frac{\partial c}{\partial \pi_h} \frac{\partial c}{\partial \pi_{h'}} \sigma_{\pi_h \pi_{h'}} \right) (1 - \gamma^g)^2 \quad (\text{C.19})$$

Diagonal terms, $h = h'$, capture the variance of inflation expectations at horizon h . The off-diagonal terms $h \neq h'$, capture the cross-horizon autocovariances of inflation expectations. The household wants to minimize this expected loss in consumption by choosing the optimal level of attention to inflation γ^g while facing a cognitive constraint $\chi \gamma^g$

$$\min_{\gamma^g} -\frac{1}{2} \frac{\partial^2 v}{\partial c^2} \sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \left(\frac{\partial c}{\partial \pi_h} \frac{\partial c}{\partial \pi_{h'}} \sigma_{\pi_h \pi_{h'}} \right) (1 - \gamma^g)^2 + \chi \gamma^g \quad (\text{C.20})$$

where $\Lambda := -\frac{1}{2} \frac{\partial^2 v}{\partial c^2} \sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \frac{\partial c}{\partial \pi_h} \frac{\partial c}{\partial \pi_{h'}} \sigma_{\pi_h \pi_{h'}}$ is, following Gabaix (2014), the cost-of-inattention factor. The solution to the optimization problem (C.20) gives the result in Proposition 1.

Proof for Lemma 1, Group-level Attention

For any two households l, k with the same educational levels s_j^1, s_j^2 their utility function and policy functions result in identical steady-state derivatives:

$$\frac{\partial^2 v}{\partial c^2} \Big|_{s_j^l} = \frac{\partial^2 v}{\partial c^2} \Big|_{s_j^k} = \frac{\partial^2 v}{\partial c^2} \quad \text{and} \quad \frac{\partial c}{\partial \pi_h} \Big|_{s_j^l} = \frac{\partial c}{\partial \pi_h} \Big|_{s_j^k} = \frac{\partial c}{\partial \pi_h} \quad (\text{C.21})$$

which are used to solve (C.20) yielding the same $\gamma^g \forall (l, k)$ with the same educational state, such that the group-level attention is equivalent to the individual household's (educational state-dependent) γ^g .

D. Appendix to Section 4

D.1. Behavioral Jacobians

Here we derive, for completeness, the Jacobians of our model with inattentive households. To solve for the behavioral Jacobians we closely follow Auclert et al. (2020) Section D.3.

Our model considers two groups of households: first, those households who have paid attention to

the macroeconomy at $t \geq \tau$ and have all information about the shock arising in $t = 0$ and second, those households who didn't pay attention and still have to learn about the shock to inflation. Learning has probability $\gamma^s(1 - \gamma^s)^\tau$, where γ^s is the attention parameter we use in (B.1). Aggregating across the households gives the following Jacobian which specifies the response of output o to input i :

$$J^{o,i} = \gamma^s \sum_{\tau=0}^{\infty} (1 - \gamma^s)^\tau J^{o,i,\tau} \quad (\text{D.1})$$

where $J^{o,i,\tau}$ is the Jacobian for the group of households learning about the shock to input i at date τ . It is the policy at t that responds to the shock to i at date s . The Jacobian for a given s is given by:

$$\begin{aligned} J_{t,s}^{o,i,\tau} &= \gamma^s \sum_{\tau=0}^s (1 - \gamma^s)^\tau J_{t,s}^{o,i,\tau} \\ &= \gamma^s (J_{t,s}^{o,i,0} + (1 - \gamma^s) J_{t,s}^{o,i,1} + \dots + (1 - \gamma^s)^s J_{t,s}^{o,i,s}) \\ &= (1 - \gamma^s)^s J_{t,s}^{o,i,s} + \gamma^s \sum_{\tau=0}^{s-1} (1 - \gamma^s)^\tau J_{t,s}^{o,i,\tau} \end{aligned} \quad (\text{D.2})$$

To derive the Jacobians for each period t we further observe:

1. for $s \geq \tau$, we assume that households respond to a news shock about the date- s -change in i similarly to a news shock about the date- $(s - \tau)$ -change in i , i.e. $J_{t,s}^{o,i,\tau} = \dots = J_{t-s-\tau,s-\tau}^{o,i,0}$.
2. for $\tau > s$ the household has already updated her information in s about the news shock to i in s and therefore is irrelevant, such that $J_{t,s}^{o,i,\tau} = J_{t,s}^{o,i,s}, \forall t$

For any $t, s > 0$ this allows us to rewrite (D.2) as:

$$\begin{aligned} J_{t,s}^{o,i,\tau} &= (1 - \gamma^s)^s J_{t,s}^{o,i,s} + \gamma^s J_{t,s}^{o,i,0} + \gamma^s \sum_{\tau=1}^{s-1} (1 - \gamma^s)^\tau J_{t,s}^{o,i,\tau} \\ &= (1 - \gamma^s)^s J_{t-1,s-1}^{o,i,s} + \gamma^s J_{t,s}^{o,i,0} + \gamma^s \sum_{\tau=0}^{s-2} (1 - \gamma^s)^{\tau+1} J_{t-1,s-1}^{o,i,\tau} \\ &= (1 - \gamma^s)^s J_{t-1,s-1}^{o,i} + \gamma^s J_{t,s}^{o,i,0} \end{aligned} \quad (\text{D.3})$$

For $s = 0, \forall t$ (D.2) simplifies to

$$J_{t,s}^{o,i,\tau} = (1 - \gamma^s) J_{t,s}^{o,i,0} + \gamma^s J_{t,s}^{o,i,0} = J_{t,s}^{o,i,0} \quad (\text{D.4})$$

For $t = 0, s > 0$, and since $t \geq \tau$, the impulse response functions are only relevant if $\tau = 0$, such that

$$J_{t,s}^{o,i,\tau} = (1 - \gamma^s)^s J_{t,s}^{o,i,s} + \gamma^s J_{t,s}^{o,i,\tau} = \gamma^s J_{t,s}^{o,i,0} \quad (\text{D.5})$$

In period $\tau = 0$ the household has full information, i.e. paid fully attention. Since $J_{t,s}^{o,i,0}$ is the Jacobian for households that learn at $\tau = 0$ about shocks and as in Auclert et al. (2020) define the FIRE Jacobian $J_{t,s}^{o,i,FI} \equiv J_{t,s}^{o,i,0}$, we get (20).

D.2. Endogenous Attention in the Quantitative Model

Here we follow the same procedure as in Section C and apply Gabaix (2014) while also extending the model to incorporate idiosyncratic income shocks as an additional state variable. In the quantitative HANK model, the household chooses consumption and labor contemporaneously. *Note: In the following we mostly suppress state arguments for brevity.*

The *rational* household pays full attention to the entire state of the economy and chooses consumption c_t^r by solving the objectively true dynamic programming problem

$$\begin{aligned} V_j^r(S_t) &= \max_{c_t, n_t, a_t^n} \{u(c_t, n_t) + \beta E_t[V_j^r(S_{t+1})]\} \\ \text{s.t. } P_t c_t + a_t^n &= s_j e_t W_t n_t + (1 + i_t^a) a_{t-1}^n \end{aligned} \quad (\text{D.6})$$

where the state is given by $S_t = (a_{t-1}^n, s_j, e_t, \{\pi_{t+j}\}_{j=0}^\infty, \{i_{t+j}^a\}_{j=0}^\infty, \{W_{t+j}\}_{j=0}^\infty)$ ²⁸, e_t is the idiosyncratic transitory income shock, s_j defines the ex-ante educational groups (which are fixed) and $V_j^r(\cdot)$ is the value function evaluating the rational policy. Once the rational agent takes their consumption, asset and labor decisions - $c_t^r, (a_t^n)^r, n_t^r$ - their realized utility evolves according to

$$\begin{aligned} V_j^r(S_t) &= u(c_t^r, n_t^r) + \beta E_t[V_j^r(S_{t+1}^r)] \\ \text{s.t. } P_t c_t^r + (a_t^n)^r &= s_j e_t W_t n_t^r + (1 + i_t^a) a_{t-1}^n \end{aligned} \quad (\text{D.7})$$

where $S_{t+1}^r = ((a_t^n)^r, s_j, e_{t+1}, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{W_{t+j+1}\}_{j=0}^\infty)$ since assets are the residuals of the household's consumption choice. The realized utility of the *rational* household can be rewritten as

$$v_{j,t}(c_t^r, n_t^r, S_t) \equiv u(c_t^r, n_t^r) + \beta E_t[V_j^r(s_j e_t W_t n_t^r + (1 + i_t^a) a_{t-1}^n - P_t c_t^r, e_{t+1}, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{W_{t+j+1}\}_{j=0}^\infty)] \quad (\text{D.8})$$

The *inattentive* agent solves the subjective dynamic programming problem subject to their budget constraint, using their subjective expectations

$$\begin{aligned} V_j^s(S_t) &= \max_{c_t, n_t, a_t^n} \{u(c_t, n_t) + \beta \tilde{E}[V_j^s(S_{t+1})]\} \\ \text{s.t. } P_t c_t + a_t^n &= s_j e_t W_t n_t + (1 + i_t^a) a_{t-1}^n \end{aligned} \quad (\text{D.9})$$

where $\tilde{E}$ is the subjective expectation operator we define in [Definition 2](#) with the attention parameter $\gamma^s \in [0, 1)$. The perceived utility of the *inattentive* household can be written as

$$\tilde{v}_{j,t}(c_t^s, n_t^s, S_t) \equiv u(c_t^s, n_t^s) + \beta \tilde{E}[V_j^s(s_j e_t W_t n_t^s + (1 + i_t^a) a_{t-1}^n - P_t c_t^s, e_{t+1}, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)]$$

respectively,

$$\tilde{v}_{j,t}(c_t^s, n_t^s, S_t) \equiv u(c_t^s, n_t^s) + \beta E_t[V_j^s(s_j e_t W_t n_t^s + (1 + i_t^a) a_{t-1}^n - P_t c_t^s, e_{t+1}, \{\gamma^s \pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{Y_{g,t+j+1}\}_{j=0}^\infty)] \quad (\text{D.10})$$

Once the inattentive agent takes their consumption, asset and labor decisions - $c_t^s, (a_t^n)^s, n_t^s$ - their actual realized utility evolves according to true macroeconomic state and the objective expectations operator E_t

$$\begin{aligned} V_j(S_t) &= u(c_t^s, n_t^s) + \beta E_t[V_j(S_{t+1}^s)] \\ \text{s.t. } P_t c_t^s + (a_t^n)^s &= s_j e_t W_t n_t^s + (1 + i_t^a) a_{t-1}^n \end{aligned} \quad (\text{D.11})$$

where $S_{t+1}^s = ((a_t^n)^s, s_j, e_{t+1}, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{W_{t+j+1}\}_{j=0}^\infty)$ and $V_j(\cdot)$ denotes the objectively true value function. Thus, the realized utility for the *inattentive* household is

$$v_{j,t}(c_t^s, n_t^s, S_t) \equiv u(c_t^s, n_t^s) + \beta E_t[V_j(s_j e_t W_t n_t^s + (1 + i_t^a) a_{t-1}^n - P_t c_t^s, e_{t+1}, \{\pi_{t+j+1}\}_{j=0}^\infty, \{i_{t+j+1}^a\}_{j=0}^\infty, \{W_{t+j+1}\}_{j=0}^\infty)] \quad (\text{D.12})$$

Taking policy c_t^s and n_t^s instead of c_t^r and n_t^r leads to a loss in the ex-ante realized utility of the household.

²⁸ Again, the state space could have been simplified to $S_t = (a_{t-1}^n, s_j, e_t, \{\pi_{t+j}\}_{j=0}^\infty)$ as we keep the inattention to inflation assumption

The utility loss $\mathcal{L}$ is defined as the expected difference between the realized utility under the rational policies c_t^r and n_t^r and the realized utility under the inattentive policies c_t^s and n_t^s

$$\mathcal{L} = \mathbb{E}(v_{j,t}(c_t^r, n_t^r, S_t) - v_{j,t}(c_t^s, n_t^s, S_t)) \quad (\text{D.13})$$

where $\mathbb{E}$ is the unconditional expected objective operator. To quantify (D.13), we follow Gabaix (2014) and replace $v_{j,t}(\cdot)$ by a second order Taylor approximation around the deterministic zero-inflation steady state. Rewrite the realized utility of the rational household (D.8) as

$$\begin{aligned} v_{j,t}(c_t^r, n_t^r, S_t) &\approx v_j + v_c \hat{c}_t^r + v_n \hat{n}_t^r + E_t \sum_{h=1}^{\infty} v_{\pi} \hat{\pi}_{t+h} + \frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^r)^2 + \frac{1}{2} \frac{\partial^2 v}{\partial n^2} (\hat{n}_t^r)^2 \\ &+ \frac{\partial^2 v}{\partial c \partial n} \hat{c}_t^r \hat{n}_t^r + E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial c \partial \pi_h} \hat{c}_t^r \pi_{t+h} + E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial n \partial \pi_h} \hat{n}_t^r \pi_{t+h} + \text{other} \end{aligned} \quad (\text{D.14})$$

where v_j is the utility in the steady state. We use the notation $\hat{x}_t = x_t - x$ as the deviation of a variable x_t from its steady state value x . We evaluate derivatives at the zero-inflation steady state $\pi = 0$, i.e., we define for $c_t = (c_t^r, c_t^s)$ and $n_t = (n_t^r, n_t^s)$ first order derivatives by $v_x \equiv \frac{\partial v_{j,t}(c_t, n_t, S_t)}{\partial x_t} \big|_{c_t=c, n_t=n, S_t=S}$, second order derivatives $\frac{\partial^2 v}{\partial x^2} \equiv \frac{\partial^2 v_{j,t}(c_t, n_t, S_t)}{\partial x_t^2} \big|_{c_t=c, n_t=n, S_t=S}$, cross-partials by $\frac{\partial^2 v}{\partial x \partial y} \equiv \frac{\partial^2 v_{j,t}(c_t, n_t, S_t)}{\partial x_t \partial y_t} \big|_{c_t=c, n_t=n, S_t=S}$ and $v_j \equiv v_{j,t}(c_t, n_t, S_t) \big|_{c_t=c, n_t=n, S_t=S}$ is the utility function evaluated at the steady state. Notice, that steady state consumption c and labor n are the same optimal choice for both types of agents. "other" collects all other first and second order partials. Rewrite the approximated utility function of the *rational* household, where $\frac{\partial c}{\partial x} \equiv \frac{\partial c_t}{\partial x_t} \big|_{c_t=c, n_t=n, S_t=S}$ and $\frac{\partial n}{\partial x} \equiv \frac{\partial n_t}{\partial x_t} \big|_{c_t=c, n_t=n, S_t=S}$,

$$v_{j,t}(c_t^r, n_t^r, S_t) \approx v_j + v_c \hat{c}_t^r + v_n \hat{n}_t^r + \frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^r)^2 + \frac{1}{2} \frac{\partial^2 v}{\partial n^2} (\hat{n}_t^r)^2 + \frac{\partial^2 v}{\partial c \partial n} \hat{c}_t^r \hat{n}_t^r \quad (\text{D.15})$$

$$+ E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial c \partial \pi_h} \hat{c}_t^r \pi_{t+h} + E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial n \partial \pi_h} \hat{n}_t^r \pi_{t+h} + \text{other} \quad (\text{D.16})$$

The approximated utility function of the *inattentive* household is given similarly by

$$v_{j,t}(c_t^s, n_t^s, S_t) \approx v_j + v_c \hat{c}_t^s + v_n \hat{n}_t^s + \frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^s)^2 + \frac{1}{2} \frac{\partial^2 v}{\partial n^2} (\hat{n}_t^s)^2 + \frac{\partial^2 v}{\partial c \partial n} \hat{c}_t^s \hat{n}_t^s \quad (\text{D.17})$$

$$+ E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial c \partial \pi_h} \hat{c}_t^s \pi_{t+h} + E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial n \partial \pi_h} \hat{n}_t^s \pi_{t+h} + \text{other} \quad (\text{D.18})$$

Notice that c and n are optimal choices when we maximize steady state utility, such that $v_c = 0$ and $v_n = 0$. Next, since we assume the utility in (16) to be additively separable, the cross-partial derivative of the value function with respect to consumption and labor evaluated at the steady state is exactly zero, $\frac{\partial^2 v}{\partial c \partial n} = 0$. The expected loss function is then given by

$$\begin{aligned} \mathcal{L} &= \mathbb{E}(v_{j,t}(c_t^r, n_t^r, S_t) - v_{j,t}(c_t^s, n_t^s, S_t)) \\ &= \mathbb{E} \left(\frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^r)^2 + \frac{1}{2} \frac{\partial^2 v}{\partial n^2} (\hat{n}_t^r)^2 + E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial c \partial \pi_h} \hat{c}_t^r \pi_{t+h} + E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial n \partial \pi_h} \hat{n}_t^r \pi_{t+h} + \text{other} \right. \\ &\quad \left. - \frac{1}{2} \frac{\partial^2 v}{\partial c^2} (\hat{c}_t^s)^2 - \frac{1}{2} \frac{\partial^2 v}{\partial n^2} (\hat{n}_t^s)^2 - E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial c \partial \pi_h} \hat{c}_t^s \pi_{t+h} - E_t \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial n \partial \pi_h} \hat{n}_t^s \pi_{t+h} - \text{other} \right) \\ &= \mathbb{E} \left(\frac{1}{2} \frac{\partial^2 v}{\partial c^2} ((\hat{c}_t^r)^2 - (\hat{c}_t^s)^2) + \frac{1}{2} \frac{\partial^2 v}{\partial n^2} ((\hat{n}_t^r)^2 - (\hat{n}_t^s)^2) \right. \\ &\quad \left. + (\hat{c}_t^r - \hat{c}_t^s) \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial c \partial \pi_h} E_t \pi_{t+h} + (\hat{n}_t^r - \hat{n}_t^s) \sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial n \partial \pi_h} E_t \pi_{t+h} \right) \end{aligned} \quad (\text{D.19})$$

where in the second to fifth line all terms collected in “other” cancel out since both types of households face the exact same true environment.

To resolve the cross-partial term in (D.19), we use the fact that $\hat{c}_t^r$ and $\hat{n}_t^r$ are the optimal policies at the rational optimum derived from the first-order condition from maximizing the rational agent’s problem, $v_c(c_t^r, n_t^r, S_t)$ and $v_n(c_t^r, n_t^r, S_t)$. Doing a first-order expansion of these first-order conditions and evaluating around the steady state gives (under additive separability) yields for consumption

$$\sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial c \partial \pi_h} E_t \pi_{t+h} = -\frac{\partial^2 v}{\partial c^2} \hat{c}_t^r \quad (\text{D.20})$$

and for labor

$$\sum_{h=1}^{\infty} \frac{\partial^2 v}{\partial n \partial \pi_h} E_t \pi_{t+h} = -\frac{\partial^2 v}{\partial n^2} \hat{n}_t^r \quad (\text{D.21})$$

Substituting condition (D.20) and (D.21) into the loss function (D.19) and using the fact that $\hat{c}_t = c_t - c$ for $c_t = (c_t^r, c_t^s)$ and $\hat{n}_t = n_t - n$ for $n_t = (n_t^r, n_t^s)$ gives

$$\mathcal{L} = \mathbb{E} \left(-\frac{1}{2} \frac{\partial^2 v}{\partial c^2} (c_t^s - c_t^r)^2 - \frac{1}{2} \frac{\partial^2 v}{\partial n^2} (n_t^s - n_t^r)^2 \right) \quad (\text{D.22})$$

To solve for the rational and inattentive policy functions, we use the Implicit Function Theorem and linearize the first-order conditions for consumption and labor of the household’s utility functions around the steady state. Expanding the policy functions around the steady state allows us to express the gap between the policy choices as

$$c_t^s - c_t^r = -(1 - \gamma^g) \sum_{h=1}^{\infty} E_t \left(\frac{\partial c}{\partial \pi_h} \right) \pi_{t+h} \quad (\text{D.23})$$

and

$$n_t^s - n_t^r = -(1 - \gamma^g) \sum_{h=1}^{\infty} E_t \left(\frac{\partial n}{\partial \pi_h} \right) \pi_{t+h} \quad (\text{D.24})$$

Substituting (D.23) and (D.24) into (D.22) yields the expected loss function

$$\mathcal{L} = -\frac{1}{2} \sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \left(\left(\frac{\partial^2 v}{\partial c^2} \frac{\partial c}{\partial \pi_h} \frac{\partial c}{\partial \pi_{h'}} + \frac{\partial^2 v}{\partial n^2} \frac{\partial n}{\partial \pi_h} \frac{\partial n}{\partial \pi_{h'}} \right) \sigma_{\pi_h \pi_{h'}} \right) (1 - \gamma^g)^2 \quad (\text{D.25})$$

Each household is characterized by the educational level s_j , which determines their income group $g \in \{Low, High\}$. The household chooses their attention level γ^g to minimize their state-dependent utility loss plus the cognitive cost of processing information $\chi \gamma^g$ by solving

$$\min_{\gamma^g} -\frac{1}{2} \sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \left(\left(\frac{\partial^2 v}{\partial c^2} \frac{\partial c}{\partial \pi_h} \frac{\partial c}{\partial \pi_{h'}} + \frac{\partial^2 v}{\partial n^2} \frac{\partial n}{\partial \pi_h} \frac{\partial n}{\partial \pi_{h'}} \right) \sigma_{\pi_h \pi_{h'}} \right) (1 - \gamma^g)^2 + \chi \gamma^g \quad (\text{D.26})$$

Solving (D.26) gives the result in [Proposition 3](#).

Proof for [Lemma 2](#), Group-level Attention

[Lemma 2](#) follows from [Lemma 1](#). Because the steady-state utility derivatives and policy gradients are identical for all agents sharing educational state $s_j \in g$, all households within group g solve the same objective and select the same optimal attention parameter γ^g , as stated in [Lemma 2](#).

Note: Aggregate Loss in the HANK model

In the decentralized HANK, the individual household at state s independently chooses their attention level $\gamma^g \equiv \gamma^g(s)$ to minimize their individual utility loss. The aggregate loss in the economy is determined by integrating the households’ individual choices across the cross-sectional distribution of households

$dD^g(e, a)$. Since households differ in their asset holdings a and idiosyncratic productivity shocks e , individual losses have to be weighted by the ergodic distribution over the state space. Thus, weighing by the ergodic distribution gives the aggregate loss function in the full HANK model

$$\min_{\gamma^g} -\frac{1}{2} \sum_{h=1}^{\infty} \sum_{h'=1}^{\infty} \left(\sum_e \int \left(\frac{\partial^2 v}{\partial c^2}(e, a) \frac{\partial c(e, a)}{\partial \pi_h} \frac{\partial c(e, a)}{\partial \pi_{h'}} + \frac{\partial^2 v}{\partial n^2}(e, a) \frac{\partial n(e, a)}{\partial \pi_h} \frac{\partial n(e, a)}{\partial \pi_{h'}} \right) dD^g(e, da) \right) (1 - \gamma^g)^2 \sigma_{\pi_h \pi_{h'}} + \chi \gamma^g \quad (\text{D.27})$$

which can be used by the policy maker.

D.3. Attention to Prices as a Function of Inflation

Using the relation between the price index and inflation $\frac{P_t}{P_{t-1}} = 1 + \pi_t$ and the expectation formation process for inflation in [Definition 2](#) we can derive an expression for the expectation operator for prices, from its relation to inflation as

$$\begin{aligned} \tilde{E}_t(P_s - 1) &= \tilde{E}_t P_t \prod_{\tau=t+1}^s (1 + \pi_\tau) - 1 \simeq \tilde{E}_t P_t (1 + \sum_{\tau=t+1}^s \pi_\tau) - 1 \\ &= P_t \sum_{\tau=t+1}^s (\gamma E_t \pi_\tau + (1 - \gamma) \tilde{E}_{t-1} \pi_\tau) + P_t - 1 = \gamma E_t (P_s - 1) + (1 - \gamma) \left(P_t \sum_{\tau=t+1}^s \tilde{E}_{t-1} \pi_\tau + P_t - 1 \right) \\ &= \gamma E_t (P_s - 1) + (1 - \gamma) \left(\frac{P_t}{P_{t-1}} \left(\tilde{E}_{t-1} P_{t-1} \sum_{\tau=t}^s \pi_\tau + P_{t-1} - 1 \right) - \frac{P_t}{P_{t-1}} \tilde{E}_{t-1} P_{t-1} \pi_t + \frac{P_t}{P_{t-1}} - 1 \right) \\ &= \gamma E_t (P_s - 1) + (1 - \gamma) \left(\frac{P_t}{P_{t-1}} \tilde{E}_{t-1} (P_s - 1) - P_t \tilde{E}_{t-1} \pi_t + \frac{P_t}{P_{t-1}} - 1 \right) \\ &= \gamma E_t (P_s - 1) + (1 - \gamma) \tilde{E}_{t-1} (P_s - 1) + (1 - \gamma) \pi_t \tilde{E}_{t-1} (P_s - 1) + (1 - \gamma) (\pi_t - P_t \tilde{E}_{t-1} \pi_t) \\ &\simeq \gamma E_t (P_s - 1) + (1 - \gamma) \tilde{E}_{t-1} (P_s - 1) + (1 - \gamma) (\pi_t - \tilde{E}_{t-1} \pi_t) \\ &\quad + (1 - \gamma) \pi_t \tilde{E}_{t-1} (P_s - 1) + (1 - \gamma) \left(-(\sum_{i=0}^t \pi_i) \tilde{E}_{t-1} \pi_t \right) \end{aligned} \quad (\text{D.28})$$

where the last two terms are second order in π . Using $\tilde{E}_{t-1} \pi_t = \gamma \sum_{j=0}^t (1 - \gamma)^j E_{t-1-j} \pi_t$ and assuming perfect foresight, the last term can be rewritten as

$$\begin{aligned} (1 - \gamma) \pi_t \left(1 - (1 + \sum_{j=0}^t \pi_j) (1 - (1 - \gamma)^t) \right) &\simeq (1 - \gamma) \pi_t \left((1 - \gamma)^t - (\sum_{j=0}^t \pi_j) (1 - (1 - \gamma)^t) \right) \\ &\simeq (1 - \gamma)^{t+1} \pi_t \end{aligned} \quad (\text{D.29})$$

Thus, the expectation process for inflation is

$$\tilde{E}_{j,t} \pi_{t+1} = \gamma^g E_t \pi_{t+1} + (1 - \gamma^g) \tilde{E}_{j,t-1} \pi_{t+1} \quad (\text{D.30})$$

and for the price level is

$$\tilde{E}_{j,t} P_{t+1} = \gamma^g E_t P_{t+1} + (1 - \gamma^g) \tilde{E}_{j,t-1} P_{t+1} + (1 - \gamma^g)^{t+1} \pi_t \quad (\text{D.31})$$

Note: In the simulations of the main quantitative model we use the correction term $(1 - \gamma)^{t+1} \pi_t$ for the price index. The difference in impulse response functions when this term is omitted is shown in [E.8](#) [Figure 19](#). The results only differ infinitesimally small.

D.4. Misperception of Prices

In the baseline model in Section 4 we assume, similar to other research papers that study non-FIRE in HANK economies, that households observe the current price level in t . However, empirical literature on expectations (see e.g., [d'Acunto et al., 2021](#); [Andre et al., 2022](#); [Montag, 2026](#)) suggests that households systematically misperceive the current macroeconomic state and therefore deviate from FIRE already in the current period t .

Following [Definition 2](#), misperception about current inflation could be introduced into our model in the following way:

$$\tilde{E}_t \pi_{t+1} = \rho \tilde{E}_{t-1} \pi_t + \rho \gamma^\delta (\pi_t \delta - \tilde{E}_{t-1} \pi_t) \quad (\text{D.32})$$

$$\tilde{E}_t \pi_{t+1} = \gamma^\delta \delta E_t \pi_{t+1} + (1 - \gamma^\delta) \tilde{E}_{t-1} \pi_{t+1} \quad (\text{D.33})$$

where $\delta < 1$ is the salience parameter, distorting household's nowcast of the economy, in addition to the attention parameter γ^δ . This allows us to derive the Jacobian under misperception as

$$J_{t,s} = (1 - \gamma^\delta) J_{t-1,s-1} + \delta \gamma J_{t,s}^{FIRE} \quad (\text{D.34})$$

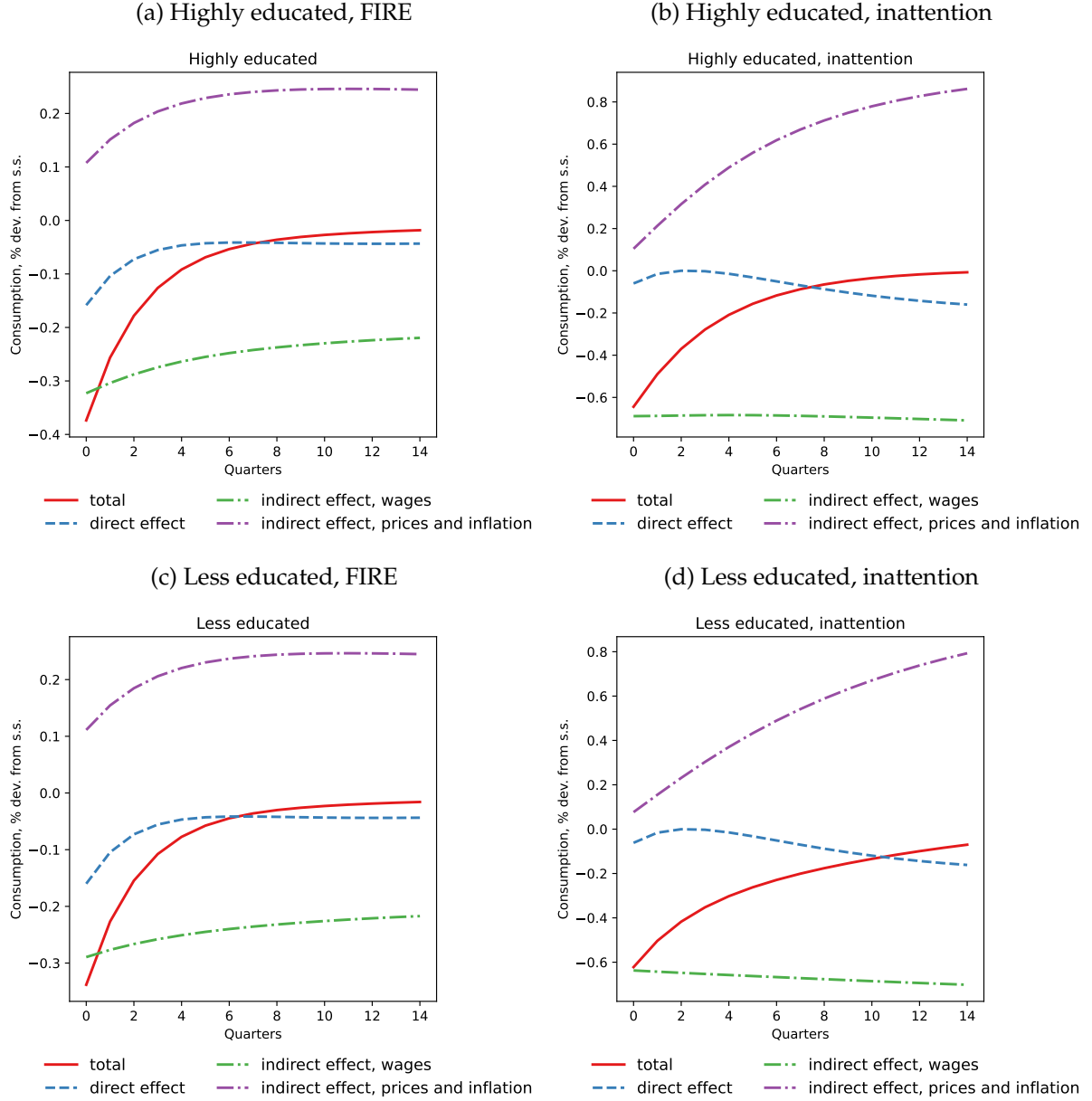
From equation (D.34), we observe that the weight put on the rational expectation Jacobian is smaller under misperception, which means that the labor supply channel that we identify potentially is even more amplified under misperception. Note, however, that the derivation in (D.34) only applies to the dynamics, but the misperception could in general also affect the solution of the steady state which we put aside in our analysis by solving the model under rational expectations like [Auclert et al. \(2020\)](#), [Pfäuti and Seyrich \(2022\)](#), [Guerreiro \(2022\)](#).

E. Appendix to Section 5: Additional Results and Figures

E.1. Direct and Indirect Effects

The aggregate impulse response of consumption in [Figure 2](#) from our quantitative HANK model in Section 5 can further be decomposed into direct responses due to the interest rate and indirect responses due to wages and prices, respectively, inflation. [Figure 8](#) shows the decomposition into direct and indirect effects for the response of consumption under FIRE for highly educated in subfigure (a) and less educated in subfigure (c) as well as under inattention for highly educated in subfigure (b) and for less educated in subfigure (d). The red line shows the total response of consumption to the interest rate that we already observe in [Figure 2](#): households decrease their consumption on impact across all models, then increase their consumption for approximately six quarters after which it then stabilizes. This total response is the sum of the direct effect of the change in interest rate (blue line) and the indirect effect of the nominal wage (green line) and the indirect effect of prices, and inflation (purple line). In line with the HANK literature ([Kaplan et al., 2018](#)) we find that indirect effects are larger than direct effects. While changes in interest rate and wage levels induce a negative response across households, changes in price level lead to a positive impact on consumption across all educational levels and expectation models. This is due to their positive effect on labor supply, which supports aggregate consumption. If agents are inattentive, price levels and inflation contribute more to the total effect on consumption. Wages have the greatest negative impact in all models, especially if households are inattentive. Both less and highly educated households decrease their consumption levels on impact and over time.

Figure 8: Direct and Indirect Effects

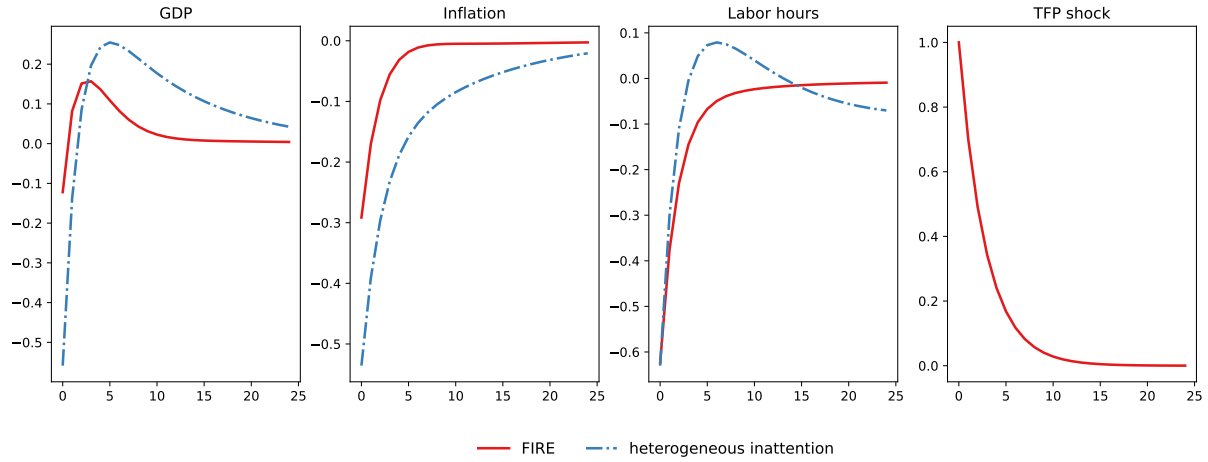


Notes: The figure shows direct and indirect effects for each type of households and for full information rational expectations and inattention cases. Indirect effects in our model include dynamics of wages and prices (inflation).

E.2. TFP and Markup Shocks

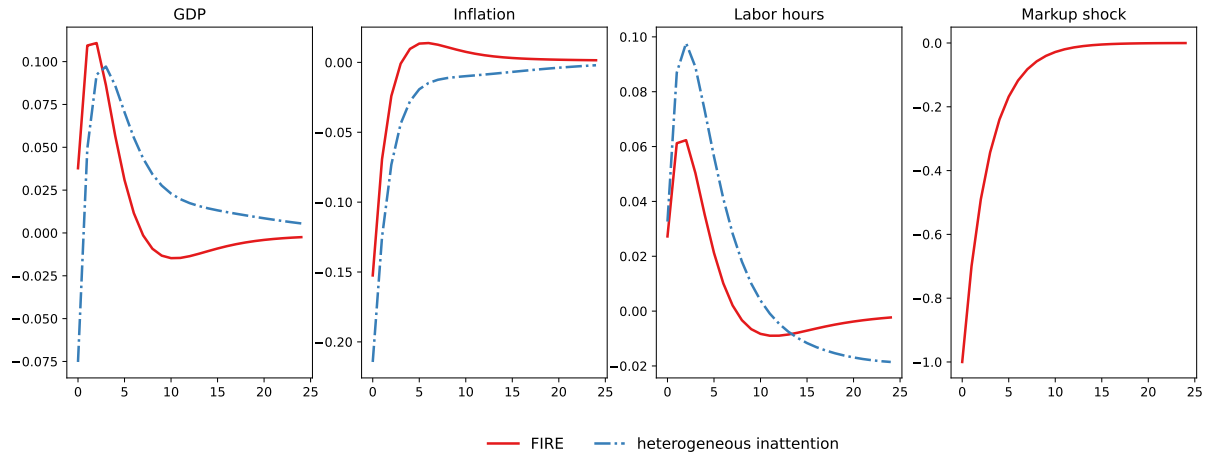
In addition to the main simulation of the HANK model under a monetary policy shock, we show here the aggregate responses of the HANK model to a 1% TFP shock in Figure 9 and a negative 1% markup shock in Figure 10 for the model under FIRE and with heterogeneous inattention. Across both expectational models, a positive TFP shock in Figure 9 leads to a decrease in inflation and labor hours which stabilize in subsequent periods. Under inattention the effect, however, is more amplified, i.e. the economy produces more output, which is driven by the positive response of labor hours and which channel we have already explored in the main quantitative model (real wages are perceived to be lower as inflation decline is not fully incorporated into the expectations and the labor supply is thus higher). The increased supply of

Figure 9: TFP Shock, Heterogeneity Across Types



Notes: The shows the impulse responses to a 1% TFP shock. The red lines show the responses in a HANK with FIRE, the dashed blue lines for the HANK with heterogeneous inattention.

Figure 10: Markup Shock, Heterogeneity Across Types



Notes: The figure shows the impulse responses to a -1% markup shock. The red lines show the responses in a HANK with FIRE, the dashed blue lines for the HANK with heterogeneous inattention.

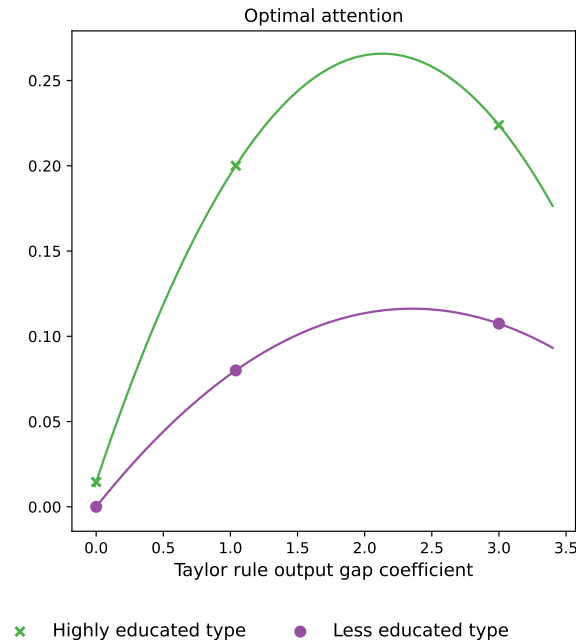
labor hours under inattention leads to output being higher in the case of inattention in comparison to the FIRE model.

A negative markup shock, as shown in Figure 10, leads to even stronger effects in the short run: the economy increases production and labor hours already on impact, while inflation strongly decreases in the short run. The supply of labor is amplified under inattention and is again driven by the household's misperception of the decrease in inflation. Under FIRE the positive supply of labor hours is more muted since households observe the decrease in inflation and their increase in real wages, while under inattention households do not fully observe the full deflationary effect and therefore increase their labor supply.

E.3. Monetary Policy and Optimal Attention across Types

To assess the role of monetary policy, complementary to studying the effect of the inflation Taylor rule coefficient in Figure 5, we additionally look at the correlation between the output gap coefficient in the Taylor rule and optimal attention across household types in Figure 11. The results here suggest

Figure 11: Trade-off depending on the Output Gap Coefficient



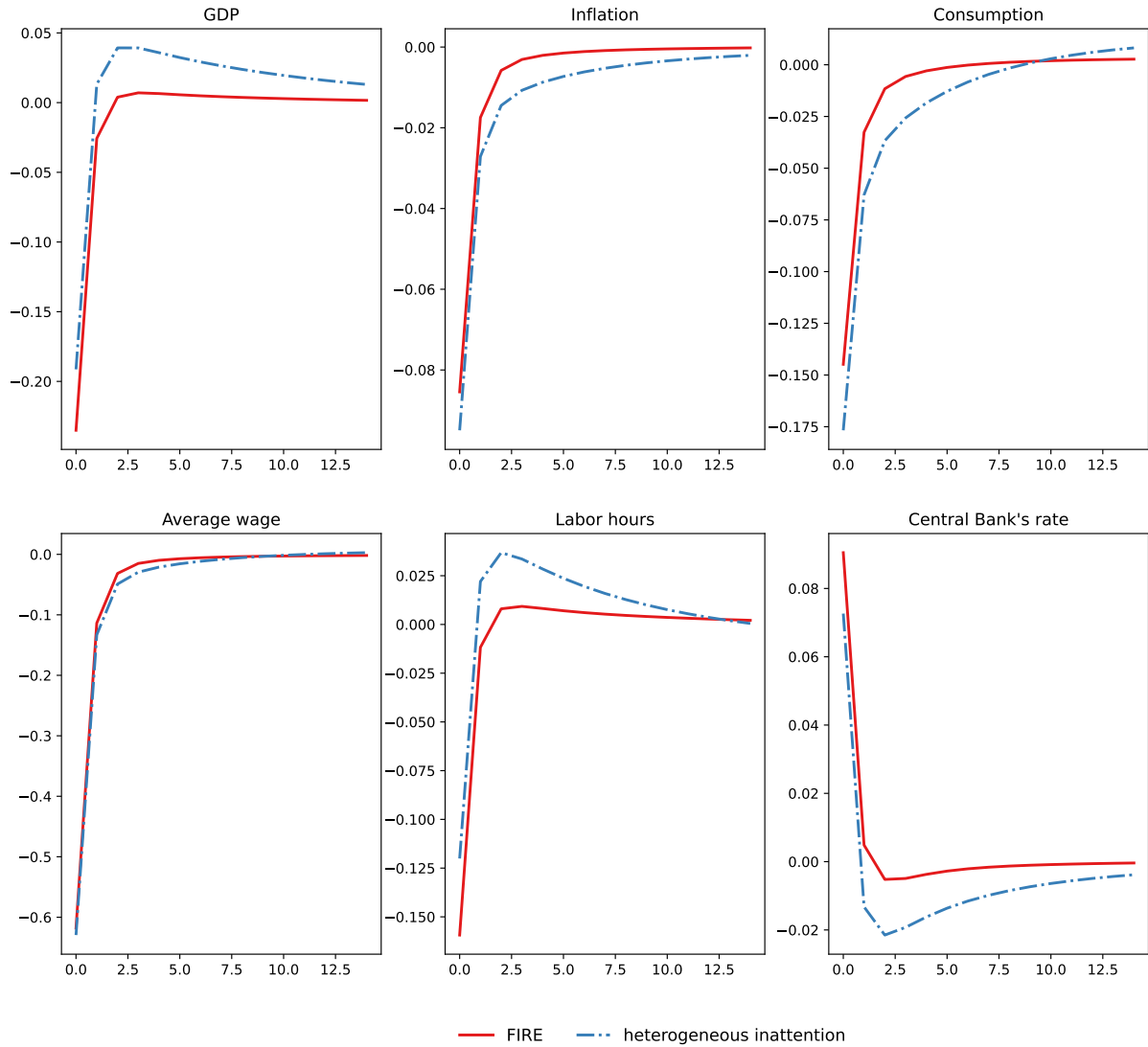
Notes: The figure shows optimal attention as a function of ϕ_y coefficient keeping other Taylor rule coefficients as in the baseline.

a hump-shaped relation between optimal attention and the output gap parameter: optimal attention increases with increasing ϕ_y . This is because the more active the policy maker reacts to changes in the output gap, the more volatile inflation is and the more attentive the households are, whereby the opposite is true for its reaction to inflation, where a more active policy towards inflation makes households less attentive since inflation gets more stabilized. Across both specifications, optimal attention is higher for the highly educated household, which is in line with our expectational framework in equation (25). A very strong output gap reaction causes attention to decline because the economy itself stabilizes ultimately also stabilizing inflation.

E.4. Flexible Wages

Contrary to Figures 1 and 2 we now assess the effect of inattention under flexible wages. Figure 12 shows the aggregate impulse response functions after a contractionary monetary policy shock in the model with flexible wages and Figure 13 shows the heterogeneous effects for both highly and less educated households. In general, the economy's trajectory under flexible wages is similar to that under sticky wages. However, under flexible wages, we observe a weaker response (in terms of magnitude) and a faster recovery from the shock than under wage rigidity. The average wage, however, evolves differently under flexible wages, which can be attributed to their quick adjustment to the new labor market equilibrium if rigidities are not present. On the other hand, under sticky wages in Figure 1, inattention and wage stickiness create two types of persistence: one from the belief channel when households misperceive

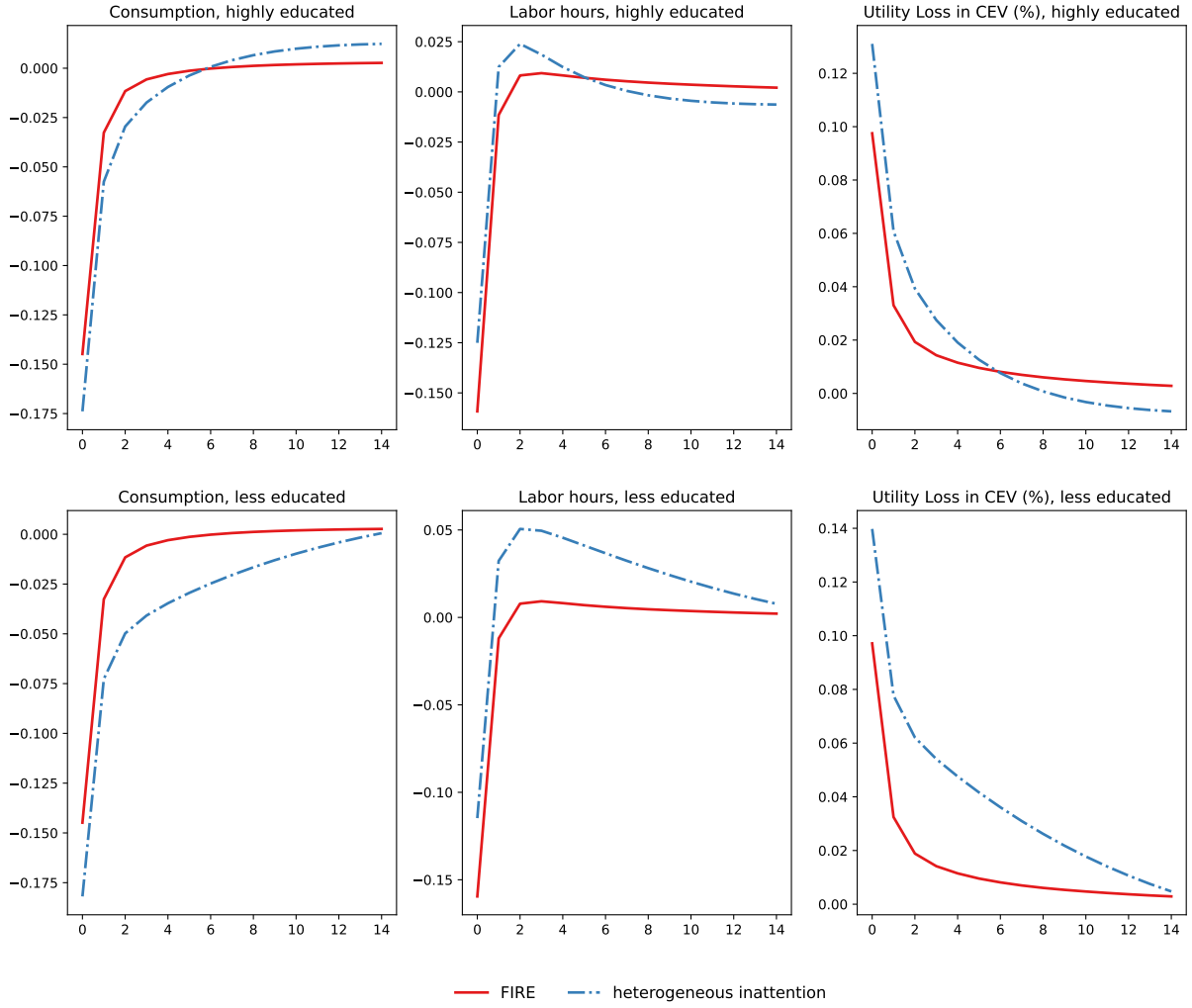
Figure 12: Flexible Wages



Notes: The figure shows the impulse responses to a contractionary 25 bps monetary policy shock in a model with flexible wages. The red line shows the responses in a HANK with FIRE and the blue dashed line shows the responses in a HANK with heterogeneous inattention.

the nominal price level and therefore distort the labor market, and the second from wage rigidity itself. If wages are flexible, the wages adjust quicker to economic distortions, so that even if household's are inattentive they misperceive the real wage to be larger than that under sticky wages since in the latter case, the economy takes longer to adjust which is reflected in the accordingly lower wage rates after a contractionary monetary policy shock. Hence, leisure becomes more costly and households are able to smooth consumption by increasing labor supply. As a consequence, as shown in Figure 13, the household's welfare recovers faster. Again, the increased labor supply is mainly driven by the less educated and, therefore, more inattentive households.

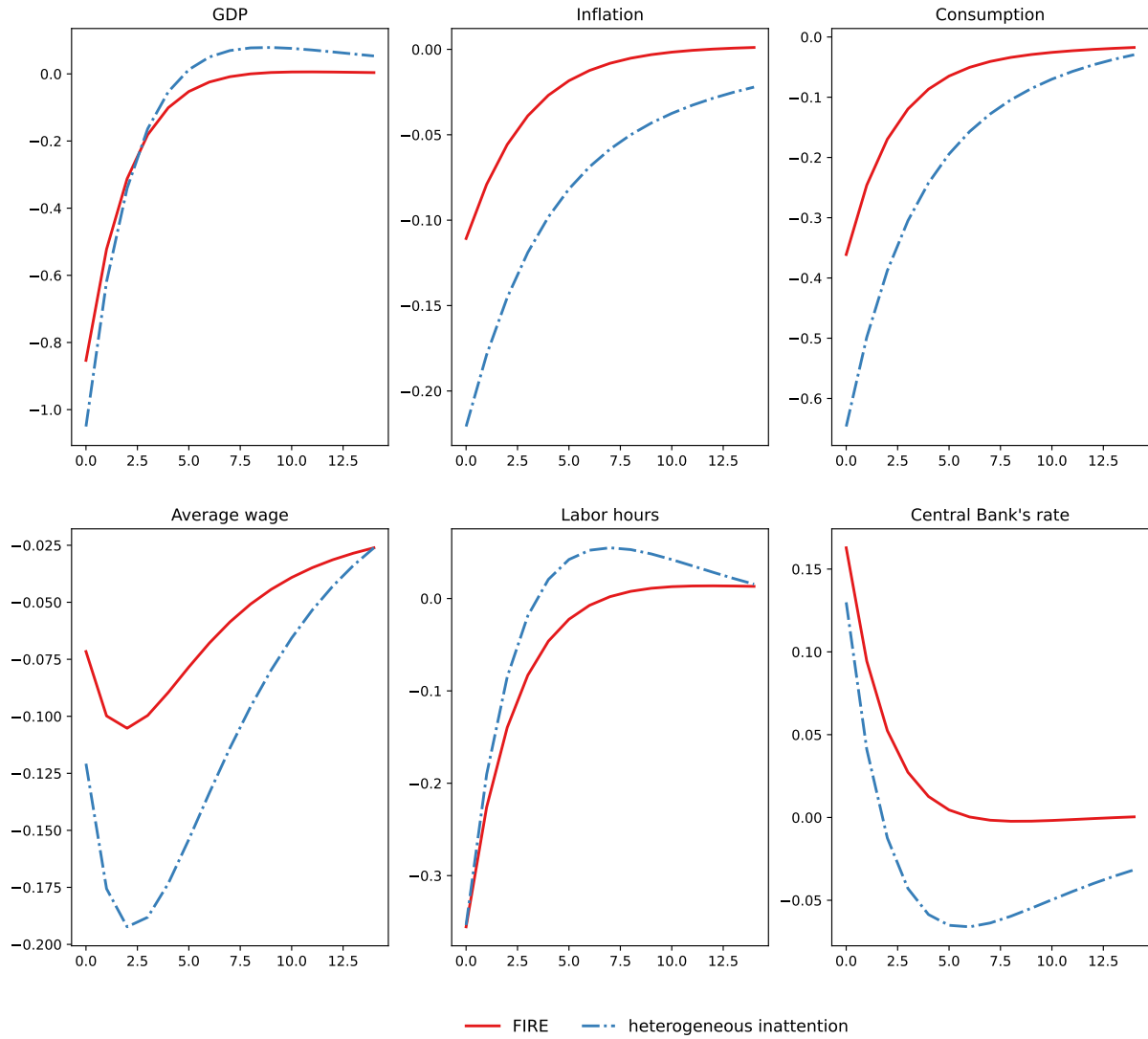
Figure 13: Flexible Wages



E.5. Inattention to the Nominal Interest Rates

Here we extend the analysis of the main model and assume that households are inattentive to the nominal interest rate, inflation, and the price level altogether. We calibrate attention to the nominal interest rate according to [Bordalo et al. \(2020\)](#) and set it to $\gamma_i^\delta = 0.8$. The results in Figure 14 and Figure 15 are close to the baseline model 1 and Figure 2. Numerically, the difference between these figures is only up to 10^{-3} units big. Additional simulations of the model for $\gamma_i^\delta < 0.8$ give the same result. If a household is inattentive to inflation, prices, and the nominal interest rate, they also misperceive the future real interest rate as $\tilde{E}_{j,t+k-1}r_{t+k} = \tilde{E}_{j,t+k-1}i_{t+k} - \tilde{E}_{j,t+k}\pi_{t+k+1}$ and form subjective expectations about it. Given that the difference and the effects from being inattentive is negligible, this provides additional evidence, that the direct channel in HANK economies has only small effects on the aggregate transmission, such that policies mainly rely on transmission through indirect channels. Ultimately, having FIRE or non-FIRE towards the nominal interest rate doesn't affect the economy, whereby having FIRE or non-FIRE towards inflation and prices does. From Figure 15 we also observe that the loss in the welfare of households and consumption-saving decisions are not affected by the additional inattention to the interest rate but rather by the increased labor supply, respectively, the inattention to inflation and price levels.

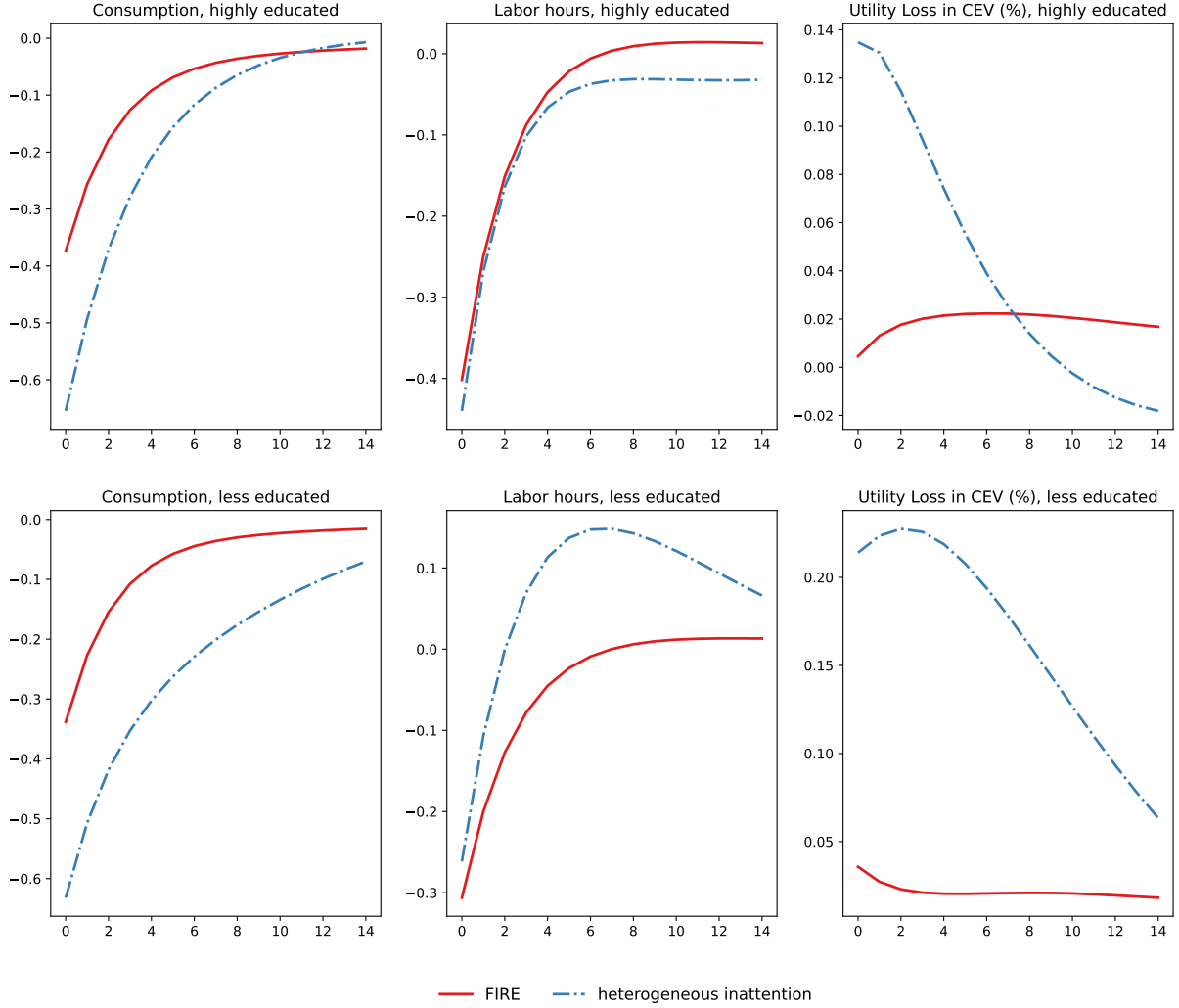
Figure 14: Inattention to the Nominal Interest Rate



E.6. Economy under Low and High Idiosyncratic Income Risk

In addition to Figure 4, we show in Figure 16 the aggregate response when households experience a high or low transitory income risk relative to the baseline risk level assumed in Section 5. The results reinforce our previous conclusions: an increase in idiosyncratic risk amplifies the effects of an increased labor supply, resulting from the belief channel. As risk levels increase, households supply more labor and thus receive more labor income and as a result have more total income, which allows them to decrease future consumption by less, so aggregate demand falls by less. This is especially true for HtM households with high MPCs who consume all their additional income in the current period. Under a high risk environment, a contractionary monetary policy shock does not induce a strong negative inflation as in the baseline or low risk environment. Most surprisingly, the average wage level is higher in the higher risk environment. This implies that labor demand and supply reinforce each other. The less negative drop in consumption, relative to the benchmark case, leads to an increase in production and simultaneously a higher demand for labor. Labor additionally is supplied more from households as they want to self-insure against business cycle changes as we have explored above. Overall, these effects mitigate the negative effect of a contractionary monetary policy, reducing deflationary effects.

Figure 15: Inattention to the Nominal Interest Rate

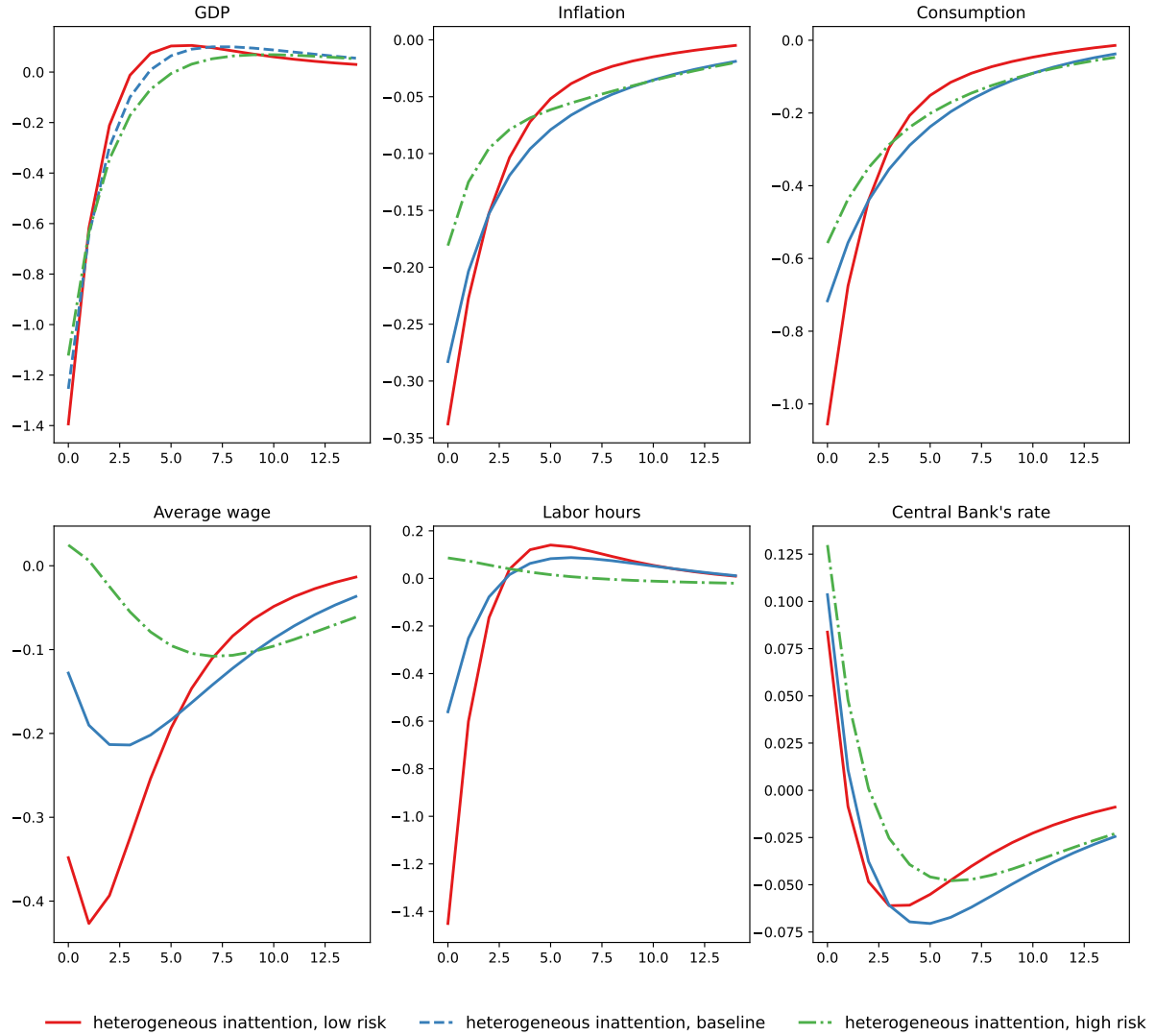


E.7. Comparison between Homogeneous and Heterogeneous Inattention

To filter the effects of heterogeneous attention, we now compare the effects from our HANK model with heterogeneous inattention to the HANK with homogeneous attention and FIRE to inflation in Figure 17 and Figure 18. Under heterogeneous attention, each household in group g has a group-specific attention parameter that depends on their educational level, i.e., $0 < \gamma^1 < \gamma^2 < 1$ where γ^1 is the attention for less educated and γ^2 the attention for highly educated households, under homogeneous attention, every household has the same attention level independent of their educational level, $0 < \gamma^g < 1 \forall g$ and under FIRE, everyone is rational and fully attentive, i.e., $\gamma^g = 1 \forall g$. The heterogeneous attention levels γ^g are calibrated to our empirical findings in Table 2, the homogeneous attention level is the weighted average between γ^1 and γ^2 and under FIRE $\gamma^g = 1 \forall g$.

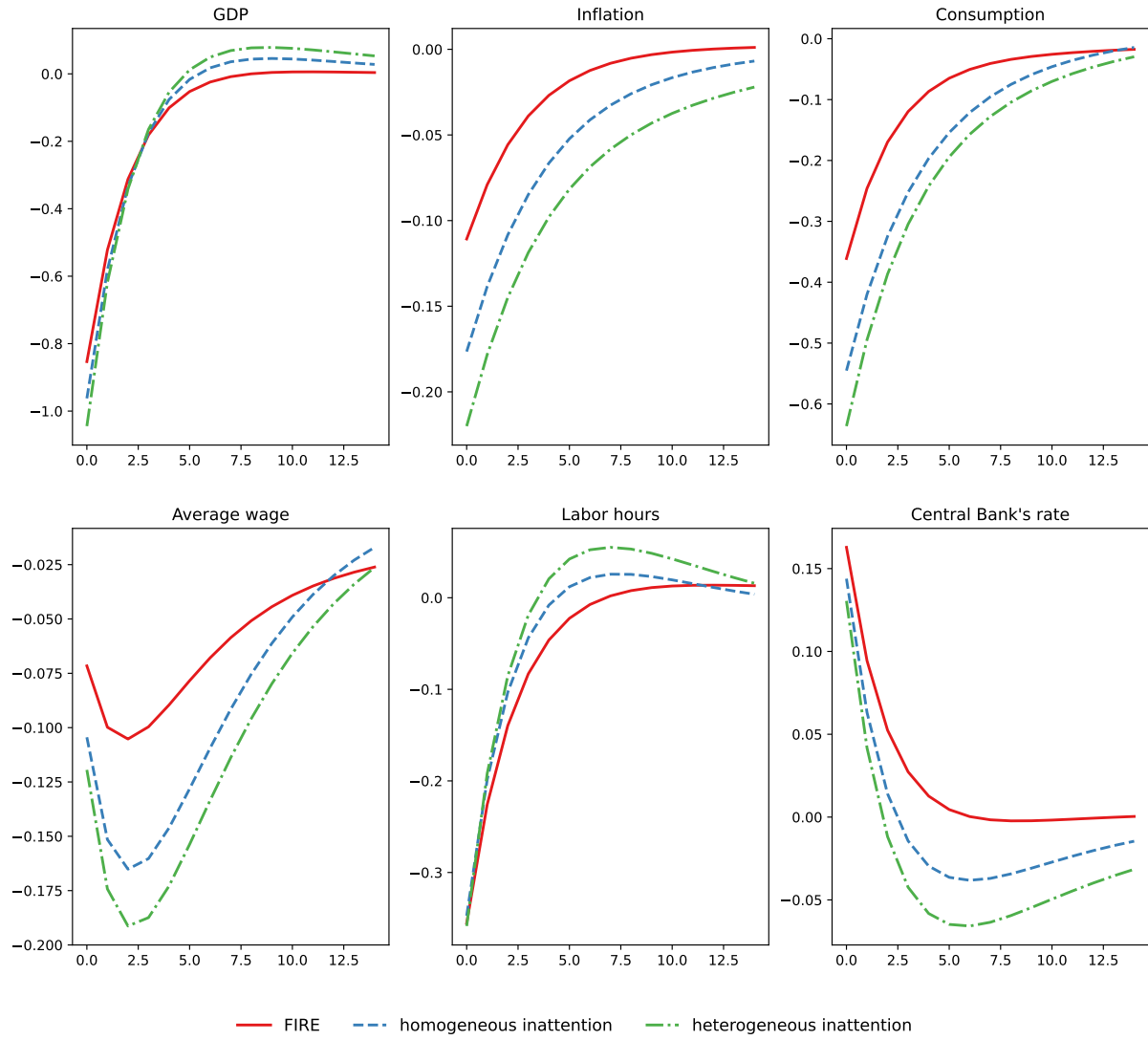
Overall, the results show similar patterns across all the expectational models specified in Figure 17 and Figure 18, however, inattention to inflation amplifies the aggregate and group-level effects. When we allow households to pay more attention to inflation, i.e., they either have FIRE or are highly educated, households increase consumption faster after the initial drop because they observe the economy more closely, allowing them to form expectations closer to FIRE and correctly adjust their perception about their increase in real income after the contractionary monetary shock. The opposite is true for less

Figure 16: Responses with Different Level of Idiosyncratic Risk



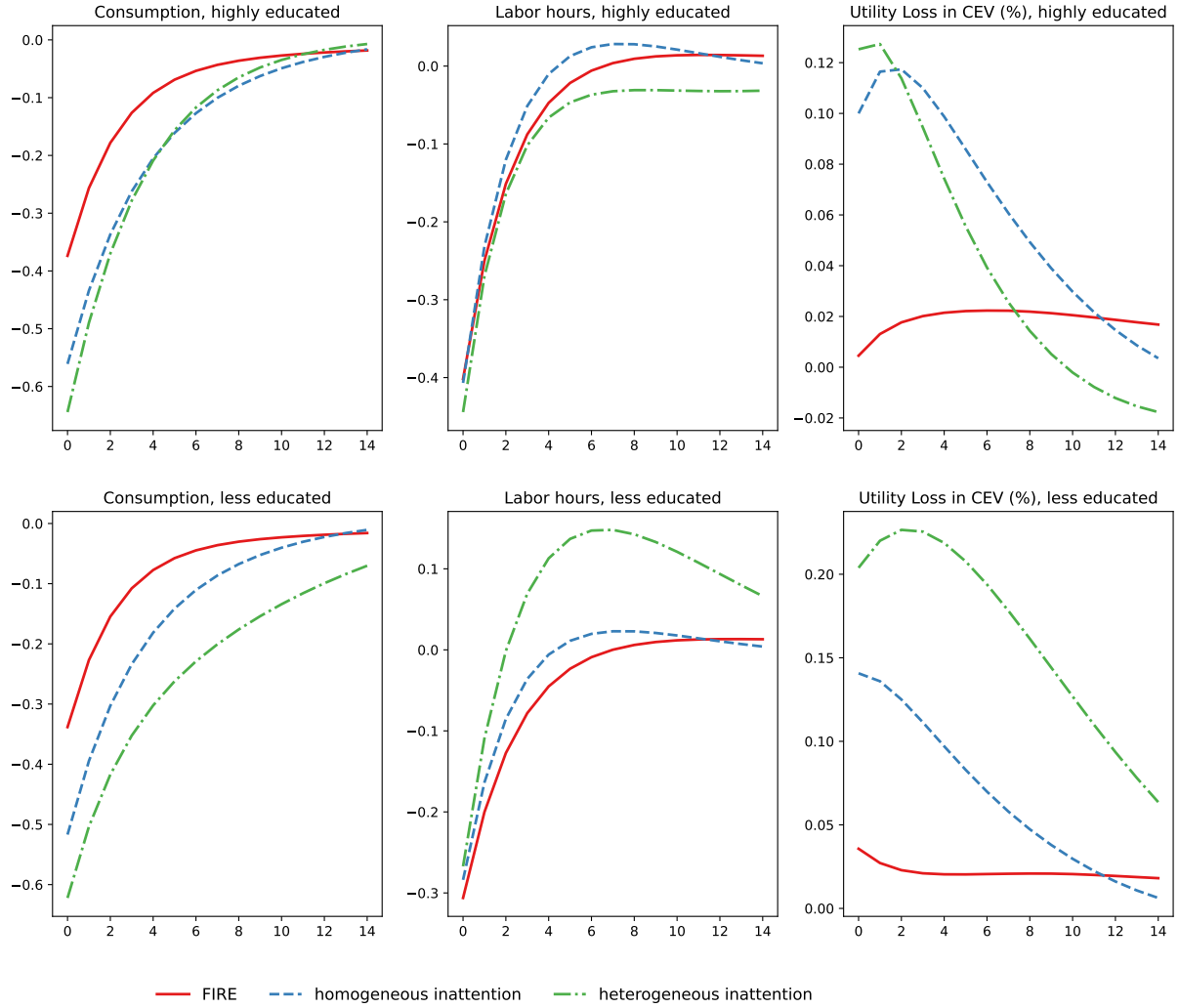
educated households who, given their greater misperception of real labor income, decrease their level of consumption for a longer period of time, under lower level of attention, as shown in Figure 18. Highly educated agents, on the contrary, even decrease their labor supply since they pay more attention relative to the homogeneous case and observe the positive change in their real labor income more precisely. Hence, as long as households misperceive their income they increase their number of hours worked. Once they adapt their expectations closer to the actual level of inflation, they decrease their labor supply as there is no need for them to work more given their increase in real income. Depending on the speed of learning these adjustments happen faster or slower. These adjustments of labor supply and consumption-saving then transmit to the aggregate in Figure 17. The nominal wage adjusts according to the new equilibria of labor supply and labor demand. Due to nominal rigidity, the wage adjusts with a lag. If all household groups have the same level of attention (homogeneous attention), consumption of highly educated households decreases slightly more in response to a contractionary monetary policy shock, compared to the less educated households. Given that highly educated households observe the changes in inflation levels more correctly than less educated households, and they also observe the general recession induced by an increase in inflation, they smooth consumption while increasing savings, benefiting off of the

Figure 17: Homogeneous and Heterogeneous Inattention



increase in interest rates. As their consumption levels are higher, in consumption equivalent terms, they lose up to 50% of what less educated households lose, as shown in Figure 18. Less educated households have lower levels of consumption and do not have enough savings to smooth their consumption, thus, they work more and experience a bigger negative welfare impact.

Figure 18: Homogeneous and Heterogeneous Inattention



E.8. Attention to Price Level as a Function of Inflation: Model Differences

As derived in Appendix D.3, the expectational processes between the price level and inflation differ by $(1 + \gamma^g)^{t+1} \pi_t$. The difference in the simulation of our main quantitative HANK model from Section 4 with and without this term is shown in Figure 19.

Compared to Figure 1, the results would slightly be amplified if the correction term were ignored. Overall, the impact is very small and negligible. Without the term GDP, inflation, consumption and the average wage level would be slightly more negative, labor hours would recover slightly less after the medium run and the central bank's reaction to inflation would be slightly less strong, especially at the medium to long run.

Figure 19: Comparison of the Quantitative Model Results With and Without the Additional Term in Price Expectation Formation Process

